Customer Support Portal Version 6 Development Specification

Developer handoff and production acceptance

Version 6 | 30 October 2026 | Review specification

Build a production support application using Option A's shared workspace and Option B's guided checks for consequential actions. Preserve the reviewed Version 6 workflows while replacing simulated identities, tab-local data and browser timers with authenticated services, durable records and reliable background processing.

This specification is for the delivery team, product owner and UAT reviewers. It defines module behaviour, roles, approvals, data controls, production requirements and observable acceptance criteria. It is a development contract for review, not a claim that the prototype already provides production security or integrations. The prototype has not been changed for this document.

The priorities are fast manual intake, controlled selection instead of repeated typing, a shared customer history, clear ownership and visible follow-ups. Consequential actions must retain exact item and amount approvals, physical receipt history and independent finance obligations. Managers need complaint and operational analysis with usable exports.

Use the module criteria with the cross-cutting criteria and closure matrix. Passing one module does not permit bypassing another module's outstanding obligations. Resolve the decision register before implementing the affected policy. No developer should infer a monetary limit, SLA target, self-approval rule, retention period or integration credential from demo data.

Baseline identity

Published commit a5b3b2b9be1fa3860382ec549beeaaad6c887cb0

Review URL: https://cs-portal-guided-review.ramesh-s325387.chatgpt.site/

No production application changes are authorised or implied by this document. Build and test against the approved requirements and the decisions recorded here.

How to use this specification

Start with the role matrix and production requirements, then follow the module relevant to the work being implemented. Each module records its Version 6 baseline, fields, workflow rules, permissions, production gaps and acceptance criteria. Module acceptance criteria without a narrower label describe the target functional contract; they do not certify that production services already exist.

Status	Meaning
Baseline	Observed Version 6 behaviour to preserve unless a documented product decision changes it.
Production requirement	Implementation required for a dependable live service; it may be absent or simulated in Version 6.
Recommendation	Proposed improvement or control requiring product or technical owner agreement where it changes policy.
Decision required	A business rule or scope choice not established by the available requirements or prototype.

Priority: preserve approved functional behaviour and implement the production requirements before live use. Items explicitly marked Recommendation or Decision required are not silently authorised policy changes. Cross-cutting criteria apply to every module. Native prototype dropdowns must become searchable selectors for large lists in production.

The document uses stable module and criterion references. Source line references identify the exact baseline commit; they may change as developers implement the live system. Word headings and PDF bookmarks provide navigation.

Module reference	Module	Page
CS-M01	My work and shared team overview	14
CS-M02	Shared customer directory and historical records	15
CS-M03	Protected corrections, legacy import and duplicate review	18
CS-M04	New Query and manager classification	20
CS-M05	All Tickets, collaboration, conversation and lifecycle	23
CS-M06	Follow-ups, deadlines and reminders	26
CS-M07	Global search and section filters	29
CS-M08	Insights, Voice of the Customer and missed follow-ups	31
CS-M09	Exports and downloadable evidence	34
OPS-SERVICE	Service approvals and controlled revisions	40
OPS-CANCELLATION	Cancellation execution and linked return fallback	42
OPS-REPLACEMENT	Three replacement arrangements and dispatch	44
OPS-RMA	RMA registration, tracking, cumulative receipt and inspection	46
OPS-WAIVER	Existing-RMA remaining collection waivers	48
OPS-PARCEL	Unmatched parcels and identity matching	50

Module reference	Module	Page
OPS-REFUND-AUTH	Manual refund authorisation and amount revisions	51
OPS-REFUND-ATTEMPT	Distinct external refund attempts and reconciliation	54
OPS-DISPUTE	Manual dispute evidence and provider outcome	56
OPS-CLOSURE	Customer confirmation, resolution, closure and reopening	57
ADM-USERS	User Management	64
ADM-ROLES	Roles, grants and effective permissions	66
ADM-SCOPE	Shared visibility, operational scope and task eligibility	68
ADM-PROFILE	Profile & Security and User Profile Requests	71
MASTER-CATEGORIES	Category Management	73
MASTER-PRODUCTS	Product Management	75
MASTER-CHANNELS	Query Channels	77
MASTER-TYPES	Query Types	78
MASTER-CAPACITIES	Query Filters (product capacities)	79
MASTER-OUTREACH	Customer Outreach	80
MASTER-STORES	Purchase Stores	81
MASTER-COUNTRIES	Countries and warehouse coverage	83
MASTER-STATUSES	Query Statuses (fixed lifecycle states)	84

Scope and source authority

The source baseline is the published Version 6 release of 30 October 2026 and its exact tracked source. User requirements take precedence over older audit notes. The final release guide explains corrected behaviour; older sections in that guide are historical and do not supersede the final release. Where the user requirement goes beyond the prototype, this document identifies production work explicitly.

Launch is a manual operating model. Agents look up a customer, select controlled classifications and record external actions with evidence. No launch requirement assumes that an email has been sent, a warehouse has received goods, a carrier has shipped goods or a provider has paid a refund merely because a portal button was clicked. All warehouses support RMA. A manager-approved exception may waive a return obligation; it does not imply that RMA is unavailable in that country.

A single shared customer directory spans the authorised support team. Brand is an attribute of tickets, products and orders, not a separate customer identity silo. All authorised support users can find shared ticket history. Permissions and operational country or warehouse assignments control mutations. Any production restriction on sensitive customer fields needs an explicit approved policy; do not silently replace shared visibility with brand silos.

Future email ingestion, automatic ticket creation, Shopify or order-system verification, carrier updates, warehouse feeds and payment execution are integration phases. Prepare stable identifiers, command contracts and external-event deduplication now. Do not present an integration as working until its independent implementation and reconciliation tests pass.

Interaction and content standards

Use plain English familiar to the Indian CS team. Use Customer, New Query, Main category, Child category, Product, Assigned to, Working now, Last change, Need information, Return waiver and Refund authorisation consistently. Keep lifecycle status separate from approval status, shipment status and payment outcome. Use Top 10 customer complaints, never compliance, for the requested rankings.

Provide keyboard-searchable selectors for every large controlled list. Typing filters authorised existing choices by label, SKU and approved aliases; it must not create a category, query type or product. Show the parent path for child categories. Preserve a selected historical inactive value visibly while preventing its selection on new work. Small finite state lists may use native selectors.

Look up email or international mobile number before showing the issue form. Show existing customers, open tickets and historical complaints or enquiries. Allow review and return without losing the intake draft. Ambiguous shared contacts require the agent to choose the correct person; never auto-merge people by phone suffix or household email.

Pre-fill verified customer and product information, generate the ticket title from selected classifications and keep discretionary narrative short. Customer words remain unedited historical evidence. Copying a reply must not mark it as sent. Confirmation copy must name the exact request, revision, items, quantities and financial amount where applicable.

Display operational dates in a consistent configured zone, initially Asia/Kolkata for the CS UI, with an unambiguous format such as 30 Oct 2026, 14:30 IST. Persist UTC instants and the applicable business timezone. Export ISO timestamps with timezone information. Show the currency code beside amounts; do not mix currency totals. Numeric and date inputs must work with keyboard navigation and retain valid draft data after errors.

Use accessible names, logical focus order, visible focus, inline error summaries and keyboard-operable dialogues. On narrow screens, keep identity, effective role and scope readable and make tables usable

through deliberate scrolling or a labelled compact view. The recommended accessibility target is WCAG 2.2 AA, subject to the delivery team's documented conformance review.

Role permissions and approval authority

Effective grants are the union of all assigned role grants. There are no deny grants, inherited role trees, brand scopes or per-user grant overrides in the prototype. Scope is one user-level string, not one scope per role.

Y means the default role has the capability; a dash means it does not. SA = Super Admin; CS = CS Executive; CM = Country Manager; Head = Head of CS; Log = Logistics; Fin = Finance; DS = Data Steward. These are default compositions, subject to the explicit launch assignment decision. Operational scope and record-state rules still apply.

Capability	SA	CS	CM	Head	Log	Fin	DS
read	Y	Y	Y	Y	Y	Y	Y
work	Y	Y	Y	Y	-	-	-
assign	Y	Y	Y	Y	-	-	-
correction	Y	-	-	-	-	-	Y
merge	Y	-	-	-	-	-	Y
importPreview	Y	-	-	-	-	-	Y
importPublish	Y	-	-	-	-	-	Y
logistics	Y	-	-	-	Y	-	-
finance	Y	-	-	-	-	Y	-
dispute	Y	-	-	-	-	Y	-
insights	Y	Y	Y	Y	Y	Y	-
export	Y	-	Y	Y	-	Y	-
masters	Y	-	-	-	-	-	-
accounts	Y	-	-	-	-	-	-
users	Y	-	-	-	-	-	-
roles	Y	-	-	-	-	-	-

Role	Approval authority and qualifications
Super Admin	All markets by reserved role ID; all grant checkboxes are locked on; name must remain Super Admin.
CS Executive	None from this role. Its read grant cannot be removed.
Country Manager	Reserved role ID cm; exact ticket country or All markets. All warehouses is not a manager-approval wildcard for cm.
Head of CS	All markets by reserved role ID head.
Logistics	None.
Finance	None.

Role	Approval authority and qualifications
Data Steward	None. Seeded role has no initially assigned user; a synthetic assigned user was used in validation.

No grant named approve exists. Final service, return-waiver and refund decisions use reserved role IDs super/head/cm plus scope; renaming a custom role to a managerial title does not confer authority.

Country Manager and Head of CS do not receive protected correction, import, merge, users, roles, accounts, masters, logistics or finance grants by default. Super Admin has all grants; Data Steward has correction/import/merge grants.

The users grant opens User Management. A roles-only custom account has no independently exposed role-management page in the normal navigation; creating/editing roles through the normal UI effectively also needs access to User Management.

Shared workspace read and bulk export are distinct capabilities. There is no separate PII mask/unmask grant in this prototype.

The builtin flag is descriptive except where code explicitly protects an ID; it does not make every built-in role immutable.

Capability definitions

Capability	Meaning
read	Shared workspace read access; ordinary ticket/customer lists are not restricted to the user's country. Required alongside the task grant by shared owner eligibility.
work	Ticket/customer intake and ticket-work actions. Does not by itself permit reopening a terminal ticket or final manager approval.
assign	Assignment and explicit reopening, subject to action-specific scope and work prerequisites.
correction	Approve/reject protected customer email and order-link corrections. Manager role alone does not add this grant.
merge	Opens the duplicate-customer review demonstration. Irreversible merge is not implemented.
importPreview	Opens a synthetic import preview. There is no arbitrary file ingestion.
importPublish	Publishes the canned reviewed demo row once. Does not implement a production import pipeline.
logistics	Logistics actions including return/parcel tasks and fulfilment. Does not grant refund-result entry.
finance	Manual external refund attempt/result entry with active/read/scope and exact authorisation checks. Does not send a payment.
dispute	Dispute work is a separate capability. Shared dispute-owner eligibility requires read plus dispute, not an extra finance grant.
insights	Access to reporting views.
export	Actual Exports screen offers Tickets, Customers, Activities, Service requests, Returns, Return waivers, Refunds and Follow-ups. The label is narrower than the implemented dataset list.
masters	Create/update/archive permitted catalogue and reference lists. Query Statuses remains read-only even for Super Admin.

Capability	Meaning
accounts	Admin reset queue and review. No invitation delivery or reset link is sent.
users	User Management entry, account changes and assignment of existing roles. There is no separate grant for assigning a privileged role.
roles	Role creation/editing. The source also permits editing most built-in role grants, with specific Super Admin and CS safeguards.

The seven seeded demo accounts illustrate compositions only. Create production accounts from an approved roster, with authenticated verified work email and operational scope. Data Steward is a seeded custom role with no initial demo assignee. A person can hold multiple roles; do not grant a capability merely because a job title sounds managerial.

Source references: dist/data.js:3-6; dist/portal.js:313-317; dist/operations.js:8-8

Production architecture and boundaries

Use an authenticated web client backed by an authorised application API and transactional persistent storage. Keep the exact technology stack a delivery-team decision. Domain services should own customer identity, ticket lifecycle, service approvals, returns, fulfilment, finance, task scheduling and catalogue access rules. Reuse the same policy functions across screens, background jobs and exports.

A successful command must durably commit its business record, audit event and next task or outbox event together. A failed validation must leave all of them unchanged. A durable worker processes reminders, escalation, export generation and integration deliveries. Worker retries must not duplicate a task, shipment, receipt, payment or customer notice.

Use protected object storage for evidence files, with file type and size controls, malware checks, download authorisation and expiring access. Store metadata, checksum, uploader and linked record identity. Evidence may be a structured reference and explanation at manual launch; do not require customers' full emails or payment details to be copied unnecessarily into logs.

Reporting can use transactional queries initially and a projection or warehouse as volume grows. Publish the report's refresh time and filters. An asynchronous projection must not be used as the authority to approve, dispatch, refund or close. Live commands always re-evaluate current authoritative balances and permissions.

Separate production, staging and development. Use synthetic data in development. Keep secrets outside the client and source repository. Version schemas and commands; apply reversible migrations where practical, and document recovery for irreversible migrations before execution.

Identity permission and concurrency controls

Replace Demo role with real authentication and server-derived identity. Recommended controls are an organisation identity provider, MFA for privileged users and finance, server-managed sessions, session revocation and step-up confirmation for high-impact administration. The prototype's named people are examples, not an instruction to create production accounts.

Evaluate active account state, effective grants, role-based approval authority, country or warehouse scope and record state on every read or command as applicable. Hiding a button is not authorisation. Custom role names must never confer manager approval authority. The role matrix defines the current default composition; multiple assigned roles form the union of grants, while the authority rule remains explicit.

Account or role changes that would strand active work must identify the affected records and require handover to eligible owners first. Recheck at commit so another session cannot create stranded work between preview and save. Keep at least one active Super Admin. Privileged changes invalidate relevant sessions or permissions caches promptly and are audited.

Use a record version or equivalent optimistic lock for ticket updates, approval decisions, receipt changes, scope revisions and master changes. If a record changes after review, refuse the stale command with a clear explanation and retain the user's draft for re-review. A final approval click must recheck the exact version and balance, not rely on the dialogue's old snapshot.

Working now is a collaboration claim distinct from assignment and task ownership. Production requires a durable claim with heartbeat or lease, explicit takeover and visible actor/time. Define lease expiry behaviour with the business owner. A stale tab cannot silently overwrite another person's work. Opening a record should display current activity without claiming ownership automatically.

Data model and integrity contracts

Use immutable internal identifiers and separate human-readable references. Retain source-system identifiers and import batch provenance. Store current labels separately from approved historical snapshots so renaming a category, product, user or role does not rewrite what was selected or approved earlier.

Customer identity must support primary and historical contact aliases, normalised email and international phone values, and deliberate shared-contact disambiguation. A verified identity correction updates lookup indexes atomically while preserving previous values and linked ticket ownership. Do not use an email string as the permanent customer primary key.

Use a composite external order namespace including store or provider, brand and external order reference. Bind each verified order to the selected customer and ticket context. Customer history does not prove that today's order or product belongs to the same case. Manual verification must record verifier, evidence, time and scope until an authoritative order integration exists.

Represent service requests and approvals as revisioned aggregates. Each approval retains immutable inbound and outbound lines, quantities, arrangement, reason, approver identity and roles, country, time, evidence and approval state. Do not mutate an approved snapshot. Changes to governed scope create a fresh revision and fresh decision; executed goods and payments cannot be rewritten.

Store physical receipts as append-only item events and adjustments with reasons, never as a replaced total. For each RMA line, show original expected, physically received, effective waived and still due. A waiver is a decision about collection, never a physical receipt. Late arrival changes current effective waived quantities without erasing the original approval. Calculate all balances with nonnegative constraints and concurrency protection.

Store money as integer minor units or an exact decimal with a currency-specific scale. Each refund authorisation has a currency and immutable approved revision. Successful and unknown or pending attempts consume or reserve available authority; failed attempts release it. Both successful and failed attempts are final and cannot be overwritten. Rejecting an amount amendment restores the previous approved authority rather than rejecting that prior approval. Never add amounts across currencies without an explicitly labelled conversion method. FX conversion and currency rounding policy require agreement.

Every business event needs occurredAt, recordedAt, actor ID, actor role snapshot, object ID, object revision, action, reason where required and correlation ID. Audit entries also contain safe before/after values. Imported history must distinguish known source facts from unknown actor/time history; do not fabricate events to make a timeline look complete.

Entity group	Required relationships and constraints
Customer and contacts	One shared customer ID; multiple primary or historical contacts; explicit shared-contact relationships; immutable correction and merge decisions.
Ticket and classification	Customer, brand, country, optional verified order, selected query type, category IDs and label snapshots, product IDs, channel, capacity, creator, current worker, lifecycle, resolution episodes.
Service request and revision	Ticket plus revision; inbound and outbound item lines; arrangement; decision history; exact remedy confirmation; no orphan executed revisions.
RMA receipt and waiver	Service revision link; expected lines; physical receipt and inspection ledger; waiver proposal snapshot; original decision and effective balance; notification evidence.
Replacement and cancellation execution	Approved scope link; unique shipment or cancellation evidence; provider reference; delivery confirmation; explicit fallback Return link when cancellation fails.
Refund and attempt	Verified order and optional remedy link; authorisation revisions; unique external attempt reference; reserved and paid amounts; evidence; reconciliation and customer notice.
Dispute	Ticket and provider reference; provider deadline; evidence receipt; submission and outcome history; independent follow-up tasks.
Task and audit	Unique active source and step; eligible owner; original and current due times; reminder time; first-miss owner; immutable change history; completion occurrence and recording times.
User role and scope	Active account, role memberships, explicit capabilities, operational scopes, change history; final active Super Admin protected.
Catalogue import and export	Stable master IDs, active state, dependencies and usage; import batches and row outcomes; export filters, snapshot, actor, row count, expiry and checksum.

Command and API contracts

The following are proposed logical commands, not claims that production endpoints exist. Each mutation accepts the authenticated actor implicitly, object ID, expected version, client request or idempotency key, payload and meaningful reason or evidence where required. The server supplies actor, role snapshot and recorded time; client-provided role claims are ignored.

Return structured validation errors tied to fields and actionable conflict responses. Use distinct unauthorised, forbidden, not found, conflict and validation outcomes without leaking records outside the allowed visibility policy. A command result should include the committed version, resulting workflow state and any created or completed task references.

For a repeated idempotency key and identical payload, return the original committed result. A changed payload with the same key must fail. Use provider event IDs and unique references independently to prevent duplicate ingestion or payment evidence. Define an idempotency retention window with the integration and finance owners.

List and export endpoints share the same canonical filter grammar. Filters must be applied before pagination. Provide stable ordering with an ID tie-breaker. Persist export query, timezone, date boundary convention, snapshot time, selected fields and row count so a downloaded file can be reconciled to the displayed result.

Command family	Essential validation and result
Lookup customer and create ticket	Normalised contact lookup first; explicit ambiguity selection; valid active controlled choices; one new ticket and one customer link; no duplicate submit.
Claim update assign and reopen	Current grant and scope; version match; explicit takeover; reopening retains resolution history; closure guard evaluates every linked obligation.
Submit revise and decide service	Eligible requester and decision owner; exact revision; quantity and order declaration; explicit exceptional assent; durable decision and next tasks.
Record receipt inspect and waive	Item-level remaining balance; immutable receipt evidence; stale waiver invalidation; manager assent; late arrival reduces effective waiver without rewriting history.
Dispatch or record cancellation	Release conditions checked transactionally; unique execution reference; no second dispatch; fallback Return covers all affected cancellation lines.
Authorise and record refund	Exact currency and amount; manager authority; reserved balance; unique external reference; immutable success; verified outcome transition and reconciliation.
Complete or reschedule task	Stage-eligible owner; explicit source transition; reason for deadline change; first missed deadline preserved; superseded work cancelled with reason.
Manage account role or master	Effective-permission preview; last-admin and stranded-work checks; dependency and historical-reference protections; complete administrative audit.
Import publish and export	Authorised preview and commit; deduplicated source IDs; row validation; safe CSV encoding; durable export status and protected download.

Background processing evidence and resilience

A durable scheduler evaluates due-soon reminders, overdue tasks and manager attention whether any browser is open. Version 6 reminds one hour before due; preserve this default until the business owner agrees another rule. Snooze affects reminder time only. Escalation does not change the task owner, deadline or first-miss record. Email or chat notifications are future channels unless explicitly included in launch.

Use a transactional outbox or equivalent reliable mechanism for workflow tasks and external notifications. Retry with bounded backoff, expose failed jobs to operations and support safe replay. Deduplicate on source record, revision and step. On recovery, process missed schedules without erasing overdue evidence or generating repeated escalation events for the same threshold.

Audit entries are append-only for ordinary application users, including Super Admin. Corrections create new events with authority and reason. Protect records against tampering and restrict audit exports. Record account/role changes, denied sensitive commands, approvals, receipts, waivers, payment attempts, exports and task ownership/deadline changes. Exclude passwords, tokens and unnecessary full payment details.

Define backups, restoration drills, monitoring, incident contacts and data recovery with the platform owner. Monitor command errors, job age, failed exports, permission denials, duplicate integration events, stuck approvals, inconsistent balances and overdue workflows. Correlation IDs must connect UI errors to safe operational logs.

Proposed starting service objectives, to validate against volumes: normal saves and searches at p95 within 2 seconds, initial usable view within 3 seconds and scheduler processing within 60 seconds of a due transition. Large exports should be asynchronous with visible progress. Agree dataset size, concurrent users, peak event rates, uptime target, recovery point and recovery time before treating these values as contractual.

Migration and integration readiness

The requested launch includes both imported history and newly entered records. Obtain the real legacy customer, ticket, catalogue, user, order and operational extracts and agree which histories are available. The old portal audits and prototype import example do not establish an actual production source schema or authorise guessing missing data.

Map stable source IDs to target IDs. Stage each import with schema validation, duplicate detection, canonical contacts, explicit date/timezone parsing, category and SKU mapping, row errors and reconciliation counts. Preview records as new, unchanged, update, stale conflict, ambiguous identity or rejected. Publish only reviewed eligible rows; retain the batch and per-row outcome.

Protect live edits during import with source version or last-modified comparisons and optimistic locking. Never overwrite a newer production edit because an older spreadsheet was reuploaded. Re-running a batch must not duplicate customers, tickets, receipts, refunds or task history. Unknown historical actors and timestamps stay labelled unknown; imported closed tickets must not automatically generate false current overdue tasks.

Dry-run the migration in staging, reconcile per-source counts and important balances, sample real-world edge cases with authorised masked data, and obtain CS, Logistics and Finance sign-off. Rehearse cutover, final delta, rollback and read-only legacy access. Preserve an import-to-source crosswalk for support and audit.

For later email ingestion, retain message and thread IDs, deduplicate provider events and use confidence-based customer matching with a human review queue for ambiguity. Do not auto-approve service actions, infer physical receipt or close a ticket from an unanswered email. For later warehouse, carrier and finance integrations, store raw provider event references and reconcile conflicting or out-of-order events before changing authoritative state.

Cross cutting acceptance criteria

All criteria in this section are production requirements. Run them on the implemented persistent application, including direct API and concurrent-session tests; the browser prototype cannot establish these guarantees.

PLAT-01 *[Production requirement]* **Given** Two sessions open the same pending approval revision. **When** One approves and the other submits a stale decision. **Then** Only the first valid command commits; the second reports the changed state, preserves its draft and creates no duplicate approval or task.

PLAT-02 *[Production requirement]* **Given** A browser modifies its visible role or sends a direct restricted API request. **When** The server evaluates the command. **Then** Authority comes from the authenticated active user and current server records; unauthorised mutations and data exports are denied and safely logged.

PLAT-03 *[Production requirement]* **Given** An account or role change would make an open task owner ineligible. **When** An administrator attempts to commit the change. **Then** The change is refused with an actionable list until eligible handover is complete; a concurrent task creation cannot bypass the protection.

PLAT-04 *[Production requirement]* **Given** No browser is open and a task reaches its reminder or due time. **When** The scheduler runs or recovers after an outage. **Then** Reminder and escalation states update durably once per event, preserving the original deadline, first-miss time and owner at the miss.

PLAT-05 *[Production requirement]* **Given** A command commits but the response is lost. **When** The client retries with the same idempotency key and payload. **Then** The original result is returned with no additional ticket, receipt, shipment, payment attempt or task; a changed payload under that key is rejected.

PLAT-06 *[Production requirement]* **Given** Receipt and waiver approval commands race on the same RMA balance. **When** Both attempt to commit. **Then** Only a decision based on the current balance can succeed; received quantities are never waived, the stale proposal is superseded or rejected and both attempts remain traceable.

PLAT-07 *[Production requirement]* **Given** Two finance sessions attempt to reserve the final available refund amount. **When** Both commands execute. **Then** The committed paid plus reserved amount cannot exceed approved authority; the second conflicting command is rejected without changing historical attempts.

PLAT-08 *[Production requirement]* **Given** A closure screen was opened before another user created an obligation. **When** Closure is submitted. **Then** The server rechecks all current requests, returns, waivers, deliveries, refunds, disputes and tasks and refuses premature closure with the blocking records identified.

PLAT-09 *[Production requirement]* **Given** A service or payment action succeeds but its background task worker fails. **When** The outbox job is replayed. **Then** The next task is created once, the action remains committed and the failure and recovery are observable without rewriting history.

PLAT-10 *[Production requirement]* **Given** A protected export is generated and the user's access is later revoked. **When** That user requests the download. **Then** Current download authorisation is rechecked and access is refused; an expired link also fails without disclosing file contents.

PLAT-11 *[Production requirement]* **Given** An import batch is published and a linked record is edited by a live agent. **When** The same or an older batch is run again. **Then** Existing source IDs do not duplicate records and the newer edit is held as a conflict rather than overwritten; reconciliation counts remain available.

PLAT-12 *[Production requirement]* **Given** A customer or master label changes after an approval. **When** Users open the request and export its history. **Then** Current lookup labels and original approved snapshots are distinguishable and the historical item, approver and amount scope remains unchanged.

PLAT-13 *[Production requirement]* **Given** A user navigates only by keyboard. **When** They look up a customer, select controlled values and submit or correct a request. **Then** All controls are reachable, focus is visible, errors identify their fields and the user can return from a dialogue without losing valid draft data.

PLAT-14 *[Production requirement]* **Given** A database and evidence-storage backup is restored in an isolated environment. **When** The team reconciles key records and resumes workers. **Then** Ticket and financial balances, attachment references, audit and task history are consistent; jobs resume without duplicate actions and measured recovery meets the agreed targets.

PLAT-15 *[Production requirement]* **Given** An approved performance dataset and concurrent-user profile. **When** Search, intake, approval, exports and scheduled jobs run under load. **Then** The agreed response-time and scheduler targets are met, no unauthorised data leaks occur and all balances and counts remain correct.

PLAT-16 *[Production requirement]* **Given** A test user types an unknown category or query type in a searchable selector. **When** They attempt to continue. **Then** No new master record is created; the user must select a valid item or the approved Needs classification path, while their customer lookup and draft remain available.

PLAT-17 *[Production requirement]* **Given** An authenticated user at 350 pixel and 390 pixel viewport widths. **When** The header and navigation render. **Then** The full current name, roles and operational scope remain readable, navigation works and no identity or essential action is clipped.

CS M01 My work and shared team overview

Give the executive a fast starting point, with direct routes to shared tickets and personal obligations without duplicating entry.

Version 6 baseline

Implemented: Five linked cards: Open team tickets, My next actions, Overdue follow-ups, Pending decisions, Rejected — action needed; All team tickets and Tickets assigned to me views; first six matching ticket rows and a route to the complete set; My upcoming actions shows up to four open tasks owned by the current person, ordered by deadline; it can include already-overdue tasks; Ticket rows distinguish Assigned to, Working now, lifecycle status, approval badges and next task. Date is shown using Indian English and IST.

Boundaries: Counts are recomputed from this browser tab's synthetic records. They are not cross-session or real-time server aggregates; Open team ticket count is a ticket count. The other four cards count tasks; one ticket may have multiple independent tasks; The team overview is shared and not restricted to the current owner. Country-specific manager attention is a different view.

Fields and validation

Field	Type	Rules and meaning
openTeamTickets	integer	Required: Yes; Description: Count non-test tickets whose status is neither Resolved nor Closed.
myNextActions	integer	Required: Yes; Description: Count open tasks with ownerId equal to the signed-in user's ID.
overdueFollowUps	integer	Required: Yes; Description: Count all open tasks with dueAt strictly before now.
pendingDecisions	integer	Required: Yes; Description: Count open tasks whose step is decision, across service, waiver and refund sources.
rejectedActionNeeded	integer	Required: Yes; Description: Count open rejection-stage tasks; not a count of every historically rejected approval.
ticketPreview	list	Required: Yes; Description: Ticket ID, customer, subject/category, assigned owner, worker, status, decisions and next action.
upcomingActions	list	Required: Yes; Description: Action title, ticket/source, deadline, overdue/upcoming indicator and Open action.

Workflow and rules

Each card must navigate with filters that reproduce its displayed count.

My next actions follows task ownership, independent of the ticket's assigned owner.

Closed/Resolved tickets are excluded from the open ticket list; prior records remain searchable.

Changing a dashboard view must not reassign records or silently complete work.

Empty states remain meaningful, including zero rejected actions.

Permissions

Operation: View overview/shared records. Required: read. Scope: Shared authorised CS workspace; not per-agent isolation.

Operation: Create query. Required: work. Scope: Action handler enforces capability.

Operation: Act on a linked task. Required: Stage-specific capability and applicable country scope. Scope: Opening a task does not grant decision authority.

Production requirements

CS-M01-P1: Requirement: Use server-side count and list queries with identical filters and an explicit snapshot/as-of time to prevent mismatched cards and destinations.

CS-M01-P2: Requirement: Refresh relevant counts after committed changes and expose stale/loading/error states; persist each user's preferred view without changing shared records.

CS-M01-P3: Requirement: Use stable user IDs for assignment and personal counts. Current prototype name snapshots remain display/audit data.

Acceptance criteria

CS-M01-AC01 *[Preserve baseline]* **Given** Non-test tickets exist in active and terminal states. **When** The user opens My work. **Then** Open team tickets counts only active non-test tickets; opening the card lists that same set.

CS-M01-AC02 *[Preserve baseline]* **Given** A user owns tasks on tickets assigned to different executives. **When** The user opens My next actions. **Then** All and only that user's open tasks are listed, independent of ticket ownership.

CS-M01-AC03 *[Preserve baseline]* **Given** An open task is due exactly now and another is past due. **When** The dashboard evaluates deadlines. **Then** Only the strictly past-due task is overdue; no task is completed automatically.

CS-M01-AC04 *[Preserve baseline]* **Given** A rejection has an unfinished customer-update task. **When** The user opens Rejected — action needed. **Then** The task is visible; after its actual completion it leaves this queue while the rejection history remains.

CS-M01-AC05 *[Preserve baseline]* **Given** More than six tickets match the selected view. **When** The user opens View matching tickets. **Then** The complete matching result is available through pagination; the six-row preview is not treated as the full dataset.

CS-M01-AC06 *[Preserve baseline]* **Given** No actions are assigned to the user. **When** The user opens the overview. **Then** Zero counts and an explanatory empty personal-actions state are shown.

CS-M01-AC07 *[Production requirement]* **Given** A shared ticket or task is changed by a second authenticated user. **When** The first user's dashboard refreshes. **Then** Counts and linked lists converge on the committed state without requiring a browser reset.

Source references: dist/portal.js:51; dist/portal.js:19; dist/portal.js:451; dist/operations.js:60

CS M02 Shared customer directory and historical records

Create each customer once and allow executives to reuse that identity and review prior complaints/enquiries across brands before creating another ticket.

Version 6 baseline

Implemented: One shared customer directory with customer ID, name, canonical email/phone, country, retained email aliases and record version; Customer list searches names/contact data plus related ticket and order references; Customer detail shows all non-test historical tickets across brands, statuses and owners, with customer details and retained aliases; New customer creation requires a trimmed non-empty name and at least an email or international phone number; Duplicate contact prompts require review. An explicitly confirmed different household member may have the same contact while keeping a separate customer ID.

Boundaries: Customer history consists of linked portal tickets and their order references; there is no connected commerce order history or email inbox; Country is a customer attribute copied to new tickets. Brand is selected per ticket rather than used to split the customer directory; No irreversible customer merge occurs in this build; The different-person confirmation is a UI validation gate; the prototype does not persist a full identity-review case for that exception.

Fields and validation

Field	Type	Rules and meaning
customerId	immutable identifier	Required: Yes; Description: Generated customer identity; all tickets keep this foreign key through approved email corrections.
name	trimmed string	Required: Yes; Description: Spaces-only is invalid; display name is not an identity key.
email	canonical email string	Required: No; Description: Trim/lowercase and strip mailto prefix; combined with phone requirement. Retained aliases are searched.
phone	international canonical string	Required: No; Description: Stored with + and digits; country code is required for new customer creation.
country	master-data reference	Required: Yes; Description: An active country selection; no country inferred from language or brand.
aliases	email array	Required: Yes; Description: Prior approved email identities, canonicalised and deduplicated.
version	integer	Required: Yes; Description: Used for stale correction checks.
historicalTickets	related records	Required: Yes; Description: Customer-linked tickets with ticket ID, issue, category, brand, order, state and assigned/working users.

Workflow and rules

Use one shared directory across brands; keep brand and store context on relevant tickets/orders.

A common phone/email may represent several people. Never merge or select a person solely because contacts match.

New customer entry is blocked on duplicate contacts unless the executive explicitly confirms this is a different person after reviewing history.

Email duplicate checks include both primary addresses and aliases. Phone identity comparison uses canonical full numbers, not suffixes.

New issue for this customer reuses customerId, but the executive must select the brand/current issue details again.

Customer history remains visible across team members; permission to read history is separate from permission to mutate or export it.

Permissions

Operation: List/view customer history. Required: read. Scope: One shared authorised directory across brands.

Operation: Create customer. Required: work. Scope: Requires validation and duplicate review.

Operation: Request correction. Required: work. Scope: Creates a pending correction; does not immediately alter identity.

Operation: Review possible duplicate. Required: merge. Scope: Review/keep-separate only in Version 6; real merge is not implemented.

Production requirements

CS-M02-P1: Requirement: Persist customer/contact identities separately from ticket snapshots. Use indexed canonical email/phone and explicit shared-contact relationships; permit legitimate household duplicates without a blanket unique-contact constraint.

CS-M02-P2: Requirement: Persist the actor, reason/evidence and timestamp for a confirmed different-person exception, and expose a reviewable identity-conflict case.

CS-M02-P3: Requirement: Apply authenticated read and field-level access on the server before returning customer data. Preserve authorised shared CS visibility; never send restricted fields and hide them only with CSS.

CS-M02-P4: Requirement: Design paginated history and retained source references for approved legacy migration and any later commerce-order history. Do not invent historical contacts, dates, ownership or order contents.

Acceptance criteria

CS-M02-AC01 *[Preserve baseline]* **Given** One customer has Brand A and Brand B tickets. **When** An authorised executive opens that customer. **Then** Both brands' histories appear under the same customer ID and retain each ticket's brand/order.

CS-M02-AC02 *[Preserve baseline]* **Given** The executive submits a name containing only spaces. **When** They save a new customer. **Then** Creation is blocked with a clear trimmed-name error and no blank record is created.

CS-M02-AC03 *[Preserve baseline]* **Given** Neither email nor phone is supplied. **When** The executive saves. **Then** Creation is blocked until at least one valid contact is provided.

CS-M02-AC04 *[Preserve baseline]* **Given** A phone is entered without an explicit international prefix for a new customer. **When** The executive saves. **Then** The form requests a complete country-coded phone instead of silently guessing the country.

CS-M02-AC05 *[Preserve baseline]* **Given** Another customer already has the canonical email/alias or phone. **When** The executive attempts ordinary creation. **Then** The match is shown and creation is blocked until the correct identity is reused or a distinct person is explicitly confirmed.

CS-M02-AC06 *[Preserve baseline]* **Given** Two household members share contact details. **When** The executive opens their histories. **Then** The customer IDs and tickets remain separate; the application does not auto-merge them.

CS-M02-AC07 *[Preserve baseline]* **Given** A previous primary email becomes a retained alias after an approved correction. **When** The executive searches that alias. **Then** The original customer and history remain discoverable.

CS-M02-AC08 *[Production requirement]* **Given** A user lacks access to a restricted customer field or record. **When** They request list, detail, search or export APIs directly. **Then** The server omits/denies restricted data; frontend hiding is not the security boundary.

Source references: dist/portal.js:59; dist/catalog.js:22; dist/cs-records.js:4; dist/cs-records.js:20

CS M03 Protected corrections legacy import and duplicate review

Correct customer/order links without silently rewriting history, and define the safe launch path for legacy records alongside newly entered work.

Version 6 baseline

Implemented: Email correction request creates Pending review with proposed canonical address, reason, customer ID, requester name and base record version; Approval requires correction permission, a still-pending request and unchanged base version; primary/alias collisions are rechecked at approval. Old email is retained as an alias; customer ID stays stable; Order-link correction uses the same review queue, a ticket version check and an approved-remedy safeguard; Approve/Reject decisions retain actor/note and an admin activity entry; Import preview displays a fixed three-row demonstration: one safe historical ticket, one stale overwrite skipped and one ambiguous identity held. Publish adds only LEGACY-901 once with original source dates; Duplicate review shows possible contact matches and Keep records separate, with an admin log entry.

Boundaries: There is no real file picker/parser, arbitrary migration pipeline, real batch validation, rollback engine or irreversible merge execution; The skipped/held import rows are illustrative outcomes, not demonstrated processing of uploaded data; Existing approved service requests block order-link changes, including old approved records; production amendment policy must preserve approved scopes; Version 6 corrects email and order links through this flow; a generic customer-field editor is not implemented.

Fields and validation

Field	Type	Rules and meaning
correction.id/status	identifier; Pending review | Approved | Rejected	Required: Yes; Description: Final decisions cannot be applied twice.
correction.customerId/ ticketId/field	references; email or order context	Required: Yes; Description: Email uses customer version; order correction uses ticket version.
correction.proposedValue/reason	string	Required: Yes; Description: Email is canonicalised, validated and collision-checked; reason is supplied by requester.
correction.baseVersion	integer	Required: Yes; Description: Reject stale approval if current customer/ticket version differs.
correction.requestedBy/ actor/note	audit snapshots	Required: Yes; Description: Prototype captures names and decision note; production adds immutable IDs/timestamps.

Field	Type	Rules and meaning
legacy.sourceSystem/ sourceRecordId/ sourceDates	production migration metadata	Required: Yes; Description: Required for real import idempotency and source provenance; only fixed sample source data exists today.
identityReview.candidates/disposition	review records	Required: No; Description: Keep-separate decision is implemented; real merge proposal, evidence and approval are production-only.

Workflow and rules

Recheck identity collisions at submission and approval; a valid proposal can become stale or conflicting before decision.

Canonical equality includes trim/case/mailto handling and retained aliases; changing letter case alone is not an email correction.

An approved correction preserves customerId and all linked ticket histories.

Protected corrections and service approvals are distinct decision types with distinct permissions.

Order changes cannot invalidate already-approved remedy scope; reopen/revise through a deliberate approved process rather than editing a linked order behind the approval.

Import existing history and retain newly entered records, but never let stale source rows overwrite newer manually confirmed data.

Shared contact is a review signal, not proof of duplicate identity.

Permissions

Operation: Request email/order correction. Required: work.

Operation: Approve/reject protected correction. Required: correction. Scope: Separate grant from Country Manager service-approval authority.

Operation: Preview history import. Required: importPreview.

Operation: Publish reviewed history import. Required: importPublish.

Operation: Review duplicate/keep separate. Required: merge. Scope: Does not grant an implemented merge execution path.

Production requirements

CS-M03-P1: Requirement: Implement correction commands transactionally with compare-and-swap versions, normalized contact-index checks, submitted/decided actor IDs and timestamps, original/proposed values and immutable reasons.

CS-M03-P2: Requirement: Implement an approved migration workflow: source mapping, preview, deterministic row dispositions, source IDs, timestamp parsing, identity-conflict quarantine, stale-update checks, reconciliation totals and idempotent publication. Confirm source files and order-history scope before implementation.

CS-M03-P3: Requirement: Treat merge as a separately designed future operation unless explicitly approved for launch. Require identity evidence, survivor/alias mapping, all linked-record reconciliation and a recoverable audit/rollback plan; do not infer authorization from the merge grant name alone.

CS-M03-P4: Requirement: Reject whitespace-only reasons and malformed references server-side. Do not log raw sensitive customer text to generic application diagnostics.

Acceptance criteria

CS-M03-AC01 *[Preserve baseline]* **Given** A CS executive proposes another customer's email using different case. **When** They submit the correction. **Then** The request is blocked by canonical primary-email conflict detection.

CS-M03-AC02 *[Preserve baseline]* **Given** The proposed email belongs to another customer's retained alias. **When** The correction is submitted or reviewed. **Then** The collision is blocked; no alias or primary identity is overwritten.

CS-M03-AC03 *[Preserve baseline]* **Given** A valid unique email correction is pending and the customer version is unchanged. **When** An authorised reviewer approves it. **Then** The new primary is canonical, the previous primary is retained once as an alias, the record version increments and ticket foreign keys remain unchanged.

CS-M03-AC04 *[Preserve baseline]* **Given** The customer or ticket changed after a correction was requested. **When** A reviewer attempts approval. **Then** Approval is rejected as stale; a fresh reviewed request is required.

CS-M03-AC05 *[Preserve baseline]* **Given** A correction already has an Approved or Rejected decision. **When** A second decision is attempted. **Then** No second mutation is applied.

CS-M03-AC06 *[Preserve baseline]* **Given** An order-link correction concerns a ticket with approved service scope. **When** The reviewer attempts to approve it. **Then** The protected scope guard blocks direct order relinking.

CS-M03-AC07 *[Preserve baseline]* **Given** The fixed safe import sample has not been published. **When** An import publisher publishes it twice. **Then** The first adds LEGACY-901 with historical source dates; the second adds no duplicate. The UI clearly labels the sample boundary.

CS-M03-AC08 *[Preserve baseline]* **Given** Two shared-contact household customers are reviewed. **When** The reviewer chooses Keep records separate. **Then** Both identities remain and the review is logged; no merge is implied.

CS-M03-AC09 *[Production requirement]* **Given** A real migration contains a safe row, a newer manual edit and an ambiguous identity. **When** A reviewed batch is published. **Then** Only safe rows commit; stale overwrites are skipped and ambiguous rows are held with reasons, totals and repeat-safe source IDs.

CS-M03-AC10 *[Production requirement]* **Given** The requester can work tickets but lacks correction/import-publish capability. **When** They call privileged APIs directly. **Then** The server refuses protected decisions and publication, regardless of visible controls.

Source references: dist/portal.js:318; dist/portal.js:384; dist/cs-records.js:10; dist/catalog.js:1

CS M04 New Query and manager classification

Minimise typing and spelling variation by checking history first, then selecting controlled issue and product values.

Version 6 baseline

Implemented: Two steps: Check customer, then Select issue details. Lookup runs as contact text is entered/pasted; One unambiguous match selects the existing customer; multiple shared-contact matches require choosing the person; a no-match path offers Add customer and continue; History warning shows total/open counts and the four latest tickets, with full history and review/resume routes; Issue form uses native dropdowns for brand, channel, query type, main category, dependent child category, product and optional capacity. Free text is limited to order reference and optional original customer message; Known product fills SKU/name/capacity; changing type/main/brand clears dependent selections. Switching customer clears the previous customer's issue draft; Save and open ticket or Save and record another; default initial review due is 24 elapsed hours and resolution target is 72 elapsed hours; Needs classification creates an explicit classification task for an active eligible manager, preferring Head of CS; save is blocked if none exists; Scoped manager Edit classification selects active controlled values, requires a reason and version match, records before/after history, completes classification tasks and creates normal review work if no open tasks remain.

Boundaries: The intake controls are native selects; they are not full server-backed searchable comboboxes; No live order validation, inbound mailbox pull, automatic email-to-ticket creation, AI classification or customer identity enrichment is connected; The 24/72-hour timings are prototype defaults, not a confirmed business-hours/SLA policy; A ticket product tied to an approved service scope cannot be replaced through classification; unchanged historical product snapshots are retained.

Fields and validation

Field	Type	Rules and meaning
lookup/phoneCode	contact string; explicit country code selection	Required: Yes; Description: Canonical exact identity lookup; local numbers require country selection unless the entered digits exactly equal a stored full international number.
customerId	confirmed identity reference	Required: Yes; Description: Required before entering issue details.
brand	controlled ticket brand	Required: Yes; Description: Brand A/B are synthetic labels; Choose brand is not valid.
channel	active Query Channels selection	Required: Yes; Description: Email, Chat, Call, Other in the seed; channel records how contact was received, not a connected integration.
queryTypeId/ mainCategoryId/ childCategoryId	controlled master references	Required: Yes; Description: Child required when the selected main category has active children; explicit Needs classification is permitted.
productId/sku/product/ capacity	catalogue reference and snapshots	Required: No; Description: Product optional if unknown/not applicable; choice restricted to active products for selected brand or Both brands. Known capacity is automatic.
order	customer-provided reference	Required: No; Description: Optional at intake; required and externally verified before relevant service actions.
received	IST occurrence timestamp	Required: Yes; Description: Cannot be in the future; created/recorded timestamps and creator are automatic.

Field	Type	Rules and meaning
description	original customer words	Required: No; Description: Preserve quotes, line breaks and Unicode; does not require retyping the entire email.
classificationHistory	before/after records	Required: Yes; Description: Actor ID/name, timestamp, reason and type/category/product snapshots after each change.

Workflow and rules

Never allow agent free text to create a query type, category, child category or product.

Require active parent/type relationships at save time, not only when populating dropdowns.

Preserve draft values while reviewing the same customer's history; clear issue-specific draft when changing identity.

Email matching is exact after canonicalisation. Phone identity matching uses a complete normalized number; do not match last digits or infer a country.

An unknown category is a tracked obligation, not a task that can be dismissed as a generic customer update.

Only active Country Manager for the ticket country, Head of CS or Super Admin with read/work can classify. Ordinary CS may create the pending issue.

Classification cannot change the original customer message or approved product snapshot silently.

Normal completion, customer-update completion and ticket closure cannot bypass unresolved classification.

Permissions

Operation: Check customer/create ticket. Required: read + work. Scope: Shared directory; prototype's creation form checks work.

Operation: Create/edit catalogue choices. Required: masters. Scope: Separate administration functions; not in agent intake.

Operation: Edit classification. Required: read + work + eligible named manager role. Scope: Country Manager restricted to ticket country; Head/Super across markets.

Production requirements

CS-M04-P1: Requirement: Use accessible searchable comboboxes over approved values, with keyboard navigation, no arbitrary text commits and parent/brand-dependent filtering. Keep original values readable when archived.

CS-M04-P2: Requirement: Validate identity, master-data activity, relation integrity and server timestamps transactionally at save; use idempotency keys to prevent double creation on retry.

CS-M04-P3: Requirement: Store catalogue IDs and immutable display snapshots, with a stable semantic query-type key for analytics. Recheck permissions and manager routing at commit.

CS-M04-P4: Requirement: Persist recoverable drafts scoped to the authenticated user and selected customer, with an explicit discard/change-customer policy and no cross-customer carryover.

CS-M04-P5: Requirement: Treat future email ingestion as a separate phase: deduplicate immutable message IDs, map threads to tickets with human review, preserve original content/attachments and received times, and never auto-approve a service remedy.

Acceptance criteria

CS-M04-AC01 *[Preserve baseline]* **Given** A returning customer's canonical email is pasted with case or whitespace variation. **When** Lookup runs. **Then** History appears before the issue form and the existing customer ID is reused.

CS-M04-AC02 *[Preserve baseline]* **Given** A local phone lacks an unambiguous country code. **When** The executive looks up the customer. **Then** The interface asks for country code instead of guessing; explicit code then permits exact canonical matching.

CS-M04-AC03 *[Preserve baseline]* **Given** Several customers share contact details. **When** Lookup returns matches. **Then** The executive chooses a distinct customer before issue entry; no automatic merge occurs.

CS-M04-AC04 *[Preserve baseline]* **Given** The executive changes query type, main category or brand. **When** Dependent selections are re-evaluated. **Then** Incompatible child/category/product/capacity selections clear and cannot be submitted as hidden stale values.

CS-M04-AC05 *[Preserve baseline]* **Given** A product with known capacity is selected. **When** The ticket is saved. **Then** Product ID, name, SKU and capacity are filled from the approved catalogue, without spelling entry by the agent.

CS-M04-AC06 *[Preserve baseline]* **Given** The executive reviews history and then changes to a different customer. **When** They return to issue entry. **Then** Same-customer review preserves the draft; different-customer selection clears old brand/order/message.

CS-M04-AC07 *[Preserve baseline]* **Given** Needs classification is selected. **When** The executive saves. **Then** An eligible manager owns a classification task; no-manager availability blocks save with an actionable message.

CS-M04-AC08 *[Preserve baseline]* **Given** A classification task is still unresolved. **When** An executive tries generic completion, sent-update completion or Resolve/Close. **Then** The unresolved classification cannot be bypassed.

CS-M04-AC09 *[Preserve baseline]* **Given** An eligible manager supplies an active type/main/child and reason on the current version. **When** They save classification. **Then** Before/after history is retained, original VOC survives, classification work finishes and ordinary issue review remains actionable.

CS-M04-AC10 *[Production requirement]* **Given** The client retries a timed-out create request or the catalogue changes during entry. **When** The server receives the command. **Then** Only one ticket is created and all selected values are revalidated against current allowed records.

Source references: dist/portal.js:65; dist/portal.js:86; dist/catalog.js:22; dist/cs-records.js:34; dist/workflow-access.js:28

CS M05 All Tickets collaboration conversation and lifecycle

Let the team locate and continue any authorised ticket with clear ownership, current worker, evidence and guarded closure.

Version 6 baseline

Implemented: All Tickets displays non-test records across executives, ten per page; search/status filters and report-drilldown filters can combine; Ticket detail separates Assigned to from Working now, shows last editor/revision, original customer words, reverse chronological activity, contact copy controls, linked service approvals and open follow-ups; Start working/Stop working records activity. Taking over from another worker requires an explicit confirmation; assignment is unchanged; Updates: Internal note, Customer reply received, Customer update sent, Remedy outcome confirmed. An idle ticket can be claimed and updated together; a busy ticket uses takeover confirmation; Occurrence time and recording time are separate. Future event time is rejected; stale ticket revisions preserve the draft and require review; Customer update sent requires confirmation of an actual external send and may complete one selected eligible follow-up. Copy status update never marks a message sent; Remedy confirmation selects an exact current approved request/revision; a Return also records No refund due or Refund required, with operational/finance guards delegated to those modules; Status changes require a reason; Resolve checks all obligations. Close requires current Resolved state and customer confirmation/reviewed closure; Resolved/Closed tickets require explicit reopen before a new follow-up or service request. Reopen preserves past resolution time, clears current resolvedAt and creates a review task.

Boundaries: Working now is simulated tab state, not a distributed lock; optimistic version checks are not tested real concurrency; Assignment and active worker are stored as display names in parts of the prototype; production must use immutable IDs; Notes and outcome records describe external communications; the portal sends no customer emails/chats; Reopening does not automatically reset the original resolutionDue target. A separate production SLA policy must define that behaviour; Pending review is system-set by service work; it is not a free manual status choice. Lifecycle states are fixed.

Fields and validation

Field	Type	Rules and meaning
ticketId/customerId/ brand/country/order	linked identity/conte xt	Required: Yes; Description: Preserve customer and order relationships; order changes use protected correction.
status	fixed lifecycle enum	Required: Yes; Description: Open, In progress, Waiting for customer, Waiting for return, Waiting for finance, Pending review, Resolved, Closed; manual status picker excludes Pending review.
owner/worker/ lastEditor/version	assignment, presence and revision	Required: Yes; Description: Different meanings; takeover must not reassign ownership.
events[]	append-only activity records	Required: Yes; Description: Actor ID/name, type, text, occurred-at and recorded-at; latest first in UI.
update.kind/text/ occurredAt	event command	Required: Yes; Description: Non-empty useful text; sent/remedy confirmations add declarations and linked task/request references.
resolvedAt/ previousResolutions[]/ resolutionDue	timestamps	Required: No; Description: Current outcome time versus retained earlier resolutions and target.
confirmation.requestId/ revision/evidence	exact remedy confirmation	Required: No; Description: Cannot confirm a superseded request scope.
draft	user text	Required: No; Description: Retained on stale update or cancelled takeover; not a sent message.

Field	Type	Rules and meaning
followUps/decisions/ evidence	linked records	Required: No; Description: Ticket next-action fields are derived from open tasks; service decision badges do not replace lifecycle status.

Workflow and rules

All authorised CS executives can read the shared queue and history; own-ticket filters are convenience views, not isolation boundaries.

A current worker claim is required for ordinary follow-up completion and update recording. A takeover records who took over from whom.

A copied status message is not an outbound event. Require explicit real-send evidence/confirmation before completing communication work.

Status Closed cannot jump directly to Resolved to manufacture a new outcome; reopen explicitly first.

Resolve/Close is blocked by unknown classification, pending service decisions/information, outstanding return quantities/holds, waiver/inspection/customer-notice conditions, unconfirmed exact remedies, finance obligations, cancellation alternatives and any open tasks.

Reopen requires ticket-update and assign/reopen grants within applicable country scope; retain prior resolutions as history and clear the current timestamp.

A change of ticket owner does not silently move each independent task. Each task has its own explicit handover.

Preserve approved scopes and immutable event history; corrections add records rather than erasing the original occurrence.

Permissions

Operation: Read/search tickets and timeline. Required: read. Scope: Shared authorised customer/ticket visibility.

Operation: Claim, add update, request correction/service. Required: work. Scope: Version 6 action gates are local; service authority and status/owner scopes add separate checks.

Operation: Assign ticket. Required: assign. Scope: Actor within ticket scope; target active, read/work capable and within scope.

Operation: Change status. Required: read + work within scope. Scope: Reopen additionally requires assign.

Operation: Approve service/remedy exception. Required: Eligible Country Manager / Head of CS / Super Admin. Scope: Not implied by work, assignment, read or ticket ownership.

Production requirements

CS-M05-P1: Requirement: Implement server-authoritative transitions and transactional close checks over every current linked obligation; retries and concurrent transitions must not bypass guards.

CS-M05-P2: Requirement: Persist authenticated actor IDs, event occurrence/recording timestamps and before/after revision data. Use compare-and-swap writes; preserve unsaved user text on conflict.

CS-M05-P3: Requirement: Implement presence/claim leases with heartbeat/expiry and explicit takeover. Show stale or disconnected presence without claiming that a lost browser tab still owns work.

CS-M05-P4: Requirement: Define scope rules for every mutation consistently; maintain required shared read visibility and distinguish CS collaboration from country-limited approval authority.

CS-M05-P5: Requirement: Store attachments in secured object storage with verified type/size, access checks and audit metadata; current evidence labels are not uploaded-file storage.

Acceptance criteria

CS-M05-AC01 *[Preserve baseline]* **Given** A matching ticket set spans several pages. **When** The executive searches or changes status filter. **Then** All matching records remain available, page resets to one on a new filter and export includes the complete filtered set.

CS-M05-AC02 *[Preserve baseline]* **Given** Another executive is Working now. **When** A second executive attempts to record an update. **Then** An explicit takeover is required; the original assigned owner remains unchanged and both actions are logged.

CS-M05-AC03 *[Preserve baseline]* **Given** No executive is currently working. **When** The executive chooses Start work and save. **Then** Claim and update are recorded in order without an unnecessary separate claim step.

CS-M05-AC04 *[Preserve baseline]* **Given** The ticket revision changes while an update form is open. **When** The executive saves the stale form. **Then** The mutation is refused and the draft is retained for review.

CS-M05-AC05 *[Preserve baseline]* **Given** A status update is copied but has not actually been sent. **When** The executive returns to the ticket. **Then** No Customer update sent event or follow-up completion is inferred from copying.

CS-M05-AC06 *[Preserve baseline]* **Given** A current approved service request has an incomplete remedy or related finance/return obligation. **When** The executive tries Resolve. **Then** Closure guards identify the outstanding obligation and the ticket remains active.

CS-M05-AC07 *[Preserve baseline]* **Given** A ticket is Resolved and all obligations are complete. **When** The executive chooses Closed with a reason and closure confirmation. **Then** The ticket closes; an attempt to close directly from an active state is blocked.

CS-M05-AC08 *[Preserve baseline]* **Given** A Resolved/Closed ticket needs a new service action or follow-up. **When** An authorised user explicitly reopens it. **Then** The former resolution is retained, resolvedAt clears and a new review task is created before new obligations are allowed.

CS-M05-AC09 *[Preserve baseline]* **Given** A ticket has a current approved remedy with a specific revision. **When** The executive records Remedy outcome confirmed. **Then** Only that exact approved scope is confirmed; superseded scopes and unsupported refund treatment are blocked.

CS-M05-AC10 *[Production requirement]* **Given** Two authenticated users submit competing updates or closure while a new obligation is being added. **When** The server commits requests. **Then** At most one valid revision transition wins; no lost event, orphan task or prematurely closed ticket is created.

Source references: dist/portal.js:57; dist/portal.js:103; dist/portal.js:322; dist/portal.js:356; dist/portal.js:444; dist/workflow-access.js:24; dist/replacements.js:77

CS M06 Follow ups deadlines and reminders

Give each operational obligation a named accountable owner, a due time and a retained history, including parallel obligations on the same ticket.

Version 6 baseline

Implemented: Follow-ups has Open, Mine, Overdue, Decisions, Rejected and History views, owner and source-type filters, search, ten rows per page, detail, source navigation and a full matching export; Task source types are ticket, request, return, waiver, refund, dispute and parcel. Parallel source/step tasks are supported. Only one open task exists per sourceType/sourceId/step key; New manual follow-ups require an active ticket, a controlled action, an eligible owner and a future deadline. Tasks can be reassigned, have their deadline revised with a reason, and have reminders snoozed by their owner; Original due, current due, first missed deadline, owner at first miss, current owner, history coverage and completion recorded time remain separate. A late deadline change or reminder snooze does not erase a recorded miss; Personal reminders appear one hour before a deadline unless rescheduled. Manager attention includes overdue tasks in the manager's scope. Reminder checks run every 30 seconds in the open page; Manual ticket-task completion requires current work claim, an allowed outcome, required note and confirmation that the action actually happened. Classification tasks cannot be completed while classification is unresolved.

Boundaries: Reminders are browser-tab notifications/toasts; there is no reliable background scheduling, push, email or SMS delivery; Migrated demonstration tasks marked current-state-only are excluded from historical missed-deadline cohorts because their earlier history is unknown; Completion records when work actually happened separately from when staff recorded it. Supported backdating can correct a provisional miss when the actual completion was on time; this is retained as a history event.

Fields and validation

Field	Type	Rules and meaning
task identity	IDs	Required: Yes; Description: Task ID, ticket ID, source type, source ID and stage/step; source links must remain valid.
title/action	controlled action or generated title	Required: Yes; Description: Manual action options: Contact customer; Check customer response; Share troubleshooting steps; Confirm delivery with customer; Confirm refund update; Review unresolved issue.
ownerId	active eligible user ID	Required: Yes; Description: Eligibility depends on source, stage, capability and country/warehouse scope; displayed name is separate.
originalDueAt / dueAt	timestamps	Required: Yes; Description: Immutable first deadline and revisable current deadline.
firstMissedAt / ownerAtFirstMiss	timestamp/ user ID	Required: No; Description: First observed breached deadline and responsible owner; retained across later changes.
remindAt	timestamp	Required: Yes; Description: Personal reminder time; does not replace operational due time.
status / completedAt / completionRecordedAt / cancelledAt	state and timestamps	Required: Yes; Description: Open/completed/cancelled with separate actual completion and recording time.
historyCoverage / history / outcome	enumeration, append-only events, text	Required: Yes; Description: Complete or current-state-only provenance; create, assignment, due, reminder, miss and completion events.

Workflow and rules

Open means status exactly open. Overdue means open and dueAt strictly earlier than now; a task due exactly now is not yet overdue.

Owner reassignment may retain an already overdue deadline. Changing the deadline requires a future time and a reason. Dispute evidence deadlines cannot be moved through generic task editing.

Record a first breach before revising or snoozing so the original miss and owner remain visible.

Reject actual completion in the future. For fully recorded tasks, reject completion more than a minute before creation; same-minute timestamp precision is clamped to creation.

Task completion operates on the relevant source/step, never blindly closes every task on the ticket.

The earliest open task drives the ticket's next-action summary; additional open tasks are counted. No open task shows No open follow-up.

Reassigning a decision or classification task must retain an active eligible manager; finance/logistics/dispute stages retain their corresponding capability.

Permissions

Operation: View shared task queues. Required: read. Scope: Shared ticket visibility; manager-attention subsets use role scope.

Operation: Create and complete a manual follow-up. Required: work. Scope: Active ticket and eligible owner; completion additionally requires current ticket claim.

Operation: Edit a task. Required: Current task owner, or assign in scope, or Super Admin. Scope: Open tasks only; destination owner must pass stage-specific eligibility.

Operation: Snooze or schedule a personal reminder. Required: Current task owner. Scope: Only own open task; no change to due time.

Production requirements

CS-P06-01: Requirement: Use a durable server task store, transactional unique open source/stage constraint and background scheduler; survive closed browsers and server restarts.

CS-P06-02: Requirement: Enforce stage-specific access and versions on every mutation, and retain immutable original due, first miss owner, actual completion and recording timestamps.

CS-P06-03: Requirement: Define approved SLA clocks, notification channels, delivery retries, escalation recipients and acknowledgement rules; do not treat the browser interval as delivery.

CS-P06-04: Requirement: Require reason and provenance for retrospective actual completion; decide the permitted backdating window and who may correct it without falsifying history.

Acceptance criteria

CS06-01 *[Preserve baseline]* **Given** An active ticket and an eligible CS user. **When** The user creates a follow-up with a selected action and future deadline. **Then** One task is created with original/current due, owner, reminder and creation history; ticket summary uses the earliest open obligation.

CS06-02 *[Preserve baseline]* **Given** A ticket is Resolved or Closed. **When** A user attempts a new follow-up. **Then** The action is blocked with explicit reopen guidance and no task is added.

CS06-03 *[Preserve baseline]* **Given** A task has become overdue. **When** Its owner snoozes a reminder or a manager changes its future deadline. **Then** First missed deadline and owner at first miss remain in history and missed-period analytics.

CS06-04 *[Preserve baseline]* **Given** A task requires a particular stage capability and country or warehouse scope; for example, a US ticket must not be routed to a UK-only owner. **When** An assignment is attempted to a user outside that scope or without the required manager/work/finance/logistics/dispute capability. **Then** The destination is unavailable and revalidated at save; eligible all-market users remain available and the task cannot become unserviceable.

CS06-05 *[Preserve baseline]* **Given** An open task belongs to a specific source/stage on a ticket with parallel obligations. **When** That source action completes. **Then** Only the applicable task finishes and the other obligations remain open.

CS06-06 *[Preserve baseline]* **Given** A manual follow-up is open. **When** A user attempts completion without work claim, outcome, note or actual-action confirmation. **Then** Completion is blocked until all prerequisites are satisfied.

CS06-07 *[Preserve baseline]* **Given** A complete-history task was recorded late but actually finished before its deadline. **When** An authorised actual completion is recorded. **Then** Actual time and recorded time remain separate; provisional miss correction is audited and the task is not falsely counted late.

CS06-08 *[Preserve baseline]* **Given** A legacy task has current-state-only history. **When** Historical missed-deadline analytics is requested. **Then** Its unknown earlier history is disclosed and excluded from the fully observed missed cohort.

CS06-09 *[Preserve baseline]* **Given** An owned task is due for a reminder and a manager has an overdue task in scope. **When** Reminder views are opened. **Then** Personal reminders and manager escalation entries appear independently; snooze does not suppress manager overdue visibility.

CS06-10 *[Production requirement]* **Given** Browsers are closed during a configured reminder window. **When** The production reminder scheduler runs. **Then** Delivery and retries occur from durable server state with an auditable delivery result.

Source references: dist/portal.js:401; dist/portal.js:416; dist/portal.js:418; dist/portal.js:421; dist/portal.js:430; dist/portal.js:438; dist/portal.js:464; dist/operations.js:55; dist/workflow-access.js:8

CS M07 Global search and section filters

Find records from customer-provided identifiers without re-entry, while keeping broad discovery distinct from identity validation.

Version 6 baseline

Implemented: Global search requires at least two trimmed characters and returns up to six non-test tickets plus six customers. It supports ticket/customer navigation; Operational lists and exports use the shared recordMatches helper: case-insensitive text and whitespace-insensitive matching over a record, linked ticket, linked customer and linked returns; Phone-shaped search queries with at least five digits can match a substring of an actual phone field after removing formatting. This enables formatted telephone search without treating arbitrary ticket/order digits as phone numbers; All Tickets combines text, owner, status, category, type, brand, country and created-date filters. Follow-ups combines queue, owner, source type and text. Customers can match their own attributes or historical tickets; Search and filter changes reset paginated views to the first page; exports use the complete matching set rather than the current ten-row page.

Boundaries: The prototype scans tab-local objects and linked records; it is not a server index or a field-level privacy implementation; Not every select is a searchable combobox and the global dropdown is intentionally capped. Section views provide the wider result set; Partial phone search is discovery only. New-ticket customer selection uses strict canonical identity matching and explicit disambiguation.

Fields and validation

Field	Type	Rules and meaning
query	string	Required: No; Description: Trimmed search term; global minimum two characters; telephone-search digits minimum five.
section filters	enumerations /date range	Required: No; Description: Only supported filters for the current section; All values mean no restriction.
result IDs	record IDs	Required: Yes; Description: Stable identifiers used for navigation rather than relying on a display label.
result context	authorised customer/ticket metadata	Required: Yes; Description: Enough context to distinguish similar people without exposing forbidden fields.

Workflow and rules

Normalise email by trimming, lowercasing and removing a leading mailto:. Phone identity requires a complete country-qualified number or an exact full stored number; do not infer identity from a suffix.

Global ticket results exclude test tickets. Apply each section's documented predicate consistently to displayed counts, pagination and export.

Search linked customer email, aliases, phone and order/ticket/return attributes in operational export datasets as well as direct row fields.

Search must not grant edit, approval, export or cross-scope write privileges merely because a record is visible.

Permissions

Operation: Global and shared operational search. Required: read. Scope: Same permitted records as the destination view.

Operation: Export matching results. Required: read and export. Scope: Same permitted dataset and filters; export permission is separate from shared read.

Production requirements

CS-P07-01: Requirement: Build an explicit server-side searchable-field whitelist and authorised projection; never return or index passwords, tokens, hidden PII, security notes or administrative secrets and then hide them in the UI.

CS-P07-02: Requirement: Use indexed, paginated APIs with stable sorting, complete counts and shared filter semantics between list and export.

CS-P07-03: Requirement: Provide accessible searchable selects for long controlled vocabularies, with keyboard navigation, clear empty states and full parent-child category path.

CS-P07-04: Requirement: Preserve the separation between broad discovery and strict identity resolution. Require agent review for multiple matches, shared phones and ambiguous country codes.

Acceptance criteria

CS07-01 *[Preserve baseline]* **Given** A customer phone is stored with a country code. **When** An agent searches using spaces, brackets or dashes. **Then** The matching record appears without creating a duplicate identity.

CS07-02 *[Preserve baseline]* **Given** A short phone suffix matches several people. **When** The suffix is used in discovery search. **Then** Results may be shown for review, but no person is automatically selected as the ticket customer.

CS07-03 *[Preserve baseline]* **Given** A refund, return, waiver, activity or service-request row is linked to a customer. **When** The agent filters the dataset by that customer's email, alias or formatted phone. **Then** The related row appears and the matching export contains the same row set.

CS07-04 *[Preserve baseline]* **Given** A result list is on page two. **When** A narrower search or section filter is applied. **Then** The list resets to page one and count, empty state and pagination remain consistent.

CS07-05 *[Preserve baseline]* **Given** More than ten matching tickets or tasks exist. **When** The authorised user exports the filtered view. **Then** The file includes every matching row rather than the visible page only.

CS07-06 *[Preserve baseline]* **Given** A user can read shared tickets but lacks export or write authority. **When** The user finds a record through search. **Then** Read navigation works while export and restricted changes remain unavailable and server-rejected in production.

CS07-07 *[Production requirement]* **Given** Restricted data exists on a record. **When** An unauthorised user searches a term that occurs only in that restricted field. **Then** No restricted content or existence signal is returned by search or export.

Source references: dist/portal.js:49; dist/portal.js:57; dist/portal.js:59; dist/portal.js:296; dist/portal.js:416; dist/cs-records.js:20; dist/catalog.js:22

CS M08 Insights Voice of the Customer and missed follow ups

Let managers trace complaint patterns and service delays back to the underlying tickets and tasks, with explicit dates and denominators.

Version 6 baseline

Implemented: Ticket analytics uses non-test tickets by original created date, inclusive IST date bounds, brand and country. Default selected range is the 30 complete days ending yesterday; Separate Top 10 complaint category panels cover the last 7, 30 and 60 complete days. These are overlapping windows ending yesterday; brand/country filters affect them but the custom comparison dates do not; Custom ranges compare with the immediately preceding period of the same inclusive number of days. Selected-period cards show tickets created, customer complaints, late resolutions and resolved-on-time rate; Resolution and friction shows median resolution duration, currently open tickets past target, enquiries, unclassified tickets and presence of created/received timestamps. Category and missed-task drilldowns link to evidence and export complete matching cohorts; Follow-up insights groups tasks by first missed deadline, retaining the responsible owner at first miss. It separates selected-period missed tasks, completed after a miss, overdue now and distinct impacted tickets; Historical status-at-cutoff is explicitly

not simulated. Metrics describe current recorded outcomes for tickets originally created in the chosen period.

Boundaries: Query types and category grouping currently use display strings; category reclassification changes the current category grouping of the original ticket cohort; No first-response SLA, customer satisfaction score, automated sentiment, topic clustering or business-hours calendar is implemented; Current cohort end comparison uses 23:59:59.000 IST; fractional timestamps in the final second require a half-open next-midnight range in production; Timestamp coverage checks presence, not full validity or imported-history completeness. Follow-up history coverage is separately disclosed; Overdue-now task count is current, not limited to selected missed-period dates; unknown-history task count is also not date-bounded. These must not be presented as historical period counts.

Fields and validation

Field	Type	Rules and meaning
from / to	IST calendar dates	Required: Yes; Description: Inclusive selected created-date or first-missed-date window; from <= to <= today; today is partial.
brand / country	controlled filter	Required: Yes; Description: All or exact selected value; same scope applies to comparison periods.
follow-up owner / source type	controlled filter	Required: No; Description: Owner refers to owner-at-first-miss when available, otherwise current owner; source type filters task records.
metric evidence	ticket/task IDs	Required: Yes; Description: Every drilldown/export must preserve exact row set and definition.
category path	full category string and stored ID	Required: Yes; Description: Full path is grouping key; short leaf labels are visual only.

Workflow and rules

Top 10 means customer complaints, not policy or SLA compliance.

Count each ticket once by its primary category; tie-break equal category counts by full path alphabetical order. Selected-period complaint mix is also capped at ten categories.

Exclude test tickets from ticket cohorts; never blend currencies into a single monetary total.

Use current resolvedAt for the current lifecycle; reopening clears it and retains earlier resolution history.

Do not substitute zero for unavailable denominator or invalid duration. Show unavailable/— where defined.

Show comparison differences as absolute count differences or percentage-point differences; they are not percentage growth.

Drilldown, exported cohort and summary count must share the same filter and date calculation.

Permissions

Operation: View ticket and follow-up analytics. Required: read and insights. Scope: Authorised shared reporting scope; production row and PII scope must be enforced on the server.

Operation: Export cohort or drilldown. Required: read and export. Scope: Exact authorised matching row set; insights access alone does not confer download permission.

Production requirements

CS-P08-01: Requirement: Implement the metric contracts below in shared server queries and automated boundary tests; use [IST start midnight, next-day IST midnight) timestamp ranges.

CS-P08-02: Requirement: Define immutable query-type identifiers and category IDs for aggregation, retain display snapshots and choose whether historic reports use current classification or classification at event time.

CS-P08-03: Requirement: Decide reporting semantics before adding resolved-in-period, historical backlog-at-cutoff, SLA business-time, first response, CSAT or sentiment metrics; collect the required events first.

CS-P08-04: Requirement: Keep unknown legacy data explicitly unknown and publish coverage/exclusions beside metrics; validate timestamp chronology at ingestion.

Acceptance criteria

CS08-01 *[Preserve baseline]* **Given** Complaint tickets exist at boundaries of the previous 7, 30 and 60 complete days. **When** Insights is opened. **Then** Each panel counts its own overlapping created-date window ending yesterday and excludes test and non-Complaint tickets.

CS08-02 *[Preserve baseline]* **Given** A custom nine-day range is selected. **When** The comparison is applied. **Then** The immediately preceding nine calendar days become the comparison; fixed 7/30/60 panels retain their independent dates.

CS08-03 *[Preserve baseline]* **Given** Equal complaint counts exist for distinct full category paths with the same leaf name. **When** A ranking is rendered and a row is opened. **Then** Stable full-path tie ordering and exact full-path drilldown avoid merging distinct categories.

CS08-04 *[Preserve baseline]* **Given** A created cohort contains resolved, unresolved and missing-due records. **When** On-time rate is calculated. **Then** Only rows with current resolvedAt and resolutionDue form the denominator; the displayed rate and percentage-point difference match that denominator.

CS08-05 *[Preserve baseline]* **Given** A previously resolved ticket is explicitly reopened. **When** The original created-date cohort is recalculated. **Then** Current resolution metrics exclude its cleared resolvedAt while previous resolution history remains available.

CS08-06 *[Preserve baseline]* **Given** No valid duration or no eligible resolution denominator exists. **When** The relevant metric is shown. **Then** The UI displays unavailable/— rather than a misleading zero-hour or zero-percent value.

CS08-07 *[Preserve baseline]* **Given** An overdue task is reassigned or has its deadline extended. **When** The first-miss period is examined. **Then** It remains attributed to its recorded first miss and owner; current overdue and completed-after-miss remain distinct measures.

CS08-08 *[Preserve baseline]* **Given** A summary or ranking produces N matches. **When** The manager opens its drilldown and exports it. **Then** The authorised full row set contains exactly N matching records even across multiple pages.

CS08-09 *[Production requirement]* **Given** A timestamp falls at 23:59:59.500 IST on the selected final day. **When** The production cohort is calculated. **Then** It is included, while the next day's 00:00:00 is excluded.

CS08-10 *[Preserve baseline]* **Given** A requested date range includes today or a legacy record lacks complete history. **When** Insights is shown. **Then** Partial-day and unknown-history limitations remain explicit rather than implying historical completeness.

Source references: dist/portal.js:286; dist/portal.js:288; dist/portal.js:292; dist/portal.js:293; dist/portal.js:446; dist/operations.js:91

CS M09 Exports and downloadable evidence

Provide complete, auditable datasets for operational review and handover without losing customer messages, decisions or task history.

Version 6 baseline

Implemented: Exports offers exactly eight datasets: Tickets, Customers, Activities, Service requests, Returns, Return waivers, Refunds and Follow-ups; Users can search linked customer/ticket attributes, preview all matching CSV rows, copy CSV, download a CSV file and use the persistent download link in this tab's export history; Export history records requester, timestamp, file name and row count. Ticket, cohort and task entry points export their complete matching list, not only the visible page; CSV uses UTF-8 BOM, CRLF row separators, every cell quoted, doubled embedded quotation marks, JSON for nested values and spreadsheet-formula neutralisation; Actual downloaded files, including the published Version 6 ticket CSV and populated return/waiver/refund history, were parsed during validation.

Boundaries: Exports are generated in the browser as data URLs and retained only in this tab. There is no server export job, durable file store or expiring signed URL; Only Tickets and Activities explicitly exclude test tickets; other datasets do not currently propagate that filter to related rows. Production must establish one coherent policy; Tickets and Follow-ups have fixed flattened columns. The other six use the union of keys present in raw selected objects; optional workflow history can add columns; Observed test row counts below describe evidence fixtures, not fixed production quantities. No records means a no-matches message rather than a header-only file.

Fields and validation

Field	Type	Rules and meaning
dataset	eight-choice enum	Required: Yes; Description: Exactly the dataset contracts below.
query / inherited filters	string and filter object	Required: No; Description: Search plus any originating list/cohort scope.
requestedAt / requester / rowCount / fileName	export audit	Required: Yes; Description: Browser-tab history at baseline; durable job metadata required in production.
CSV payload	UTF-8 CSV	Required: Yes; Description: Complete authorised result with linked IDs and original timestamp offsets.

Workflow and rules

Both read and export grants are required. Download generation and access must not bypass row/field scope in production.

Nested objects/arrays remain JSON text inside quoted CSV cells. Embedded commas, quotation marks and line breaks must round-trip through a standards-compliant CSV parser.

Prefix an apostrophe for potential spreadsheet formula values beginning after optional whitespace/BOM with =, +, @ or -, or beginning with a tab/newline/carriage return. This intentional safety prefix may appear on phone values beginning +.

Keep stable record IDs and original time-zone offsets. Never substitute unknown actor/time/history with invented values.

Include full original customer message, classification audit, previous resolution history, refund attempts, waiver and receipt evidence rather than exporting only a summary.

Current-customer fields and historical snapshots have different meanings and must be named clearly in a future stable schema.

Permissions

Operation: Preview, generate and download CSV. Required: read and export. Scope: Authorised rows and fields for the dataset; customers shared visibility alone does not confer export permission.

Production requirements

CS-P09-01: Requirement: Define and version eight explicit export schemas; use stable columns even for empty/optional data. Decide flattened child tables versus nested JSON without discarding evidence.

CS-P09-02: Requirement: Generate large exports on the server, recheck authorisation at job creation and download, use short-lived protected files, audit filters/columns/row count, and apply retention policy.

CS-P09-03: Requirement: Use the same authorised query as the list/cohort and consistently exclude test/archive data according to an agreed policy for linked datasets.

CS-P09-04: Requirement: Retain formula-injection protection and test real downloaded files with multiline messages, Unicode, quotes, delimiter characters, empty fields and nested histories.

Dataset contracts

Tickets

File: support-tickets.csv

Grain: One row per matching non-test ticket.

Selection: db.tickets.filter(!test), shared ticket search; list/cohort/single-ticket entry points pass their exact filtered rows.

Schema: Fixed flattened contract

Column names: ticket_id, customer_id, customer, email, phone, customer_aliases, brand, country, order_number, rma_numbers, carrier_tracking, type, query_type_id, category, category_id, product, product_id, sku, capacity, channel, priority, status, owner, working_now, subject, customer_message, created_by, created_at, received_at, recorded_at, resolved_at, previous_resolutions, next_action, task_owner, task_due, resolution_due, last_editor, ticket_revision, evidence, classification_history, source

Nested: customer_aliases; previous_resolutions; evidence; classification_history

Notes: RMA and carrier tracking identifiers are joined with ' | '; original message is preserved. source defaults to Manual entry. linked customer attributes reflect current shared customer record.

Evidence: hosted-tickets-downloaded.csv

Observed rows: 61

Customers

File: support-customers.csv

Grain: One row per matching shared customer record.

Selection: db.customers filtered by linked recordMatches.

Schema: Union of enumerable keys present in selected raw customer objects

Column names: id, name, email, phone, country, aliases, version

Nested: aliases

Notes: No automatic split by brand; records are shared. No blanket test flag predicate is applied.

Evidence: customers-downloaded.csv

Observed rows: 6

Activities

File: support-activities.csv

Grain: One row per ticket timeline event.

Selection: Events of non-test tickets; each row adds ticket_id and customer_id then event keys; linked recordMatches.

Schema: Union of ticket_id, customer_id and event keys present in the result

Column names: ticket_id, customer_id, at, actor, type, text

Additional fields when recorded: recordedAt; actorId

Nested:

Notes: Older fixture events can lack actorId/recordedAt; do not manufacture them. Actual event versus recorded time are distinct when supplied.

Evidence: activities-downloaded.csv

Observed rows: 13

Service requests

File: support-service-requests.csv

Grain: One row per matching service-request record in db.requests, not one row per item or approval event.

Selection: Raw db.requests filtered by linked recordMatches.

Schema: Union of enumerable keys present; fields expand with populated transitions

Column names: id, ticketId, kind, revision, status, reason, condition, items, owner, due, actor, final, itemLines, replacementLines, requesterId, approverId, approvedScope

Nested: itemLines; replacementLines; approvedScope; history/revision/confirmation objects when present

Notes: Approval and revision history remain nested rather than flattening away prior decisions; row does not itself execute a cancellation, return or replacement.

Evidence: service-requests-downloaded.csv

Observed rows: 5

Returns

File: support-returns.csv

Grain: One row per matching return/RMA record, with computed balance.

Selection: db.returns.map(r=>({...r,balance:returnBalance(r)})), linked recordMatches.

Schema: Union of raw return keys plus balance

Column names: id, ticketId, requestId, rma, warehouse, stage, tracking, owner, due, items, proof, sent, dispatch, receiptHistory, waivers, receiptVersion, balance, condition, receivedAt, inspectionScope, dispatchLines, dispatchEvidence, dispatchedAt

Nested: items; receiptHistory; waivers; balance; inspectionScope; dispatchLines; dispatchEvidence

Notes: Keep receipt and waiver histories; computed balance reflects late receipt without rewriting approved historical waiver evidence.

Evidence: operations-returns-downloaded.csv

Observed rows: 5

Return waivers

File: support-return-waivers.csv

Grain: One row per waiver across returns, including rejected and approved records.

Selection: allReturnWaivers(db) adds return/ticket/RMA links; linked recordMatches.

Schema: Union of flattened waiver and link keys present

Column names: id, returnId, rma, ticketId, requestId, revision, scope, snapshot, lines, reason, evidence, requesterId, requestedBy, requestedAt, ownerId, dueAt, status, history, owner, due, actorId, actor, roles, decidedAt, decisionNote, customerNotice

Nested: snapshot; lines; evidence; history; roles; customerNotice

Notes: Retain original receipt snapshot, revision, need-information/resubmission, rejection/approval and customer-notice evidence.

Evidence: operations-return-waivers-downloaded.csv

Observed rows: 2

Refunds

File: support-refunds.csv

Grain: One row per matching refund case; attempts and authorization remain nested.

Selection: Raw db.refunds filtered by linked recordMatches.

Schema: Union of raw refund keys present, including legacy compatibility fields

Column names: id, ticketId, currency, goods, shipping, paid, attempt, history, order, customerId, brand, requestId, status, final, revision, requesterId, attempts, authorization, sourceHistoryIncomplete

Nested: attempt; attempts; authorization; history

Notes: Preserve failed attempts, successful retries, payment proof, approval and imported-history coverage. Legacy attempt is retained for compatibility; do not double-count it as a second payment when attempts includes it.

Evidence: operations-refunds-downloaded.csv

Observed rows: 1

Follow ups

File: support-follow-ups.csv

Grain: One row per matching task, including open, completed and cancelled in the global dataset.

Selection: taskExport(db.tasks); section/drilldown exports pass exact matching task list.

Schema: Fixed flattened contract

Column names: task_id, ticket_id, source_type, source_id, stage, action, owner_id, owner, original_due, current_due, first_missed_deadline, owner_at_first_miss, completed_at, completion_recorded_at, cancelled_at, status, reminder_at, history_coverage, outcome, history

Nested: history

Notes: Original/current due and first-miss owner are separate. Preserve unknown history coverage and actual completion/recording times.

Evidence: follow-ups-downloaded.csv

Observed rows: 27

Acceptance criteria

CS09-01 *[Preserve baseline]* **Given** Each of the eight datasets contains matching records. **When** An authorised user downloads each dataset. **Then** A real file is saved, parsed successfully and contains the exact matching row grain and complete fields, not just a preview.

CS09-02 *[Preserve baseline]* **Given** A ticket message contains commas, quotes, line breaks and Indian-script text. **When** The ticket is exported and parsed. **Then** The original message round-trips as one cell and one ticket row without truncation.

CS09-03 *[Preserve baseline]* **Given** A field starts with a spreadsheet formula-triggering character. **When** It is exported. **Then** The CSV neutralises formula execution while preserving the intended data with the documented safety prefix.

CS09-04 *[Preserve baseline]* **Given** A refund includes an imported success, a failed attempt and a successful retry. **When** Refunds is exported. **Then** All attempt records, authorisation, proof references and history remain available; totals do not double-count legacy compatibility fields.

CS09-05 *[Preserve baseline]* **Given** An RMA has partial receipt, approved waiver and a later physical receipt. **When** Returns and Return waivers are exported. **Then** Original receipt and waiver snapshots/history remain, while current balance correctly reflects the late receipt.

CS09-06 *[Preserve baseline]* **Given** A filtered result spans more than one page. **When** The user exports it. **Then** The file row count equals the full matching set, with no extra out-of-filter records.

CS09-07 *[Production requirement]* **Given** A user lacks export permission or loses scope before download. **When** The user requests a file. **Then** The action is denied; production download authorisation is rechecked rather than trusting a previously rendered link.

CS09-08 *[Preserve baseline]* **Given** The search has no matches. **When** Export is requested. **Then** The user receives a clear no-matches result and no misleading file or successful row count is recorded.

CS09-09 *[Production requirement]* **Given** Optional histories differ between records. **When** The production schema version is exported. **Then** Column order and semantics remain stable with explicit empty values and schema version.

Source references: dist/portal.js:294; dist/portal.js:418; dist/cs-records.js:66; dist/cs-records.js:76

OPS SERVICE Service approvals and controlled revisions

Request one approved cancellation, return or replacement scope, with a single eligible decision and a named next action.

Version 6 baseline

Published V6 synthetic, tab-local behavior; one valid Country Manager, Head of CS or Super Admin decision suffices. No multi-level approval chain is implemented.

Request submission creates Pending decision / final=false; ticket becomes Pending review, a workflow-generated value not offered in the manual status menu. Decision review has Review to Decision to Confirm steps.

Need information remains non-final at the same revision. CS answers and resubmits the same revision. Approve and Reject are final for that record. Rejection creates a customer-explanation task; it does not silently resolve the ticket.

A scope revision creates a new AR record with previousId and revision+1, marks the old record supersededBy, cancels its request tasks and holds ancestor RMAs without dispatch. Approval migrates those RMA lines by stable lineId/productId and preserves physical receipts.

Submitted order-verification checkbox is required by the form. No independent persisted assent object for that checkbox exists; decision actor/time/note and ticket activity carry prototype evidence.

Fields and validation

Field	Type	Rules and meaning
kind	enum	Required: Yes; Rule: Return, Replacement or Cancellation; revision cannot change kind.
itemLines / replacementLines	line arrays	Required: Yes; Rule: Stable lineId, productId, snapshot name/SKU, expected whole quantity 1–999; one row per product. Inbound empty only for no-return replacement; outbound required for replacements.
condition / exceptionReason	enum / text	Required: conditional; Rule: Three replacement arrangements; Return uses Warehouse receipt required; Cancellation uses Confirm warehouse stop. No-return reason required and trimmed.
requesterId / approverId / due / reason	IDs, timestamp, text	Required: Yes; Rule: Active eligible manager; future IST deadline; linked order required; reason required by form.
revision / previousId / supersededBy / final / approvedScope	version metadata	Required: Yes; Rule: Execution must use current approved scope. approvedScope fingerprints request ID, revision, kind, replacement path, inbound/outbound and exception reason.
decision result / next owner / due / note / assent	decision payload	Required: Yes; Rule: Approve routes to Logistics; Need information and Reject route to eligible CS. No-return approval requires explicit exception checkbox.

Workflow and rules

Products newly selected must be active, not deleted and match ticket brand or Both brands; unchanged historical lines retain snapshots even if archived.

Cannot revise after shipment/dispatch. Cannot remove a physically received line or lower expected below received. Traverse all request ancestors to protect earlier RMAs.

Pending/Need information/Approved collection waivers block scope revision: withdraw undecided waiver first, or create separate service work. Existing RMA cannot be converted to a no-return request; use separate remaining-return waiver.

Final decision checks current authority, non-final/non-superseded record, eligible next owner and future due time. Approve binds approvedScope before setting status.

Approved revision with existing RMA normally continues in return task; advance replacement retains independent dispatch fulfilment task. Old revision never authorizes current execution.

Permissions

CS work grant requests and answers; all CS can follow team approvals. Manager identity is based on built-in super/head/cm roles, not a generic approval grant.

managerMayDecide: Active Super Admin/Head of CS across markets; Active Country Manager only its country or All markets. UI approver lists also require read permission.

Production must apply active/read/grant/market checks at every command, regardless of button visibility; some V6 handlers use can(grant) only.

Production requirements

Persist immutable request/decision/revision records with database optimistic locking and a unique accepted final decision per revision.

Atomically commit scope/decision/tasks/activity; reject stale confirmations if revision, role, status, owner eligibility or due time changes.

Persist order verification and exception assents with actor, timestamp, displayed scope version and evidence rather than treating HTML required validation as authorization.

Version SKU snapshots and retain all superseded history; do not automatically import operational truth from an order integration until independently authorized and specified.

Acceptance criteria

SERVICE-01 *[Module acceptance]* **Given** An active linked-order ticket. **When** CS submits valid lines, reason, manager and future deadline. **Then** Create one pending revision and manager decision task; close superseded ticket follow-up; perform no external action.

SERVICE-02 *[Module acceptance]* **Given** A UK ticket and US-only Country Manager. **When** Manager opens or submits decision. **Then** Reject authority; Head of CS/Super Admin remain eligible.

SERVICE-03 *[Module acceptance]* **Given** A pending request. **When** Manager asks for information and CS answers. **Then** Remain same revision, transition CS information task back to manager decision, retain question and answer history.

SERVICE-04 *[Module acceptance]* **Given** A pending request. **When** Manager rejects with CS next owner. **Then** Final Rejected record and CS explanation task; no RMA/shipment generated.

SERVICE-05 *[Module acceptance]* **Given** An approved scope. **When** CS submits valid revision before shipment. **Then** Retain old decision, mark it superseded, hold linked ancestor returns pending new approval.

SERVICE-06 *[Module acceptance]* **Given** A received product line. **When** Revision removes it or reduces expected below received. **Then** Reject without mutating receipt or old scope.

SERVICE-07 *[Module acceptance]* **Given** Any ancestor RMA or completed shipment. **When** User attempts no-return conversion or shipped-scope revision. **Then** Reject with separate-waiver/separate-request guidance.

SERVICE-08 *[Module acceptance]* **Given** A no-return pending revision. **When** Manager omits explicit exception assent. **Then** Do not approve or create fulfilment task.

SERVICE-09 *[Module acceptance]* **Given** Two actors reviewing the same revision. **When** One final decision is accepted first. **Then** Second final decision must be rejected; production concurrency acceptance required, not proven by tab-local simulation.

Source references: dist/portal.js:120–159,398,403–411; dist/operations.js:6–37; dist/replacements.js:8–37; output/final-validation-2026-10-30/operations/cases.json:OPS-REG-011–016, OPS-DOM-019

OPS CANCELLATION Cancellation execution and linked return fallback

Record whether an approved cancellation actually succeeded and keep its obligation open until a valid alternative remedy is complete.

Version 6 baseline

Approval alone is not cancellation. Logistics records external warehouse-stop and cancellation outcomes separately.

Ordinary completion requires Stopped before dispatch + Confirmed cancelled, reviewed refund treatment, then CS customer update.

Already dispatched records Not possible after dispatch against exact cancellation scope, without rewriting its approval. An alternative Return must be on the same ticket and cover every cancellation product/quantity.

A pending alternative can be linked, but does not retire the obligation. After its approved receipt/waiver, inspection and remedy, Logistics revisits cancellation checks; the cancellation task then becomes CS customer update.

Fields and validation

Field	Type	Rules and meaning
stop	enum	Required: Yes; Rule: Unconfirmed / Stopped before dispatch / Already dispatched.
cancel	enum	Required: Yes; Rule: Unconfirmed / Confirmed cancelled; Already dispatched + Confirmed cancelled is rejected.
alternativeRequestId	request ID	Required: conditional; Rule: Current same-ticket Return; every cancellation product covered with expected>=cancelled quantity.
refundTreatment	enum	Required: Yes; Rule: No refund due / Refund required; placeholder is invalid.
proof / due	text / timestamp	Required: Yes; Rule: Trimmed meaningful external evidence and future next check.

Field	Type	Rules and meaning
execution / executionHistory	audited snapshots	Required: Yes; Rule: Outcome, exact scope fingerprint, link, actor/time; prior stop/cancel/execution stored per save.

Workflow and rules

No cancellation writes on Resolved/Closed tickets; explicit reopen first.

Alternative is allowed only when Already dispatched. Wrong ticket/type, superseded scope or insufficient quantities rejected.

Closure of fallback requires current approved/final linked Return, RMA, no remaining quantity/hold, acceptable inspection of actual receipts, no waiver issue and exact request/revision remedy confirmation.

Refund required accepts same-ticket/order/brand/customer non-rejected refund authorisation: ticket/order-only, exact cancellation revision, or its current alternative Return revision. Full settlement remains a separate finance closure gate.

Incomplete normal checks remain Logistics fulfilment; dispatched failure becomes CS alternative task; settled outcome becomes CS customer-update task.

Permissions

Execution requires shared eligibleUser Logistics: active, read+logistics and ticket market scope; All warehouses is eligible.

CS creates/obtains alternative Return approval and records customer notice; eligible manager decides Return and refund authorisations separately.

Production requirements

Store external cancellation references and outcome changes immutably; never treat an approval response as an external order cancellation API result.

Perform fallback eligibility, refund linkage and task retirement in one transaction; revalidate alternative version at final closure.

If real integrations are added later, use idempotent commands and reconcile uncertain external outcomes before allowing conflicting action.

Acceptance criteria

CANCEL-01 *[Module acceptance]* **Given** Approved cancellation, open ticket. **When** Logistics records stopped-before-dispatch and confirmed cancelled with reviewed refund treatment. **Then** Create CS customer-update obligation; no fake money movement.

CANCEL-02 *[Module acceptance]* **Given** Already dispatched order. **When** Logistics also selects Confirmed cancelled. **Then** Reject inconsistent combination.

CANCEL-03 *[Module acceptance]* **Given** Already dispatched without alternative. **When** Outcome recorded. **Then** Keep obligation open and create Request separate return approval task.

CANCEL-04 *[Module acceptance]* **Given** Wrong-ticket or under-quantity Return. **When** User links it. **Then** Reject link, retain original cancellation obligation.

CANCEL-05 *[Module acceptance]* **Given** Linked pending/rejected/superseded Return. **When** User tries retiring cancellation. **Then** Remain blocked until current valid approved remedy exists.

CANCEL-06 *[Module acceptance]* **Given** Approved alternative with return due or damaged inspection. **When** User saves checks or resolves. **Then** Keep cancellation outstanding.

CANCEL-07 *[Module acceptance]* **Given** Completed alternative and required refund fully settled. **When** Logistics revisits checks, CS records update, resolves and closes. **Then** Retire old obligation with preserved execution/approval history; reach Closed.

CANCEL-08 *[Module acceptance]* **Given** Terminal ticket or ineligible Finance-only actor. **When** Attempt cancellation execution. **Then** Reject until authorized reopen/Logistics execution.

Source references: dist/cancellations.js:1–29; dist/portal.js:161–169; dist/finance.js:62–67; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-007,008; OPS-DOM-019

OPS REPLACEMENT Three replacement arrangements and dispatch

Execute approved outbound items while preserving any separate inbound return obligations.

Version 6 baseline

Return first, then replace: approval to RMA to each due quantity received or manager-waived to acceptable inspection of actual receipts to dispatch to CS delivery/remedy confirmation.

Replace now, return later: approval to RMA registration to independent dispatch and inbound tasks; shipment/customer delivery do not remove return balance.

Replace without a return: empty inbound list and stated exception reason; explicit eligible-manager assent bound to exact request/revision/scope; no RMA created for this request. Other returns still block ticket closure.

Dispatch records an external shipment already arranged; it does not book carrier service. One shipment per approved replacement scope is modeled; partial outbound shipments are not implemented.

Fields and validation

Field	Type	Rules and meaning
replacementLines	SKU quantity snapshot	Required: Yes; Rule: Approved outbound scope, independent of inbound lines.
waiver	no-return approval snapshot	Required: no-return only; Rule: requestId, revision, full scope, actorId/name/roles, at, exception reason and decision note.
shipment	external dispatch record	Required: Yes; Rule: Unique case-insensitive reference, trimmed proof, approved scope/lines, actor/time; destination and SKUs checkbox required.
customer delivery due	timestamp	Required: Yes; Rule: Future IST; default48h for dispatch follow-up.
confirmation	remedy evidence	Required: Yes; Rule: Exact requestId/revision, actor, occurredAt, recordedAt, evidence; separate from shipment.

Workflow and rules

Reject dispatch for non-current/non-final/non-approved request or changed fingerprint.

Return paths require registered current RMA without hold; return-first additionally requires zero outstanding balance and acceptable actual-receipt inspection.

No-return requires matching approved exception, not merely a selected arrangement.

Case-insensitive shipment reference uniqueness checks all request shipments and return dispatch fields. Duplicate dispatch cannot overwrite history.

Dispatch creates CS remedy task; advance keeps inbound task open. Settlement after advance shipment must not create a second dispatch task.

Remedy confirmation validates current request/revision and exact shipment scope; advance delivery confirmation can precede inbound settlement, but ticket resolution still blocks.

Permissions

Manager approves; Logistics grant records dispatch; CS work records delivered/remedy outcome.

V6 recordShipment checks active Logistics grant; production must add uniform server-side read/market/terminal-state guard to every dispatch command, not rely on UI routing.

Production requirements

Store shipment and line identities immutably with atomic uniqueness and version checks.

Retain independent inbound/outbound state machines and independent task owners.

Do not introduce partial shipment or substitute-item rules without an explicit policy; record another approved service request for new work in baseline.

Acceptance criteria

REPLACE-01 *[Module acceptance]* **Given** Return-first with RMA outstanding. **When** Logistics attempts dispatch. **Then** Block until receipts/approved waivers settle every line.

REPLACE-02 *[Module acceptance]* **Given** Return-first with received items marked damaged. **When** Dispatch attempted. **Then** Block pending acceptable inspection; collection waiver does not cure damage.

REPLACE-03 *[Module acceptance]* **Given** Return-first with zero actual receipts fully waived. **When** Dispatch attempted after approval. **Then** Allow without inventing inspection of nonexistent receipts; customer waiver notice remains required for closure.

REPLACE-04 *[Module acceptance]* **Given** Advance replacement without RMA. **When** Dispatch attempted. **Then** Block and request RMA registration/owner.

REPLACE-05 *[Module acceptance]* **Given** Advance replacement with RMA. **When** Dispatch and delivery confirmed before inbound completion. **Then** Allow outbound progression but block ticket resolution on inbound balance.

REPLACE-06 *[Module acceptance]* **Given** No-return with exact approved assent. **When** Logistics records valid shipment, CS verifies delivery. **Then** Permit no-RMA path through closure when other obligations clear.

REPLACE-07 *[Module acceptance]* **Given** No-return exception on one request and unrelated open return. **When** Ticket resolution attempted. **Then** Block on unrelated return, preserve it.

REPLACE-08 *[Module acceptance]* **Given** Shipment reference already used or shipment already recorded. **When** Second dispatch attempted. **Then** Reject; retain prior record.

REPLACE-09 *[Module acceptance]* **Given** Scope changed after approval. **When** Dispatch or remedy attempted. **Then** Reject and require current approved revision.

Source references: dist/replacements.js:7–75; dist/portal.js:173,236–244,108–111; output/final-validation-2026-10-30/operations/cases.json:OPS-REG-011–017

OPS RMA RMA registration tracking cumulative receipt and inspection

Separate identity/tracking from physical received quantities, and move remaining work to the proper stage.

Version 6 baseline

Every warehouse supports RMA; UI registers an RMA created outside the portal. Warehouse list is active Countries master, defaulting to ticket country.

One return per request is opened if already registered; no-return replacement rejects RMA registration. Case-insensitive RMA uniqueness is checked.

Carrier delivered or instructions sent never means warehouse received. Receipt uses cumulative per-line physical quantities; received cannot decrease or exceed expected.

Existing synthetic receipts are labelled imported snapshots with earlier event history unavailable. New events retain before and after values, inspection, actor, occurred and recorded timestamps.

After settlement: Return creates CS remedy task; undispatched Replacement creates Logistics release task; already-dispatched advance does not create duplicate release.

Fields and validation

Field	Type	Rules and meaning
rma / requestId / ticketId	references	Required: Yes; Rule: Current approved non-superseded request and external RMA reference.
warehouse / tracking / sent / proof	master, text, boolean, text	Required: Yes; Rule: Tracking optional; instructions Not yet sent / Confirmed sent / Delivery failed. Portal sends nothing.
items[].received	cumulative integer	Required: Yes; Rule: Each >= previous received and <=expected, at least one positive physical quantity.
receivedAt / condition / proof / next check	timestamp, enum, text	Required: Yes; Rule: Occurred time valid, not future or earlier than previous receipt; acceptable or damaged-review inspection; meaningful trimmed proof; future task deadline.
receiptVersion / inspectionScope / receiptHistory	version and immutable events	Required: Yes; Rule: Inspection fingerprint covers all cumulative physical quantities.
stage / hold	derived state	Required: Yes; Rule: Awaiting/partial receipt, Inspection required, Collection settled with waiver or Matched receipt; hold still counts outstanding.

Workflow and rules

currentReturn requires non-terminal ticket, approved/final/current request and no hold. Late receipt after Closed/Resolved requires authorized explicit reopen first.

Receipt or inspection update increments receiptVersion and supersedes Pending decision/Need information waivers, cancelling their decision/information tasks. Approved waivers remain as history.

Received quantities and effective waived quantities are displayed separately; late receipt reduces effective waived balance without rewriting original approval.

All physically received items require acceptable inspection. Zero-receipt fully waived return needs no physical inspection.

Follow-up accepts tracking, result, evidence and eligible Logistics owner; CS may record follow-up but cannot mark physical receipt. Instructions confirmed sent sets sent=true only.

Permissions

RMA and receipt use Logistics grant; follow-up is CS or Logistics, routed to active scoped Logistics owner.

Production must enforce consistent active/read/market access and active-ticket checks for every RMA mutation; V6 handlers vary in depth.

Production requirements

Use server-generated immutable event IDs and append-only receipt evidence; correction of an erroneous receipt needs a separate authorized correction event, not a quantity overwrite.

Separate warehouse IDs from country IDs for production while preserving all-warehouse RMA availability.

Carrier, WMS and label integrations are future scope; until then show explicit manual external evidence labels.

Commit receipt, waiver supersession, inspection and task changes atomically; preserve late-arrival and reopened-ticket evidence.

Acceptance criteria

RMA-01 *[Module acceptance]* **Given** Approved return and unique external reference. **When** Logistics registers RMA. **Then** Create zero-received lines, owned return task and Waiting for return ticket; record instruction state without sending.

RMA-02 *[Module acceptance]* **Given** Duplicate reference or no-return scope. **When** RMA submitted. **Then** Reject duplicate/no-return registration.

RMA-03 *[Module acceptance]* **Given** Carrier reports delivery. **When** CS records tracking follow-up. **Then** Keep physical quantities unchanged and receipt obligation open.

RMA-04 *[Module acceptance]* **Given** Expected2, received1. **When** Warehouse records cumulative2 with valid time and evidence. **Then** Append history and update current balance; preserve first receipt.

RMA-05 *[Module acceptance]* **Given** Received1. **When** User submits0, expected+1, fraction or future occurrence. **Then** Reject without mutating receipt history.

RMA-06 *[Module acceptance]* **Given** Unsettled RMA with pending waiver. **When** Receipt/inspection changes. **Then** Mark pending waiver Superseded by receipt, cancel its tasks and require fresh decision snapshot.

RMA-07 *[Module acceptance]* **Given** Settled Return. **When** Acceptable inspection saved. **Then** End receipt task and create CS agreed-remedy task.

RMA-08 *[Module acceptance]* **Given** Already-dispatched advance replacement. **When** Remaining receipt settles. **Then** End inbound work without recreating dispatch; retain delivery confirmation.

Source references: dist/portal.js:173,207–244,336–344; dist/rma-waivers.js:3–20,60–80; output/final-validation-2026-10-30/operations/cases.json:OPS-REG-011,014,017; OPS-HOST-025

OPS WAIVER Existing RMA remaining collection waivers

Allow manager-approved collection exceptions without marking goods received or erasing original return evidence.

Version 6 baseline

Separate ret.waivers collection; not the no-return replacement assent. Can apply to partially received or zero-receipt existing RMAs.

One undecided waiver per RMA. Requested quantities cannot exceed each line still due; only whole numbers, at least one positive.

Snapshot binds return scope, request/revision, current received quantities, receiptVersion, inspection, hold and existing applicable approvals. Stale decision must be withdrawn and requested afresh.

Decision flow is review then confirm; explicit collection assent required on Approve only. Need information remains pending and resubmits unchanged snapshot; Reject leaves collection due.

Approved waiver creates CS customer notice independently of receipt/inspection/dispatch/remedy. Waiver never authorizes refund or proves delivery.

Fields and validation

Field	Type	Rules and meaning
lines	per-line requested quantities	Required: Yes; Rule: lineId/product/SKU, quantity, expected/received/remaining-at-request snapshots.
scope / snapshot / requestId / revision	version fingerprints	Required: Yes; Rule: Must match current RMA at decision or information resubmission.
reason / evidence / ownerId / dueAt	text, IDs, timestamp	Required: Yes; Rule: Trimmed substantive reason/evidence; eligible manager and future due time.
status / actor / roles / decidedAt / history	decision audit	Required: Yes; Rule: Pending decision, Need information, Approved, Rejected, Withdrawn or Superseded by receipt.
customerNotice	notice evidence	Required: approved only; Rule: Actor, occurredAt, recordedAt and evidence after explicitly recorded sent customer update.

Workflow and rules

effective waived=min(sum applicable approved quantity,max(0,expected-received)); remaining=max(0,expected-received-effective waived). Applicable requires Approved, matching return scope, actorId and decidedAt.

Approved history remains even if late receipt reduces effective waiver to zero. An approved waiver with mismatched scope or missing customer notice blocks closure.

Withdraw only Pending decision/Need information with reason; withdrawal does not waive collection.

Approved/Rejected waiver next owner is CS; Need information permits CS or Logistics; answer chooses eligible manager.

If partial waiver leaves balance, keep existing receipt task/deadline; fully waived receipt task is cancelled as collection no longer required, not falsely completed as receipt.

Pending/Need information waiver, unacceptable actual-receipt inspection or missing approved-waiver customer notice blocks closure.

Permissions

CS/Logistics requests or answers; eligible manager decides; CS/Logistics/eligible manager may withdraw undecided request.

Production must revalidate market/read/grant and snapshot on every write, independent of displayed owner choices.

Production requirements

Persist original approved quantity and current effective quantity separately; never mutate approval history to hide late arrival.

Decision uniqueness and snapshot concurrency must be database enforced.

Persist explicit assent text/version, actor and exact line scope; immutable customer communication evidence with external channel reference when available.

Acceptance criteria

WAIVER-01 *[Module acceptance]* **Given** Partial receipt and one unit due. **When** CS requests waiver of that unit with evidence. **Then** Pending snapshot created; received unchanged.

WAIVER-02 *[Module acceptance]* **Given** One unit due. **When** Request quantity2, fraction, duplicate line or allzero. **Then** Reject invalid waiver.

WAIVER-03 *[Module acceptance]* **Given** Pending waiver. **When** Manager selects Need information; CS answers. **Then** Return to Pending decision on same scope with retained history and manager task.

WAIVER-04 *[Module acceptance]* **Given** Pending waiver. **When** Manager rejects. **Then** Keep quantities due; create CS rejection explanation.

WAIVER-05 *[Module acceptance]* **Given** Pending waiver. **When** Manager approves without assent. **Then** Reject; no waived quantity applied.

WAIVER-06 *[Module acceptance]* **Given** Pending waiver with changed receipt/version. **When** Manager decides or CS answers. **Then** Reject stale snapshot; fresh request required.

WAIVER-07 *[Module acceptance]* **Given** Zero actual receipts, full approved waiver. **When** Dispatch/closure evaluated. **Then** No invented physical inspection; still require waiver notice and actual remedy/delivery.

WAIVER-08 *[Module acceptance]* **Given** Partial actual receipts and approved remainder waiver. **When** Received items damaged. **Then** Block release/resolution until acceptable inspection.

WAIVER-09 *[Module acceptance]* **Given** Closed waived return. **When** Authorized reopen then late receipt recorded. **Then** Append physical receipt, retain original waiver/history, reduce effective waiver, complete reopen work and allow reclosure when gates clear.

Source references: dist/rma-waivers.js:1–81; dist/portal.js:179–222,108–111; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-005; OPS-REG-015–017

OPS PARCEL Unmatched parcels and identity matching

Track unidentified warehouse parcels without inventing a customer receipt or refund.

Version 6 baseline

Logistics records warehouse/reference/evidence and a future deadline; creates an unmatched parcel plus independent parcel task with no ticket ID yet.

Badge recomputes count of status Unmatched after save/link. Queue retains linked records.

Link parcel to verified RMA records identity match evidence, marks Linked, finishes parcel task, appends parcel proof to return proof and logs ticket event. Physical quantities remain unchanged.

Fields and validation

Field	Type	Rules and meaning
reference / warehouse / proof	text / master / text	Required: Yes; Rule: Trimmed parcel reference and receipt evidence; warehouse Countries selector.
owner / due / status	user, timestamp, enum	Required: Yes; Rule: Current Logistics actor; future deadline; Unmatched to Linked.
rma / identity evidence	reference/text	Required: Yes; Rule: Select existing RMA; explain verified identity; receipt recorded separately.

Workflow and rules

Identity match never marks received, inspected, refunded or remedied.

V6 does not provide complete duplicate-parcel, cross-warehouse matching, stale-state or terminal-ticket validation; treat these as production requirements, not already enforced behavior.

Parcel task clickthrough opens unmatched queue; linked row shows RMA.

Permissions

Logistics action grant records and links parcels; task reassignment uses warehouse as country scope when no ticket exists.

Production requirements

Use stable parcel IDs, reference uniqueness policy and evidence attachments; validate selected RMA belongs to permitted warehouse/customer scope.

Recheck current unmatched state, current approved RMA and any terminal-ticket restriction before linking; persist match evidence and reversal/correction audit.

Coordinate parcel-link and receipt transactions; avoid duplicate physical receipt if multiple parcel scans relate to one RMA.

Acceptance criteria

PARCEL-01 *[Module acceptance]* **Given** Unidentified physical parcel. **When** Logistics saves reference/evidence and future due. **Then** Create owned unmatched task; badge rises immediately; no customer receipt inferred.

PARCEL-02 *[Module acceptance]* **Given** Unmatched parcel. **When** Logistics selects verified RMA and evidence. **Then** Mark Linked, finish parcel task, log identity evidence; badge falls immediately.

PARCEL-03 *[Module acceptance]* **Given** Linked parcel. **When** Warehouse quantities inspected. **Then** Quantities change only through separate receipt action.

PARCEL-04 *[Module acceptance]* **Given** Whitespace reference or receipt proof. **When** Save unmatched parcel. **Then** Reject meaningful-data omission.

PARCEL-05 *[Module acceptance]* **Given** Production concurrent link attempts. **When** Two users link same parcel. **Then** Exactly one accepted current match; later attempt conflicts rather than silently relinks.

PARCEL-06 *[Module acceptance]* **Given** Production wrong warehouse or inaccessible RMA. **When** Match submitted. **Then** Reject and retain unmatched obligation; additional server-side acceptance requirement.

Source references: dist/portal.js:170,362–364; dist/workflow-access.js:9–12,19; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-009

OPS REFUND AUTH Manual refund authorisation and amount revisions

Record the total approved refund promise for a verified customer/order and optional exact service revision.

Version 6 baseline

This is authorisation/evidence recording; no payment gateway call exists. CS or Finance requests; one eligible manager decides.

One non-rejected authorisation per case-insensitive order + brand + customer, across tickets. Currency is not part of this duplicate namespace; revision keeps original currency/order.

Request may link exact current approved service revision or Ticket / order only. Requested scope includes ticket/customer/brand/order/service revision/currency/goods/shipping.

UI currencies GBP,USD,EUR,CAD,INR; domain accepts any uppercase three-letter code syntactically. Prototype uniformly uses two decimal places, not a general currency-exponent system.

Initial rejection is final Rejected. Rejected amount amendment restores previous goods/shipping/authorisation/requested scope and Approved status, preserving attempts. Revision number and proposal history remain; reason/evidence shown may still be the rejected proposal.

Fields and validation

Field	Type	Rules and meaning
ticket/order/customer/ brand/requestId/ requestRevision	linked identifiers	Required: Yes; Rule: Active linked-order ticket; optional service must be current approved/final on same ticket.
currency / goods / shipping	currency and amounts	Required: Yes; Rule: Positive combined total, nonnegative components, <=2 decimal places; UI checks external order/previous refunds.
reason / evidence / approverId	text and manager ID	Required: Yes; Rule: Trimmed nonblank reason and order eligibility proof; eligible decision owner.
revision / requestedScope / authorization / amendment	version snapshots	Required: Yes; Rule: authorization carries exact scope, actor, roles, at and note; amendment.before retains prior approved values.
status / final / history	decision audit	Required: Yes; Rule: Pending decision/Need information nonfinal; Approved/initial Rejected final; answer resubmits unchanged amount.

Workflow and rules

All new finance actions reject Resolved/Closed ticket until explicit reopen. Customer/brand/order changes or non-current linked approval block action.

Need information routes to eligible CS, not payment. Answers append history and return to Pending decision.

Revision allowed only Approved+final, no pending/unknown reserved amounts, valid original scope, changed amount and revised total>=already paid. New manager decision required.

Domain cents parser accepts unsigned integer or decimal with1–2 digits, uses safe integer amount*100, rejects exponent/negative/overprecision and per-component value above10,000,000,000 minor units. This bound is a technical prototype cap, not a business refund policy.

Authorisation tasks: decision to manager; information/rejection to CS; approved unpaid to Finance reconcile; fully paid/no reservation to CS customer update. Default internal due24h; payment-result form can set reconciliation owner/due.

Return remedy explicitly selects No refund due or Refund required. Required checks non-rejected matching authorisation exists; full approval/payment enforced later by ticket resolution.

Permissions

Requests/revisions/answers require shared eligibleUser work or finance in ticket scope: Active, read+action and market. All warehouses scope is accepted by shared rule.

Manager authority: Active super/head globally or cm in country/All markets. UI owner options also require read. No separate Finance approval substitutes for eligible manager.

Production requirements

Store monetary values as integer minor units (or exact decimal) with currency exponent metadata; reject unsupported currencies rather than accepting any three letters. No FX conversion implied.

Validate refundable total against authoritative order, previous refunds/credits and agreed shipping/tax policy; prototype relies on staff evidence/assent and has no connected order cap.

Persist order-check assent and immutable proposal/approved versions; distinguish requested total from currently authorised total in pending amendment displays.

Atomic approval and owner/task routing with idempotency and optimistic concurrency; ensure no mutation survives if eligible next owner is unavailable.

Specify controlled currency/order relinking and multi-ticket/order handling rather than bypassing duplicate-authorisation guard.

Acceptance criteria

AUTH-01 *[Module acceptance]* **Given** Active verified linked order. **When** CS requests positive total with exact eligible service link. **Then** Create Pending decision authorisation and manager task, no money transfer.

AUTH-02 *[Module acceptance]* **Given** Same customer/brand/order already has non-rejected authorisation. **When** New record attempted, even on another ticket/currency. **Then** Reject duplicate and direct to revision/attempt history.

AUTH-03 *[Module acceptance]* **Given** Different customer or brand with same order string. **When** Valid authorisation submitted. **Then** Namespace permits distinct authorisation after eligibility review.

AUTH-04 *[Module acceptance]* **Given** Pending authorisation. **When** Eligible manager requests information then CS answers. **Then** Remain nonfinal until resubmitted and explicitly approved/rejected.

AUTH-05 *[Module acceptance]* **Given** Wrong-market manager or changed linked scope. **When** Decision attempted. **Then** Reject without authorising execution.

AUTH-06 *[Module acceptance]* **Given** Approved total90 paid90. **When** Propose100, Need information, answer, Reject. **Then** Restore Approved90/paid90/available0; retain original attempt and proposal audit.

AUTH-07 *[Module acceptance]* **Given** Restored90 authorisation. **When** New total100 approved. **Then** Permit only10 additional balance, not another100.

AUTH-08 *[Module acceptance]* **Given** Paid90 or unknown reservation. **When** Revision to89 or any revision while reserved. **Then** Reject without altering paid/history.

AUTH-09 *[Module acceptance]* **Given** Return remedy marked Refund required without matching authorisation. **When** CS confirms remedy. **Then** Reject before saving confirmation; with valid record allow confirmation but resolution still requires full settlement.

AUTH-10 *[Module acceptance]* **Given** Terminal ticket. **When** New authorisation/revision/decision/attempt attempted. **Then** Require authorized reopen.

Source references: dist/finance.js:6–43,62–67; dist/portal.js:246–275,108–111; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-004–006; OPS-DOM-020–022; OPS-HOST-024–025

OPS REFUND ATTEMPT Distinct external refund attempts and reconciliation

Record verified external outcomes without duplicate payment evidence or overpayment.

Version 6 baseline

Outcome unknown and Pending reserve funds. Succeeded counts as paid. Failed frees its amount but remains immutable attempt history.

Paid=sum Succeeded; reserved=sum Pending/Outcome unknown; available=max(0,total-paid-reserved), computed in integer hundredths then displayed in currency units.

New attempt requires distinct reference and no other unresolved attempt. Existing unresolved attempt may be reconciled without changing amount/reference identity.

Imported seed paid amount becomes explicitly incomplete historical Succeeded entry; sourceHistoryIncomplete marks missing earlier decision/attempt details. Do not invent original actors/timestamps.

Fields and validation

Field	Type	Rules and meaning
attempt ID / reference / amount	identity and money	Required: Yes; Rule: Positive amount; new reference trimmed/nonblank; case-insensitive reference uniqueness across all refund records.
result / proof	enum/text	Required: Yes; Rule: Outcome unknown,Pending,Succeeded,Failed; trimmed meaningful verification proof.
owner / next reconciliation	user/ timestamp	Required: Yes; Rule: Active scoped Finance owner, future IST check; persists on attempt and open task.
history / actor / at	audit	Required: Yes; Rule: Each update retains before outcome/reference/proof; refund-level audit also references attempt.

Workflow and rules

Only current approved/final exact authorisation scope permits external attempt.

Reject amount+other Succeeded/Pending/Unknown >authorised total. A new attempt while any unresolved other attempt exists is blocked even if nominal unused balance remains.

Existing Succeeded/Failed cannot be overwritten. Changed reference on existing identified attempt rejected. A failed retry must be a new distinct attempt.

An imported unresolved attempt without reference may acquire one when verified; UI labels provider reference optional for existing records.

Payment result save records evidence only; success creates CS notice task when paid==total and reserved==0. Failed with unpaid balance leaves Finance next-attempt obligation; unknown/pending keeps reconciliation.

Every non-rejected refund record linked to ticket must be valid/current/approved and fully paid, zero reserved, before resolution.

Permissions

Shared eligibleUser Finance authorization in market; active read+finance, including custom role and All warehouses; inactive/no-read/wrong-scope rejected.

Production requirements

External payment integration is not authorized/built. If later added, distinguish initiate-payment commands from record/reconcile commands, with idempotency keys and gateway transaction IDs.

Transactional reservation ledger with per-authorisation locking; race tests must prove concurrent attempts cannot overpay.

Retain provider evidence securely; use reconciliation webhooks/jobs only with separately specified authentication and signature verification.

Do not retry uncertain financial action automatically. Escalate expired reconciliation while keeping funds reserved.

Acceptance criteria

ATTEMPT-01 *[Module acceptance]* **Given** Approved132 with paid60 and unknown72. **When** Finance attempts new payment record. **Then** Reject/block new attempt; available0 reserved72.

ATTEMPT-02 *[Module acceptance]* **Given** Unknown72 existing attempt. **When** Meaningful Failed outcome recorded. **Then** Preserve failed attempt, reserved0 available72; allow distinct new reference.

ATTEMPT-03 *[Module acceptance]* **Given** Available72. **When** New attempt73 submitted. **Then** Reject exceeds-authorisation; no financial mutation.

ATTEMPT-04 *[Module acceptance]* **Given** Available72. **When** Distinct new unknown72 recorded. **Then** Reserve72 and prohibit another attempt.

ATTEMPT-05 *[Module acceptance]* **Given** Unknown72. **When** Succeeded with whitespace-only proof submitted. **Then** Reject; paid/reserved unchanged.

ATTEMPT-06 *[Module acceptance]* **Given** Unknown72 with meaningful provider evidence. **When** Succeeded recorded. **Then** Paid132/reserved0; create CS notice; retain imported60 and earlier failed history.

ATTEMPT-07 *[Module acceptance]* **Given** Final Succeeded or Failed attempt. **When** User overwrites outcome/reference. **Then** Reject; use separate permitted attempt after failure.

ATTEMPT-08 *[Module acceptance]* **Given** Reference used on any refund. **When** New record with same reference differing case. **Then** Reject duplicate identity.

ATTEMPT-09 *[Module acceptance]* **Given** Custom All warehouses Finance user with read. **When** Valid result recorded; compare inactive/no-read/wrong-market variants. **Then** Eligible custom user succeeds; invalid variants reject.

ATTEMPT-10 *[Module acceptance]* **Given** Approved100 paid90. **When** Attempt11 then valid10. **Then** Reject11, accept10 once; retain two attempts90+10; ticket can close after other gates.

Source references: dist/finance.js:12–14,44–67; dist/portal.js:246–255,266–283; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-001–003; OPS-DOM-021–022; OPS-HOST-024–025

OPS DISPUTE Manual dispute evidence and provider outcome

Track evidence preparation and the externally confirmed provider result without claiming a bank submission.

Version 6 baseline

Existing seeded dispute can be updated; V6 has no general new-dispute creation UI or provider integration.

Record types: Evidence pack draft; Evidence actually submitted; Provider confirmed closed. Submission sets Submitted; awaiting decision and creates provider-outcome task; confirmed closure finishes dispute tasks.

External provider deadline is initialized from original due date and remains separate from next internal check. Internal rescheduling does not extend provider deadline.

Submitted evidence cannot be downgraded to draft; provider-confirmed closed record cannot be edited further by this form.

Current proof/notes fields are replaced on update; ticket activity and task histories retain some progression. No complete immutable evidence-pack version collection or won/lost outcome enumeration exists.

Fields and validation

Field	Type	Rules and meaning
providerDeadline	external timestamp	Required: Yes; Rule: Displayed independently of next internal check; no edit control in dispute form.
result / proof / notes	enum/text	Required: Yes; Rule: Trim proof/notes; submission and closure require nonblank provider receipt/evidence; draft proof may be absent.
submitted / submittedAt / status	state metadata	Required: Yes; Rule: submittedAt set once; provider closure terminal for this form.
nextCheck	timestamp	Required: Yes; Rule: Future IST even when recording closure; default24h. Current actor becomes provider-outcome owner.

Workflow and rules

Requires dispute grant and active ticket; terminal ticket must reopen.

Draft alone does not mark submitted or finish evidence obligation.

Provider closure is manual evidence; no network submission occurs. Baseline allows directly recording verified provider closure without requiring a locally recorded prior submission.

Ticket closure depends on outstanding dispute task; replacements.resolutionIssue has no independent scan of dispute state. Production must add explicit dispute state gate so missing/corrupt tasks cannot bypass unresolved dispute.

Permissions

Finance/dispute permission updates existing dispute; shared task ownership requires scoped dispute grant. V6 handler uses can(dispute), so uniform server market authorization remains required.

Production requirements

Immutable evidence-pack versions, attachment hashes, external receipt/reference, recorded actor/time and provider result taxonomy (e.g. won/lost/withdrawn) require design.

Create/import/link dispute workflow and explicit reopen/correction rules for provider-confirmed outcomes must be specified before production.

Separate provider deadline escalation from internal reminders; persistent background scheduling needed.

Do not claim bank submission until confirmed external receipt; retain original deadline if task snoozed/rescheduled.

Acceptance criteria

DISPUTE-01 *[Module acceptance]* **Given** Existing dispute. **When** Evidence pack draft saved. **Then** Remain unsubmitted; preserve external deadline and evidence obligation.

DISPUTE-02 *[Module acceptance]* **Given** Draft dispute. **When** Submission with whitespace proof. **Then** Reject without submitted state.

DISPUTE-03 *[Module acceptance]* **Given** Draft dispute. **When** Meaningful actual submission receipt recorded. **Then** Set submittedAt once; create Check dispute provider outcome task.

DISPUTE-04 *[Module acceptance]* **Given** Submitted dispute. **When** User selects draft. **Then** Reject downgrade.

DISPUTE-05 *[Module acceptance]* **Given** Submitted dispute. **When** Internal follow-up rescheduled. **Then** Provider submission deadline unchanged.

DISPUTE-06 *[Module acceptance]* **Given** Verified provider closure evidence. **When** Record Provider confirmed closed. **Then** Complete dispute tasks; other ticket obligations remain.

DISPUTE-07 *[Module acceptance]* **Given** Provider confirmed closed. **When** Another update attempted. **Then** Reject terminal form mutation.

DISPUTE-08 *[Module acceptance]* **Given** Production unresolved dispute with missing task. **When** Resolve/close attempted. **Then** Explicit dispute-state gate blocks; this is additional production acceptance, not V6 domain behavior.

Source references: dist/portal.js:285; dist/operations.js:39–43; dist/replacements.js:77–89; output/final-validation-2026-10-30/operations/cases.json:OPS-REG-010

OPS CLOSURE Customer confirmation resolution closure and reopening

Allow final status only when all current operational obligations are independently settled.

Version 6 baseline

CS must be current worker to record update; Start work and save claims ticket then records update. Copy status update never means sent.

Customer update sent requires explicit external-sent checkbox and may finish exact linked customer task. Approved waiver notice writes separate customerNotice/history.

Remedy outcome confirmed requires exact current approved request/revision and verified-actual-remedy checkbox. Return additionally selects refund treatment; replacement requires exact approved shipment.

Resolved is allowed only after resolutionIssue and classification checks. Closed additionally requires prior status Resolved and customer confirmation/reviewed-closure checkbox.

Authorized reopen clears resolvedAt, retains prior resolution timestamps, creates Review reopened customer issue task and preserves approvals/receipts/waivers/attempts/history.

Fields and validation

Field	Type	Rules and meaning
update type / text / occurredAt	enum, text, timestamp	Required: Yes; Rule: Internal note,Customer reply received,Customer update sent,Remedy outcome confirmed; actual occurrence separate from recorded time.
linked task / approved request / refund treatment / checkboxes	explicit selection	Required: conditional; Rule: Only intended task/request is completed; exact revision bound to confirmation.
new status / reason / closure assent	enum, text, boolean	Required: Yes; Rule: Reason trimmed by status guard; Closed requires prior Resolved and assent.
resolvedAt / previousResolutions	timestamps	Required: Yes; Rule: Current resolution date set once; reopen preserves old timestamp then clears current.

Workflow and rules

Gate order: finance validity/payment; any outstanding return/hold; waiver pending/stale/notice/inspection; Return refund-treatment completeness; cancellation execution/fallback; nonfinal service requests; each approved Return/Replacement exact remedy; all open follow-ups.

Every RMA on ticket counts, not only latest selected request; no-return exception on one request never erases other obligations.

Need information is unresolved across service/refund/waiver; Rejected request has no fulfilment duty but its customer explanation task must complete.

Imported/legacy evidence gaps must stay visible. Do not fabricate historic assents or completion timestamps.

Open follow-up list is authoritative for generic work, but production must enforce domain facts independently so task deletion/corruption cannot bypass finance/dispute/logistics obligations.

Permissions

Status requires shared eligible work in country. Reopen additionally requires assign/reopen grant. Closed to Resolved directly rejected; reopen to an active status first.

All CS team visibility is a read requirement; operational mutation authority still depends on action/market. Demo role menu is review simulation, never a production login mechanism.

Production requirements

Persist closure-assent actor/time/reason and exact assessed domain versions. V6 closure checkbox is not a separate durable attestation object.

Server-computed gates with transactionally consistent snapshots, durable tasks and notifications; no browser refresh should lose history in production.

Explicit late-arrival workflow retains closed history, requires authorized reopen and re-evaluates inspection/remedy before reclosure.

Acceptance criteria

CLOSE-01 *[Module acceptance]* **Given** All operational facts done but open follow-up remains. **When** CS resolves. **Then** Block and show remaining action.

CLOSE-02 *[Module acceptance]* **Given** Pending/Need information service or refund/waiver. **When** Resolve attempted. **Then** Block regardless of ticket display status.

CLOSE-03 *[Module acceptance]* **Given** Required refund authorisation absent or not fully paid. **When** Resolve attempted. **Then** Block with missing/unfinished finance reason.

CLOSE-04 *[Module acceptance]* **Given** Advance replacement delivered but inbound due. **When** Resolve attempted. **Then** Block on remaining return.

CLOSE-05 *[Module acceptance]* **Given** Approved waiver without notice or actual receipts damaged. **When** Resolve attempted. **Then** Block on notice/inspection.

CLOSE-06 *[Module acceptance]* **Given** All facts and tasks complete. **When** CS records Resolved. **Then** Set current resolvedAt and retain activity.

CLOSE-07 *[Module acceptance]* **Given** Open/In progress or unconfirmed Resolved. **When** User selects Closed. **Then** Require prior Resolved plus explicit reviewed closure/customer confirmation.

CLOSE-08 *[Module acceptance]* **Given** Resolved ticket with assign/reopen permission. **When** User reopens with reason. **Then** Clear current resolution, preserve earlier timestamp and create new review follow-up.

CLOSE-09 *[Module acceptance]* **Given** Closed waived return reopened and late item received. **When** Receipt/inspection/reopen task completed then Resolved to Closed. **Then** Retain all original waivers/receipts; new physical evidence appended; pass gates again.

Source references: dist/portal.js:108–111,322–334; dist/replacements.js:77–89; dist/workflow-access.js:24–41; output/final-validation-2026-10-30/operations/cases.json:OPS-FIX-005–008; OPS-REG-010–017; OPS-HOST-024–025

Approval and task transition matrix

One eligible manager decision is the launch baseline. A transition must create or complete the named owned task atomically with its source record. Operational evidence records actual external work; it never performs an external action by itself.

Service request

Transition	Next responsibility	Guard
New -> Pending decision Submit verified scope	Eligible manager Decide service request	Active linked-order ticket; checked SKU quantities and future due
Pending decision -> Need information Need information	Eligible CS Provide requested information	Non-final current record; manager note; same revision
Need information -> Pending decision Answer/resubmit	Eligible manager Decide service request	Same revision/scope; answer, owner and future due
Pending decision -> Approved/final Approve	Logistics Cancellation checks / RMA and conditions / no-return shipment	One eligible decision; no-return assent if applicable
Pending decision -> Rejected/final Reject	CS Explain rejected request	No fulfilment; customer task remains
Current before shipment -> Old superseded; new Pending decision Submit new scope revision	Eligible manager Decide new revision	Preserve received lines; hold ancestor RMAs; no pending/approved collection waiver

Cancellation

Transition	Next responsibility	Guard
Approved and incomplete -> Not possible after dispatch Record already dispatched	CS Request or complete separate Return	Valid same-ticket full-quantity alternative; obligation not retired by link alone
Externally cancelled or completed fallback -> Outcome settled Revisit valid execution checks	CS Confirm outcome/refund eligibility with customer	Refund treatment reviewed and fallback actual remedy complete

RMA

Transition	Next responsibility	Guard
Registered or partial -> Partial / Inspection required / Settled Receive/waive remaining quantities	Logistics or CS Remaining receipt / inspection / replacement dispatch / Return remedy	Separate balances and inspection; do not duplicate dispatch on advance path

Replacement

Transition	Next responsibility	Guard
Approved -> Independent outbound + inbound work Register RMA for advance	Logistics Request fulfilment plus return receipt	RMA before outbound; neither task removes the other obligation

Transition	Next responsibility	Guard
Release eligible -> Shipment recorded Record external dispatch	CS Verify delivery and remedy	Unique reference and exact approved scope; inbound may remain for advance

Collection waiver

Transition	Next responsibility	Guard
New -> Pending decision Request against current balance	Eligible manager Decide waiver	One pending per RMA; nonzero whole remaining quantities
Pending decision -> Approved Approve	CS Tell customer waived quantities	Explicit assent and unchanged snapshot; sync inbound work independently
Pending decision -> Need information / Rejected Need information / Reject	CS or Logistics for information; CS for rejection Provide information / explain rejection	No waived quantity applied
Pending decision or Need information -> Superseded by receipt Receipt/inspection changes	Logistics for remaining physical work Fresh waiver only if still required	Cancel stale waiver tasks; retain original history

Refund authorisation

Transition	Next responsibility	Guard
New or approved amendment -> Pending decision / Need information / Pending decision Submit / Need info / answer	Manager / CS / manager Decision / information / decision	No execution while authorisation nonfinal
Pending decision -> Approved / Rejected Approve or initial Reject	Finance if unpaid; CS if rejected or already paid Reconcile/record attempt / explain rejection / customer notice	Approved exact scope; defaults 24h

Refund amendment

Transition	Next responsibility	Guard
Pending decision -> Previous approved values restored Reject	Finance if balance remains else CS Next attempt/reconciliation or customer notice	Keep paid attempt history; proposal audit retained

Refund attempt

Transition	Next responsibility	Guard
New or unresolved -> Reserved Record unknown/pending	Finance Verify provider outcome	No new attempt or amount amendment until resolved
Unresolved -> Immutable Failed / Succeeded Record Failed / Succeeded	Finance if unpaid; CS if total paid Next distinct attempt / customer notice	Meaningful proof; cannot overpay; reserve released

Dispute

Transition	Next responsibility	Guard
Draft/existing -> Submitted; awaiting decision Record actual external submission	Current dispute-permitted actor Check provider outcome	Provider receipt required; provider deadline unchanged
Not provider closed -> Provider confirmed closed Record confirmed provider closure	No further dispute task Other ticket obligations continue	Evidence required; no actual bank action

Ticket

Transition	Next responsibility	Guard
Active -> Resolved Resolve after all gates	No open operational tasks Review final closure	Classification complete; domain obligations settled
Resolved -> Closed Reviewed closure assent	None None	Must already be Resolved; gates rechecked
Resolved/Closed -> Chosen active status Authorized reopen	Reopening eligible CS/manager Review reopened customer issue	Work+assign/reopen permission; history retained

Resolution and closure matrix

Every applicable row must be clear. A no-return replacement or a collection waiver never clears another return, delivery, decision, refund, dispute or task. Reviewed closure is not an override of these obligations. V6 dispute blocking is task-based; production must also check dispute state independently as specified in the dispute module.

Obligation	Blocks when	Cleared by
Service decision	Any current request final=false, including Need information.	Eligible final decision; rejection explanation task also completed.
Physical return	Any RMA has scope hold or positive remaining expected-received-effectiveWaived.	Valid receipt or explicit manager waiver; hold cleared by approved current scope.

Obligation	Blocks when	Cleared by
Inspection	Any actual received items lack acceptable inspection for current receipt scope.	Valid acceptable inspection event; no inspection required for zero actual receipts.
Collection waiver	Pending/Need information; approved scope mismatch; approved notice missing.	Fresh final decision/withdrawal/supersession as appropriate; CS customerNotice for approved waiver.
Replacement remedy	Current approved replacement lacks valid exact shipment or request/revision confirmation; no-return assent invalid.	Approved exact dispatch and CS remedy confirmation; inbound and other RMAs separately settled.
Return remedy and refund eligibility	No current confirmation; unsettled/unchecked physical return; missing refund treatment; Refund required without matching non-rejected authorisation.	Exact remedy confirmation and explicit No refund due/Refund required; full refund settlement separately checked.
Cancellation	Warehouse stop/external cancel unconfirmed and no complete valid alternative Return; refund treatment missing.	Ordinary external cancellation or complete linked fallback; revisit checks and CS update.
Refund settlement	Non-rejected authorisation undecided/Need information, stale scope, paid!=total or reserved>0.	Exact eligible approval and successful external evidence for entire promise; zero unresolved funds; customer task completed.
Dispute	Open dispute evidence/provider-outcome task; V6 has no additional standalone dispute-state check.	Verified provider closure finishes tasks; production additionally checks dispute state independently.
Customer notice/other follow-ups	Any open task on ticket, even after physical/financial completion.	Valid task-specific evidence/action; arbitrary task deletion must not bypass production domain gates.
Final status	Needs classification; Closed requested before Resolved or without reviewed/customer-confirmation assent.	Authorized classification, Resolved, then explicit Closed assent with reason.

ADM USERS User Management

Create staff identities, assign role combinations and operational scope, and change account state without stranding work.

Version 6 baseline

Users, Roles & permissions and Admin activity tabs share a text search. User Management is visible only with users grant.

New users default to Invited and All markets. The demo identity selector contains Active accounts only. Invited does not send an invitation.

Editing a name updates matching legacy owner/worker/taskOwner strings on tickets and matching owner strings on requests/returns. Task owner IDs and prior actor snapshots are separate records.

There is no user deletion UI. The persistence boundary is the current browser tab.

Fields and validation

Field	Type	Rules and meaning
id	string	Baseline: Generated U plus session counter; immutable in form.
name	string	Baseline: Required HTML text field; no explicit trimming, whitespace-only rejection or uniqueness check in editUser.
email	email string	Baseline: Required HTML email input; case-insensitive duplicate check across users. Unlike reset matching, the save check does not explicitly trim.
roles	role ID array	Baseline: At least one assigned role. Effective-grant preview updates on form change.
scope	single string	Baseline: All markets, All warehouses, or any Countries name, including archived country entries.
status	enum	Values: Active; Invited; Inactive

Workflow and rules

Reject duplicate work email and no-role submissions before mutation.

Reject removal of super or non-Active state on the last Active Super Admin.

Before saving an existing user, check all open tasks owned by ID and active ticket owner/worker assignments. Block effective loss of login/read/work/scope/manager/logistics/finance/dispute ability that would strand currently eligible work.

Both Active to Invited and Active to Inactive require handover when open work remains. Changing roles while keeping Active is not a bypass.

Changing a shared role evaluates every affected user before changing the role.

The guard requires a separate eligible handover. It does not automatically reassign tasks, reset deadlines or clear missed commitments.

Retaining an equivalent grant through another role avoids a false handover block. Scope narrowing is allowed if remaining work is still eligible.

Guard nuance: read loss or Active-to-nonactive blocks any owned open tasks; other capability loss uses current-eligible to proposed-ineligible comparison. Pre-existing invalid assignments are not universally repaired by this guard.

Permissions

Entry: users

Create edit: users

Create role button: roles in addition to User Management entry

Default authorised role: Super Admin

Delegation caveat: Any assigned users grant can assign any listed role, including super. Production delegation restrictions are a decision, not implemented policy.

Production requirements

Authenticate the actor and enforce user/role changes on the server; never use the demo identity selector as authorisation.

Perform eligibility, last-active-admin and uniqueness checks atomically against current records, including concurrent edits.

Use immutable staff IDs for every assignment and retain historical actor names as snapshots. Do not migrate ownership through display-name string matching.

Normalize and validate work email and nonblank staff name server-side; use unique normalized identity constraints.

Persist before/after roles, scope and account state, actor, reason and timestamp. Revoke sessions when access is removed.

Implement a handover review of all open responsibilities, using their source-specific eligibility and preserving original/current deadlines and overdue attribution.

Implement actual invitation delivery and verification separately from account status, with explicit failure/expiry handling.

Acceptance criteria

ADM-USERS-01 *[Preserve baseline]* **Given** An actor without users grant. **When** They attempt User Management. **Then** The navigation entry is absent and the page denies access.

ADM-USERS-02 *[Preserve baseline]* **Given** A valid staff form with no selected role. **When** Save is requested. **Then** At least one role is required and no user is created.

ADM-USERS-03 *[Preserve baseline]* **Given** An existing work email. **When** Another user is saved using a case variant. **Then** The duplicate is rejected without changing the existing account.

ADM-USERS-04 *[Preserve baseline]* **Given** Only one Active Super Admin remains. **When** Its super role is removed or state becomes Invited/Inactive. **Then** The change is rejected.

ADM-USERS-05 *[Preserve baseline]* **Given** An Active person owns open tasks. **When** Their account is changed to Invited or Inactive. **Then** Save requires handover and lists representative work references; task history is unchanged.

ADM-USERS-06 *[Preserve baseline]* **Given** An eligible work owner remains Active. **When** The required grants or country scope are removed. **Then** Save is blocked until relevant work is handed over.

ADM-USERS-07 *[Preserve baseline]* **Given** A classification or decision task owner. **When** The reserved managerial role is removed while work remains. **Then** Save is blocked even if ordinary work permission remains.

ADM-USERS-08 *[Preserve baseline]* **Given** All ticket and follow-up responsibilities have been handed over to eligible users. **When** The previous account is inactivated. **Then** Save succeeds and the user disappears from the demo identity menu; deadlines and missed-owner history remain.

ADM-USERS-09 *[Preserve baseline]* **Given** A second assigned role supplies the same needed capability. **When** Another role is removed. **Then** Equivalent effective access does not cause a false handover block.

ADM-USERS-10 *[Preserve baseline]* **Given** All assigned work is in one country. **When** Scope narrows to that same country. **Then** The change may succeed if every responsibility remains eligible.

ADM-USERS-11 *[Production requirement]* **Given** Concurrent admins attempt to remove the final Active Super Admin. **When** Both transactions are processed. **Then** The server preserves at least one Active Super Admin and records the rejected attempt.

ADM-USERS-12 *[Production requirement]* **Given** A staff display name changes. **When** The edit commits. **Then** Immutable ownership IDs still resolve correctly and historical action snapshots remain attributable.

Source references: dist/portal.js:313-315; dist/portal.js:44-46; dist/admin.js:4-22; tests/admin.test.mjs:11-69

ADM ROLES Roles grants and effective permissions

Maintain role definitions and show the union of capabilities assigned to each user.

Version 6 baseline

Role editor has a required name and one checkbox for each of the 16 grants. Zero selected grants are permitted.

Role names are checked case-insensitively for duplicates. They are not explicitly trimmed by the save function.

Super Admin retains its exact name and all grants. Built-in CS cannot lose read. Other built-in role grant lists may be edited.

No role deletion, role hierarchy, deny rule, temporary role assignment or per-user grant override is implemented.

Fields and validation

Field	Type	Rules and meaning
id	string	Baseline: Reserved seed ID or generated role-session-counter.
name	required text	

Field	Type	Rules and meaning
grants	array of IDs from the exact grant matrix	
builtin	boolean	Baseline: Seed metadata; not a universal immutability flag.

Workflow and rules

Effective grants are a set union, so adding a role cannot negate another role's permission.

The role list shows assigned-user count and whether the reserved ID has service-decision authority.

Every assigned user is checked against the proposed shared-role definition before save; one stranded-work conflict blocks the whole role edit.

Custom role names do not create Country Manager, Head of CS or Super Admin authority. Approval is tied to IDs, not labels or number of grants.

A users grant controls account/role assignment; a roles grant controls role definitions. The normal UI combines them through User Management.

No explicit self-privilege-escalation or maker/checker control exists in role editing.

Permissions

Edit role: roles

Normal navigation: users

Review own effective grants: Profile & Security without a separate grant

Service approver ids: super; head; cm

Production requirements

Use an explicit server permission catalogue and reject unknown grant IDs.

Evaluate the effective union and source-task eligibility transactionally before applying shared-role changes.

Cache invalidation/session enforcement must take effect for all users inheriting an edited role.

Keep privileged reserved identities independent of editable display labels.

Persist role-definition before/after history and impacted-user audit. Decide delegated-admin boundaries before enabling production custom role administration.

Acceptance criteria

ADM-ROLES-01 *[Preserve baseline]* **Given** A user has two roles with overlapping grants. **When** Effective permissions are displayed. **Then** Each grant appears once as the union.

ADM-ROLES-02 *[Preserve baseline]* **Given** An existing role name. **When** A new role uses a case variant. **Then** Save is refused.

ADM-ROLES-03 *[Preserve baseline]* **Given** Super Admin is edited. **When** Grant removal or a different name is attempted. **Then** All grants remain enabled and the reserved name is preserved.

ADM-ROLES-04 *[Preserve baseline]* **Given** The CS role is edited. **When** read is deselected. **Then** Save is refused even if no task would be stranded.

ADM-ROLES-05 *[Preserve baseline]* **Given** A shared role supplies work to active task owners. **When** Its work grant is removed. **Then** The edit is blocked for the affected user and no partial grant mutation occurs.

ADM-ROLES-06 *[Preserve baseline]* **Given** Every affected user retains equivalent grants through another role. **When** The first role loses those duplicate grants. **Then** The edit can succeed without forcing unnecessary handover.

ADM-ROLES-07 *[Preserve baseline]* **Given** A new custom role with every ordinary grant and a managerial-looking name. **When** Its user attempts final service approval. **Then** The custom role alone does not confer reserved manager authority.

ADM-ROLES-08 *[Preserve baseline]* **Given** A custom role has no grants. **When** It is saved and assigned. **Then** The role exists but does not unlock shared workspace capabilities.

ADM-ROLES-09 *[Production requirement]* **Given** A role changes while an affected user has an open session. **When** The next protected action is submitted. **Then** The server uses current grants rather than stale browser state.

ADM-ROLES-10 *[Production requirement]* **Given** A role-change request contains an unknown permission ID. **When** It reaches the server. **Then** The request is rejected and no unrecognised authority is stored.

Source references: dist/data.js:3-6; dist/portal.js:313-315; dist/portal.js:390-390; dist/admin.js:18-22

ADM SCOPE Shared visibility operational scope and task eligibility

Separate the ability to see shared customer work from the ability to perform a scoped action or own its follow-up.

Version 6 baseline

Customer directory and ticket visibility remain shared across brands and the CS team; country scope does not filter normal read access.

userInScope treats reserved super/head roles and scopes All markets or All warehouses as global. Otherwise the user's one scope string must equal ticket.country.

All warehouses is global in the shared operational eligibility helper, including a custom Finance user. It is not limited to a logistics warehouse entity.

managerMayDecide requires Active and super/head, or cm with All markets/exact ticket country. A cm user scoped All warehouses does not gain global manager authority.

A missing read grant blocks ordinary workspace pages. Users, resets, profile and masters are exceptions to the outer read gate and have their own action/page checks.

Some raw action functions use can(grant) while shared eligibility uses Active + read + grant + scope. Treat this browser implementation as an interaction baseline, not proof that every endpoint is securely enforced.

Fields and validation

Field	Type	Rules and meaning
user.scope	single scope label	
ticket.country	country name snapshot	
task.ownerId	staff ID	
task.sourceType/step	routing classifiers	
unmatched.warehouse	country-name label	Baseline: Used as scope country for unmatched parcel task eligibility.

Workflow and rules

Normal work/finance/logistics eligibility requires Active, read, the needed grant and in-scope ticket when supplied.

Decision tasks for request/waiver/refund require read + Active + scope + reserved manager authority, with no separate approve grant.

Classification tasks additionally require work.

Waiver information tasks require work OR logistics after Active/read/scope checks.

Customer update, rejection, remedy and alternative steps require work. Refund information also routes to work.

Other refund tasks require finance; dispute tasks require dispute; return/parcel and fulfil steps require logistics; remaining tasks default to work.

A ticket assignment is separate from each follow-up assignment. Handover must address both when applicable.

Read is separate from export. There is no distinct contact/PII visibility grant or brand-level isolation in the prototype.

Permissions

Shared read: read

Manager authority: Reserved role ID plus account state and manager scope

Owner eligibility: Source-specific taskOwnerEligible mapping

All warehouse nuance: Global operational wildcard; not a CM approval wildcard

Production requirements

Implement the same explicit permission/scope checks on the server for all reads, mutations, ownership changes and exports; use user IDs from authenticated sessions.

Use stable country and warehouse IDs rather than display strings and model warehouse scope separately if the business requires that distinction.

Keep shared team visibility independent from operational country authority; do not introduce hidden per-agent ticket silos.

Define export/PII treatment as its own policy without assuming that visibility automatically grants bulk extraction.

Specify whether reserved manager authority requires current read access in every server decision endpoint, avoiding inconsistent raw helper use.

Implement task-owner checks in both pickers and transaction-time validation. Retain eligibility checks when user/role changes are applied.

Acceptance criteria

ADM-SCOPE-01 *[Preserve baseline]* **Given** A country-scoped CS account with read. **When** It views shared customers and team tickets. **Then** Visibility is not reduced to only its owned or in-country records.

ADM-SCOPE-02 *[Preserve baseline]* **Given** An Active user lacks read but has work. **When** Ordinary workspace access or task eligibility is evaluated. **Then** The user cannot access the workspace or become an eligible work owner.

ADM-SCOPE-03 *[Preserve baseline]* **Given** An Active custom user has read and finance with All warehouses scope. **When** An authorised external refund result is recorded. **Then** Shared eligibility permits the action; this positive path was also browser-tested on Version 6.

ADM-SCOPE-04 *[Preserve baseline]* **Given** A UK-only Country Manager. **When** A US service request requires a final decision. **Then** The manager cannot decide it; Head of CS or Super Admin can.

ADM-SCOPE-05 *[Preserve baseline]* **Given** A cm account has All warehouses but not head/super. **When** Manager approval eligibility is checked. **Then** All warehouses does not substitute for All markets or exact country.

ADM-SCOPE-06 *[Preserve baseline]* **Given** A custom role has a managerial title but a generated role ID. **When** Manager eligibility is checked. **Then** The name does not confer decision authority.

ADM-SCOPE-07 *[Preserve baseline]* **Given** A classification follow-up. **When** Owner choices are built. **Then** Only Active in-scope users with read, work and manager authority qualify.

ADM-SCOPE-08 *[Preserve baseline]* **Given** A dispute follow-up. **When** A read + dispute custom user is checked. **Then** No additional finance grant is required by task-owner eligibility.

ADM-SCOPE-09 *[Preserve baseline]* **Given** An unmatched parcel task with a warehouse country. **When** Owner eligibility is evaluated. **Then** The warehouse country is used for scope and logistics is required.

ADM-SCOPE-10 *[Preserve baseline]* **Given** An overdue follow-up is reassigned. **When** An eligible new owner is saved. **Then** Original due time, first missed time and missed owner remain in task history.

ADM-SCOPE-11 *[Preserve baseline]* **Given** A CS user has read but no export. **When** Exports is opened. **Then** The page explains the missing export permission and does not offer dataset downloads.

ADM-SCOPE-12 *[Production requirement]* **Given** A crafted request bypasses UI visibility controls. **When** The server receives a protected mutation. **Then** It revalidates current state, effective grants and applicable scope before any change.

Source references: dist/workflow-access.js:4-22; dist/operations.js:8-8; dist/portal.js:44-47; dist/portal.js:404-406; dist/portal.js:419-425; dist/portal.js:444-444; dist/finance.js:42-44

ADM PROFILE Profile Security and User Profile Requests

Show the current staff identity and permissions, and review the legacy password-reset queue without misrepresenting a delivered reset.

Version 6 baseline

Profile shows name, email, role names, fixed IST time zone and scope. It has no editable profile preference controls despite its display-preferences subtitle.

Request password reset adds a queue row for the current user's email with Pending verification. Repeated clicks can create multiple request rows.

Queue records list request ID, email and state. accounts grant is required to review them.

Review resolves the current email identity each time using trimmed lowercase matching. Exactly one Active staff match is required for eligibility.

The verified action records Verified - link not sent (demo). The source stores the longer punctuation form displayed in the interface. It never stores passwords, authenticates the requester or sends a link.

Unknown, ambiguous and nonactive identities have explicit reviewed states. Saving repeats the identity check; previous status text cannot confer eligibility.

Fields and validation

Field	Type	Rules and meaning
reset.id	string	
reset.email	string	Baseline: Captured from the requesting demo identity or seed.
reset.status	state label	Baseline: Seed labels Expired/Not found; new Pending verification; reviewed no account/ambiguous/account not active; verified link not sent.
reset.accountId	nullable staff ID	Baseline: Set from current matching result, not from state label.
reset.reviewHistory	array	Baseline: at, actorId, previousStatus, status, accountId; append only when state or linked account changes.

Workflow and rules

No match yields Reviewed — no account and null accountId.

More than one normalized match yields Reviewed — ambiguous account and null accountId.

A unique non-Active match yields Reviewed — account not active with the matched ID.

A unique Active match yields Verified — link not sent (demo), without credential or email actions.

Saving requires an Active reviewer with accounts and an existing request from the queue.

An unchanged repeated review does not append duplicate reviewHistory entries, although the UI still writes an admin-log review entry.

Profile permissions are the union of assigned roles. The displayed final-service-decision hint is role-ID based and says eligible within scope; it is not an approval action.

Permissions

Profile: Available to the current demo identity, including without read

Self request: No separate grant in the current action

Review queue: accounts; saving also checks Active

Default reviewer: Super Admin

Production requirements

Replace the demo selector with authenticated sessions and establish actual identity proof before recovery actions.

Use secure expiring single-use recovery or identity-provider flows; do not represent review as delivery. Track requested, verified, sent, delivery failure, expired, consumed and cancelled states where implemented.

Protect recovery endpoints against enumeration and excessive repeated requests; define rate limits and verification methods with the security owner.

Persist request, actor, verification evidence/reference and delivery outcome without storing passwords or reusable recovery secrets in the audit.

Revalidate account eligibility at processing and token consumption; invalidate sessions or recovery artefacts according to the approved security policy.

Acceptance criteria

ADM-PROFILE-01 *[Preserve baseline]* **Given** A current demo user. **When** Profile is opened. **Then** Name, email, assigned roles, IST and scope match the selected identity.

ADM-PROFILE-02 *[Preserve baseline]* **Given** The profile reset action. **When** A request is created. **Then** A new queue row uses that account email and Pending verification; no link is sent.

ADM-PROFILE-03 *[Preserve baseline]* **Given** An actor without accounts. **When** They attempt reset-queue review. **Then** Access is refused.

ADM-PROFILE-04 *[Preserve baseline]* **Given** No staff identity matches an email. **When** The reset is reviewed twice. **Then** It remains Reviewed — no account and never becomes verified.

ADM-PROFILE-05 *[Preserve baseline]* **Given** Two staff identities have the same normalized email. **When** Review is saved. **Then** The ambiguous state is recorded with no verified account link.

ADM-PROFILE-06 *[Preserve baseline]* **Given** A unique Invited or Inactive identity. **When** Its reset is reviewed. **Then** The account-not-active state is recorded and no verified action is offered.

ADM-PROFILE-07 *[Preserve baseline]* **Given** A unique Active identity with case/space differences in the request email. **When** Review is saved. **Then** Normalized matching identifies that account and explicitly records that no link was sent.

ADM-PROFILE-08 *[Preserve baseline]* **Given** A reset was previously eligible but the account is now inactive. **When** The review is saved again. **Then** Current identity/state wins and the new review history records loss of eligibility.

ADM-PROFILE-09 *[Preserve baseline]* **Given** Identity and reviewed state are unchanged. **When** The same review is repeated. **Then** No duplicate reviewHistory transition is appended.

ADM-PROFILE-10 *[Production requirement]* **Given** An expired, consumed or revoked production recovery token. **When** It is submitted. **Then** The server refuses reuse and no credential change occurs.

Source references: dist/portal.js:316-317; dist/portal.js:387-390; dist/admin.js:24-43; dist/data.js:16-16

MASTER CATEGORIES Category Management

Maintain a controlled query-type, main-category and child-category hierarchy for intake and classification.

Version 6 baseline

Category rows are grouped by main and child; usage counts and Active, Archived or Parent archived state are visible.

The UI supports two category levels under a query type. A main category without children is selectable; when active children exist, intake requires a child.

Needs classification is the permanent fallback CAT-15 added by syncCategories; it is not an editable categoryNodes row.

Existing tickets preserve their saved category path when master names change. New choices use the current path.

Fields and validation

Field	Type	Rules and meaning
id	immutable category ID	
name	text	Baseline: Required; trim/collapse whitespace, reject < or >; HTML maxlength 80.
queryTypeId	query-type ID	Baseline: Active type required; existing node's type is fixed in editor.
parentId	nullable main category ID	Baseline: Child parent must be a root of the same type.
active/deleted	booleans	Baseline: Delete is a soft flag; archived/deleted nodes are unavailable to new intake.

Workflow and rules

Duplicate category name is forbidden case-insensitively within the same query type and parent; deleted nodes are excluded from the duplicate check.

New children require an active usable root. Editing a child may retain its existing archived root but cannot newly attach to another unusable root.

Archiving a root makes descendants unusable without necessarily flipping each child's own active flag. Tree-archive from delete review explicitly archives every affected node.

Restoring a child under an unusable root is blocked; restore the root first.

Delete review lists the affected subtree. Any direct or descendant ticket use blocks deletion and offers Archive instead.

Delete confirmation checks use again to protect against changes since the dialog opened.

Usage checks ticket category IDs and matching current category-path strings. The prototype does not implement a universal production foreign-key dependency scan.

An existing category cannot be edited while its query type is archived, because save requires an active type.

Permissions

View: Master data is view-only without masters

Mutate: masters

Agent intake: work, using controlled dropdowns; no category free text creation

Production requirements

Use stable IDs, parent/type foreign keys and transactional dependency checks. Preserve immutable category snapshots on historical tickets.

Enforce depth, cycle prevention, type consistency, length and normalized sibling uniqueness on the server, not just HTML controls.

Recheck current active availability when a stale intake form is submitted.

Record category mutations with actor, before/after values and affected subtree; resolve query-type retirement policy before rollout.

Acceptance criteria

MASTER-CATEGORIES-01 *[Preserve baseline]* **Given** An authorised administrator and an active query type. **When** A unique main category is created. **Then** It appears under that query type and becomes an approved intake choice.

MASTER-CATEGORIES-02 *[Preserve baseline]* **Given** An active main category. **When** A child is created. **Then** It inherits the same type and appears only beneath that parent.

MASTER-CATEGORIES-03 *[Preserve baseline]* **Given** A sibling category name already exists. **When** A case variant is saved at the same level. **Then** Save is rejected; the same label under a different parent is not automatically a collision.

MASTER-CATEGORIES-04 *[Preserve baseline]* **Given** A parent has active children. **When** An agent records its category. **Then** A controlled child choice is required rather than free text.

MASTER-CATEGORIES-05 *[Preserve baseline]* **Given** A root is archived. **When** Its children are inspected and intake is reopened. **Then** Children show Parent archived and are excluded from new selection.

MASTER-CATEGORIES-06 *[Preserve baseline]* **Given** An archived child under an archived root. **When** Restore child is selected. **Then** The administrator is instructed to restore the main category first.

MASTER-CATEGORIES-07 *[Preserve baseline]* **Given** An unused main and child subtree. **When** Deletion is confirmed. **Then** All listed nodes are soft-deleted and disappear from management/intake.

MASTER-CATEGORIES-08 *[Preserve baseline]* **Given** A ticket references a category or descendant. **When** Delete is requested. **Then** Deletion is blocked and archive is offered without rewriting ticket history.

MASTER-CATEGORIES-09 *[Preserve baseline]* **Given** No approved category suits the issue. **When** The agent chooses Needs classification. **Then** The fallback remains available and routes the ticket for manager classification.

MASTER-CATEGORIES-10 *[Production requirement]* **Given** A category becomes referenced after a delete dialog opens. **When** Delete reaches the server. **Then** The transaction refuses deletion and retains all references.

Source references: dist/catalog.js:3-19; dist/portal.js:301-307; dist/portal.js:373-375; dist/portal.js:72-78

MASTER PRODUCTS Product Management

Maintain product identities, brand eligibility and capacities used by tickets and service line items.

Version 6 baseline

List shows product name, SKU, brand, fixed capacity, state and management actions. Search includes product record fields.

Products are optional in initial intake, with Not known / not applicable. Service requests require explicit controlled product lines under their own validation.

New Query filters active, nondeleted products by selected brand or Both brands. A product with fixed capacity populates and disables the capacity input.

Existing ticket and request line snapshots remain readable after rename/archive; the catalogue product ID remains the link.

Fields and validation

Field	Type	Rules and meaning
id	immutable product ID	
name	text	Baseline: Required; trim/collapse whitespace; HTML maxlength 100.
sku	text	Baseline: Required; trimmed and uppercased; HTML maxlength 50; unique among nondeleted records, including archived ones.
brand	enum	Values: Both brands; Brand A; Brand B
capacity	capacity label or empty	Baseline: Not applicable maps to empty; other choices come from active Query Filters.
active/deleted	booleans	

Workflow and rules

Duplicate SKU checks are case-insensitive; duplicate product names are not prohibited.

Archiving removes new selections without deleting historical snapshots. Restore makes the product available again subject to brand.

Delete is blocked if a ticket.productId, request inbound/outbound line productId or return item productId references the product. The same check runs at confirmation.

An unused product is soft-deleted. Because deleted records are excluded from the SKU duplicate check, the demo allows later SKU reuse.

Brand choices are hardcoded labels. Capacity is stored as a label, not a capacity-master ID. Server validation and durable referential policy are not implemented.

Permissions

View: Read-only master view without masters

Mutate: masters

Intake select: work

Service line select: Action-specific service permissions plus approved line checks

Production requirements

Keep stable product/SKU identities and versioned snapshots on each ticket/service line; do not rewrite an approved remedy when master data changes.

Apply server-side brand and active-product validation on save and on revision; validate product quantities and references in the service domain.

Use transactional in-use checks across every dependent record and an approved SKU reuse/retirement policy.

Separate brand IDs and capacity values from editable display labels. Validate all lengths and allowed values server-side.

Acceptance criteria

MASTER-PRODUCTS-01 *[Preserve baseline]* **Given** A valid new product with unique SKU. **When** The administrator saves it. **Then** An active catalogue record is created with normalized name/SKU and selected brand.

MASTER-PRODUCTS-02 *[Preserve baseline]* **Given** A SKU exists on an active or archived product. **When** A case variant is submitted. **Then** Creation is blocked.

MASTER-PRODUCTS-03 *[Preserve baseline]* **Given** A Brand B-only product. **When** The agent selects Brand A. **Then** The product is not offered for that ticket.

MASTER-PRODUCTS-04 *[Preserve baseline]* **Given** A selected product has a fixed capacity. **When** Intake renders. **Then** Capacity is populated from the product and disabled for manual selection.

MASTER-PRODUCTS-05 *[Preserve baseline]* **Given** A product is archived. **When** New Query choices refresh. **Then** It disappears while existing recorded snapshots remain readable.

MASTER-PRODUCTS-06 *[Preserve baseline]* **Given** An unused product. **When** Deletion is confirmed. **Then** It is flagged deleted and removed from new choices.

MASTER-PRODUCTS-07 *[Preserve baseline]* **Given** A ticket, return or inbound/outbound service line references a product. **When** Delete is requested. **Then** Delete is blocked and archive is offered.

MASTER-PRODUCTS-08 *[Preserve baseline]* **Given** A user lacks masters. **When** They review products. **Then** They have no create/edit/archive/delete controls.

MASTER-PRODUCTS-09 *[Production requirement]* **Given** A product is retired or becomes referenced while an editor is open. **When** The stale mutation reaches the server. **Then** Current availability and dependency checks run transactionally; approved line snapshots remain unchanged.

Source references: dist/portal.js:308-310; dist/portal.js:376-377; dist/portal.js:72-78; dist/operations.js:9-17; dist/catalog.js:12-14

MASTER CHANNELS Query Channels

Control the channel label used when recording how a customer contacted the team.

Version 6 baseline

Seed values: Email, Chat, Call, Other.

Generic master list supports add, edit, archive, restore and search; no delete action.

New Query Received through offers active channel names. Channel is stored on the ticket as a label.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	required display text	
active	boolean	

Workflow and rules

Names are case-insensitively unique within the list, including archived entries. Generic save does not explicitly trim or reject whitespace-only strings beyond HTML required.

Archive affects new intake choices; historical channel labels are not migrated.

Email is an intake default label; changing it is not an email-integration configuration.

Permissions

View: Master data view

Mutate: masters

Production requirements

Use stable channel IDs and stored historical labels; enforce trimmed nonblank names and unique normalized values server-side.

Define a valid active default and behavior when the previous default is retired. Email ingestion remains future integration.

Acceptance criteria

MASTER-CHANNELS-01 *[Preserve baseline]* **Given** A unique channel. **When** It is added. **Then** It appears in the active Received through dropdown.

MASTER-CHANNELS-02 *[Preserve baseline]* **Given** An existing channel name. **When** A case variant is added. **Then** It is rejected.

MASTER-CHANNELS-03 *[Preserve baseline]* **Given** A channel used by an old ticket. **When** It is archived. **Then** It is absent from new intake while the stored historical channel remains readable.

MASTER-CHANNELS-04 *[Production requirement]* **Given** Only whitespace or an invalid default is submitted. **When** The server validates the change. **Then** A valid nonblank name and active default are required.

Source references: dist/data.js:15-15; dist/portal.js:75-78; dist/portal.js:297-311; dist/portal.js:382-382

MASTER TYPES Query Types

Control the top-level query classification used by categories and tickets.

Version 6 baseline

Seed values: Complaint, Enquiry, Communication. Add/edit/archive/restore are enabled.

Active type IDs drive New Query and category creation. Existing category type is fixed in category editor.

Some downstream semantics, including complaint reporting, depend on labels such as Complaint. Arbitrary custom type semantics are not universally configurable.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	required display text	
active	boolean	

Workflow and rules

Generic duplicate-name check is case-insensitive across active and archived records; trimming is not explicit.

Archived types are excluded from new intake and category creation. Existing category save requires its type to be active.

Renaming a type rebuilds current master category paths through syncCategories, but historical ticket type/path snapshots are not rewritten.

No generic in-use protection is implemented for type archive/rename; production must define consequences for active categories and reporting.

Permissions

View: Master data view

Mutate: masters

Production requirements

Separate semantic type codes from display labels so Complaint reporting is not broken by label changes.

Define retirement behavior for linked category trees and open tickets. Validate active linked type on intake and classification save.

Do not promise new workflow/report semantics merely because a new type label can be added.

Acceptance criteria

MASTER-TYPES-01 *[Preserve baseline]* **Given** An active type. **When** New Query or new-category editor opens. **Then** The type is offered as a controlled choice.

MASTER-TYPES-02 *[Preserve baseline]* **Given** A type is archived. **When** New intake choices are rebuilt. **Then** It is excluded without rewriting saved ticket snapshots.

MASTER-TYPES-03 *[Preserve baseline]* **Given** An existing category's type is archived. **When** That category is saved. **Then** The active-type requirement refuses the save until the type is restored.

MASTER-TYPES-04 *[Production requirement]* **Given** A display label changes. **When** Historical and current complaint reporting runs. **Then** Semantic type identity, not the mutable label, determines complaint membership.

Source references: dist/data.js:15-15; dist/catalog.js:16-19; dist/portal.js:72-78; dist/portal.js:303-304; dist/portal.js:311-311

MASTER CAPACITIES Query Filters product capacities

Maintain capacity options such as 6, 8 and 12 litres; this module is not saved ticket search filters.

Version 6 baseline

Seed values: 6 litres, 8 litres, 12 litres. Add/edit/archive/restore use the generic master editor.

Active capacity names feed optional intake capacity and product-capacity choices. Not applicable is a separate fixed UI option.

A selected product's fixed capacity is copied and locked in intake; product capacity is stored as a label.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	display label	
active	boolean	

Workflow and rules

Generic case-insensitive duplicate checking applies; no numeric litre parser or unit-conversion logic exists.

Archiving removes ordinary new choices but does not migrate existing product/ticket capacity strings.

There is no capacity-master in-use delete flow; generic records are archived/restored, not deleted.

Permissions

View: Master data view

Mutate: masters

Production requirements

Decide whether capacity is a structured numeric quantity/unit or a controlled label. Preserve old product/ticket snapshots.

Define handling for products whose fixed capacity references an archived/renamed option; do not silently replace their capacity.

Acceptance criteria

MASTER-CAPACITIES-01 *[Preserve baseline]* **Given** A new active capacity label. **When** Product editor and intake are opened. **Then** The label is offered in their controlled capacity choices.

MASTER-CAPACITIES-02 *[Preserve baseline]* **Given** A product has a fixed capacity. **When** It is selected in intake. **Then** Its stored capacity takes precedence over optional manual capacity.

MASTER-CAPACITIES-03 *[Preserve baseline]* **Given** A capacity is archived. **When** Ordinary capacity choices rebuild. **Then** It is absent from new selectable options while stored history remains.

MASTER-CAPACITIES-04 *[Production requirement]* **Given** An archived capacity is referenced by an active product. **When** The administrator changes that master value. **Then** The approved retirement policy preserves product identity/history and explains required remediation.

Source references: dist/data.js:15-15; dist/portal.js:75-76; dist/portal.js:297-311

MASTER OUTREACH Customer Outreach

Retain the legacy New/Existing Customer reference classification, not an outbound campaign module.

Version 6 baseline

Seed values: New Customer, Existing Customer, Unknown.

Generic add/edit/archive/restore/search are present. No current New Query field or customer-master assignment persists an Outreach choice.

There are no campaigns, mailing lists, message dispatches or automated outreach functions.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	display label	
active	boolean	

Workflow and rules

Generic duplicate-name and mutation rules apply. The list is reference-only in Version 6.

Adding or archiving an entry must not be described as changing intake behavior or sending communication.

Permissions

View: Master data view

Mutate: masters

Production requirements

Keep this explicitly reference-only unless the business confirms a persisted customer classification and its derivation rules.

If activated later, define the entity, default, history and meaning of New versus Existing; do not infer outbound communication consent or campaign behavior.

Acceptance criteria

MASTER-OUTREACH-01 *[Preserve baseline]* **Given** The reference list. **When** It is opened. **Then** New Customer, Existing Customer and Unknown are visible with their states.

MASTER-OUTREACH-02 *[Preserve baseline]* **Given** An authorised master editor. **When** A unique reference is added or edited. **Then** The list updates and an admin-log entry is recorded.

MASTER-OUTREACH-03 *[Preserve baseline]* **Given** An Outreach entry is archived/restored. **When** The state changes. **Then** Only its reference-list availability changes; no message is sent and no customer classification is automatically rewritten.

MASTER-OUTREACH-04 *[Production requirement]* **Given** No policy has approved operational Outreach linkage. **When** The production module is delivered. **Then** Its interface clearly identifies reference-only scope and offers no implied campaigns.

Source references: dist/data.js:15-15; dist/portal.js:61-61; dist/portal.js:75-78; dist/portal.js:297-311

MASTER STORES Purchase Stores

Maintain legacy brand/store references without implying an integrated order-source configuration.

Version 6 baseline

Seed references: Brand A / UK store, Brand A / US store, Brand B / EU store, Brand B / Canada store.

Generic add/edit/archive/restore/search are present. No current intake Purchase Store field binds to this list.

Brand choice in intake is separately hardcoded to Brand A/Brand B; a store label does not establish a brand-country mapping or external connector.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	display label	
active	boolean	

Workflow and rules

Case-insensitive duplicate-name checks apply to the list; generic archive has no order dependency scan.

Store labels are reference data only. They do not authorize order changes, imports or external commerce access.

Permissions

View: Master data view

Mutate: masters

Production requirements

If store selection becomes operational, define stable store IDs, brand/country mapping, order namespace and account permissions before connecting it.

Do not display connector status or synchronization success for a reference-only entry.

Acceptance criteria

MASTER-STORES-01 *[Preserve baseline]* **Given** The legacy store master. **When** It is opened. **Then** Each reference has ID, name, state and permitted management controls.

MASTER-STORES-02 *[Preserve baseline]* **Given** An existing store name. **When** A case variant is created. **Then** The duplicate is blocked.

MASTER-STORES-03 *[Preserve baseline]* **Given** A store entry is renamed or archived. **When** The master is saved. **Then** The reference list changes but no order or intake brand mapping is automatically changed.

MASTER-STORES-04 *[Production requirement]* **Given** No operational store mapping has been approved. **When** The production reference module is shown. **Then** It remains labelled as reference-only with no claimed external synchronization.

Source references: dist/data.js:15-15; dist/portal.js:75-78; dist/portal.js:297-311

MASTER COUNTRIES Countries and warehouse coverage

Maintain customer market labels used by intake, operational scopes and the prototype's country-based RMA warehouse choices.

Version 6 baseline

Seed countries include United Kingdom, United States, Canada, France and India. India is not excluded.

New customer country choices filter active countries. Staff scope and unmatched-parcel warehouse selectors map all country names without an active filter.

RMA warehouse coverage is available across the country list. There is no separate warehouse entity, warehouse address or per-warehouse staff map in this prototype.

Country labels are used in user scope, ticket/customer records and return/parcel scope comparisons. Generic rename does not migrate those stored strings.

Fields and validation

Field	Type	Rules and meaning
id	master ID	
name	country display label	
active	boolean	

Workflow and rules

Generic case-insensitive duplicate checking applies. Add/edit/archive/restore do not include country-reference dependency protection.

Archive removes new customer choices but does not imply a universal block on RMA or scope pickers.

All markets and All warehouses are fixed scope options outside the Countries list.

Permissions

View: Master data view

Mutate: masters

Assign staff scope: users

Rmaactions: logistics with action-specific checks

Production requirements

Represent countries and warehouses with stable IDs, explicit active status and separate business concepts.

Preserve all-warehouse RMA availability while defining operational eligibility and retirement behavior. Do not automatically exclude India.

Apply a consistent approved active/retired policy across intake, assignment, warehouse selection and existing work; prevent label changes from silently breaking scope checks.

Acceptance criteria

MASTER-COUNTRIES-01 *[Preserve baseline]* **Given** An active new country. **When** A new customer is created. **Then** It is available in the country dropdown.

MASTER-COUNTRIES-02 *[Preserve baseline]* **Given** A country is archived. **When** New customer choices are opened. **Then** It is excluded there; current source still includes it in staff scope and unmatched-parcel warehouse choices.

MASTER-COUNTRIES-03 *[Preserve baseline]* **Given** The initial master list. **When** Coverage is reviewed. **Then** India is present and is not automatically excluded from the portal.

MASTER-COUNTRIES-04 *[Production requirement]* **Given** A country or warehouse is renamed/retired while referenced by open work. **When** The production change is applied. **Then** Stable IDs preserve access and history, and the approved retirement policy handles every affected workflow consistently.

Source references: dist/data.js:15-15; dist/portal.js:61-61; dist/portal.js:314-314; dist/portal.js:364-364; dist/workflow-access.js:5-13

MASTER STATUSES Query Statuses fixed lifecycle states

Expose the supported ticket lifecycle labels without allowing master edits to break state transitions or filtering.

Version 6 baseline

Fixed states: Open, In progress, Waiting for customer, Waiting for return, Waiting for finance, Pending review, Resolved, Closed.

The list is read-only for every role, including Super Admin. No Add/Edit/Archive controls are offered.

Manual Change status offers seven states, excluding Pending review; approval workflow sets Pending review. Service decision status is separate from ticket lifecycle.

Fields and validation

Field	Type	Rules and meaning
id	seed master ID	
name	fixed lifecycle label	
active	fixed availability in the UI	

Workflow and rules

Master toolbar, editor and toggle dispatch all guard Query Statuses against mutation.

Terminal lifecycle states are Resolved and Closed. Reopening requires work plus assign, applicable scope and a reason, with prior resolution retained.

The status list is not an administrator-defined workflow engine. Custom states and aliases are unsupported.

Permissions

View: Master data view

Mutate: None, including Super Admin

Ticket transition: work and appropriate lifecycle guards; assign additionally for reopening

Production requirements

Persist lifecycle codes as stable enums; map display labels independently if localization is introduced.

Use an explicit transition service with business guards and immutable transition history. Do not re-enable arbitrary status master editing without a designed workflow migration.

Acceptance criteria

MASTER-STATUSES-01 *[Preserve baseline]* **Given** A Super Admin opens Query Statuses. **When** Available actions are inspected. **Then** The fixed eight labels are visible and Add/Edit/Archive are unavailable.

MASTER-STATUSES-02 *[Preserve baseline]* **Given** A direct master edit/toggle action targets Query Statuses. **When** It is dispatched. **Then** The fixed-state guard prevents mutation.

MASTER-STATUSES-03 *[Preserve baseline]* **Given** Open tickets exist. **When** The Open ticket filter is selected. **Then** Existing stable lifecycle values match the correct tickets.

MASTER-STATUSES-04 *[Preserve baseline]* **Given** A work-only custom user opens a Closed ticket. **When** They attempt to reopen it. **Then** The assign/reopen permission is required; approval decisions remain separate from lifecycle labels.

Source references: dist/data.js:15-15; dist/admin.js:45-48; dist/portal.js:297-311; dist/portal.js:382-382; dist/workflow-access.js:24-38

Analytics metric contracts

MET01 Tickets created

Grain: ticket

Baseline definition: Count of selected ticket cohort.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Custom from/to inclusive; default today-30 through yesterday. Today may be selected but is partial; Production must replace final-second bound with <next IST midnight.

Source refs: file: dist/portal.js; line: 288; symbol: cohort; file: dist/portal.js; line: 292; symbol: insightsPage

MET02 Customer complaints

Grain: ticket

Baseline definition: Count selected cohort where type === 'Complaint'.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Display string equality is current baseline; future immutable type IDs recommended; One ticket counts once regardless of number of messages or items.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage

MET03 Enquiries

Grain: ticket

Baseline definition: Count selected cohort where type === 'Enquiry'.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Other query types remain in Tickets created but not Complaint or Enquiry.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage

MET04 Top 10 complaints 7 30 60 days

Grain: primary full category path

Baseline definition: For N in 7,30,60: count Complaint tickets created from IST date today-N through yesterday; group by current full category string; descending count then localeCompare full path ascending; first ten.

Filters: Non-test tickets; original created timestamp between each fixed rolling window's IST start 00:00:00 and end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Independent overlapping windows; custom comparison range does not alter these dates; Brand/country filters do apply; Leaf labels may repeat; grouping/drilldown uses full path; Progress bar is category count divided by largest category count, not share of complaints; Zero rows yield no ranking items; caption still states zero complaints.

Source refs: file: dist/portal.js; line: 289; symbol: ranking; file: dist/portal.js; line: 292; symbol: insightsPage

MET05 Selected period complaint mix

Grain: primary full category path

Baseline definition: Same ranking function as Top 10 but over selected-period Complaint cohort; capped at ten.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Not a full category distribution if more than ten categories occur; Current reclassification changes grouping; no historical classification snapshot query.

Source refs: file: dist/portal.js; line: 289; symbol: ranking; file: dist/portal.js; line: 292; symbol: insightsPage

MET06 Late resolutions

Grain: ticket

Baseline definition: Count selected cohort with truthy resolvedAt and parsed resolvedAt > parsed resolutionDue.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: No requirement that resolution happened inside selected dates; it is created-cohort reporting; A reopened ticket has cleared current resolvedAt and is not a current resolution; Invalid/missing due does not pass comparison; production should explicitly exclude and report missing coverage.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage

MET07 Resolved on time

Grain: resolved ticket with due

Baseline definition: Denominator = selected rows where both resolvedAt and resolutionDue are truthy. Numerator = denominator rows where parsed resolvedAt <= parsed resolutionDue. Rate = numerator/denominator*100; display one decimal.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Empty denominator returns null and shows —; Exact deadline equality is on time; Baseline truthiness does not validate malformed dates; production must validate/exclude and disclose invalid inputs; Not resolved-in-period rate and not all-ticket closure rate.

Source refs: file: dist/portal.js; line: 290; symbol: rate

MET08 Median resolution duration

Grain: valid current resolved ticket

Baseline definition: For selected rows with truthy resolvedAt, duration hours=(resolvedAt-created)/3600000; retain values >=0, sort ascending; middle for odd count, mean of two middle values for even; display one decimal.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: No valid durations shows —; Negative and NaN durations excluded; missing resolution excluded; Elapsed clock hours, not business hours. Current resolution only; previous resolution cycles are not separately aggregated.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage

MET09 Open tickets past resolution target

Grain: active ticket in created cohort

Baseline definition: Count selected cohort where status is neither Resolved nor Closed and resolutionDue < current time.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Current backlog among the chosen created cohort, not historical backlog at period end; Equality at current time is not yet overdue; invalid dates do not pass.

Source refs: file: dist/portal.js; line: 19; symbol: active; file: dist/portal.js; line: 292; symbol: insightsPage

MET10 Needs classification

Grain: ticket

Baseline definition: Count selected cohort with category exactly 'Needs classification'.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Current metric uses label equality; the completion guard also recognises CAT15. Production should use immutable classification state/ID consistently; Unclassified tickets remain in total tickets.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage; file: dist/cs-records.js; line: 34; symbol: needsClassification

MET11 Timestamp data coverage

Grain: ticket

Baseline definition: Round(100 * count(selected rows with truthy created and received) / count(selected rows)).

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current outcomes of the original created-date cohort, not status as of cutoff.

Edge cases: Empty cohort shows —; Presence-only measure: not a full validity, source history or audit-completeness measure; Created-date filtering already excludes records that cannot parse into the cohort.

Source refs: file: dist/portal.js; line: 292; symbol: insightsPage

MET12 Equal period comparison

Grain: metric pair

Baseline definition: D=round((date(to)-date(from))/86400000)+1. Prior range=from-D through from-1. Counts show current-prior; rate shows current-prior percentage points to one decimal.

Filters: Non-test tickets; original created timestamp between selected IST start 00:00:00 inclusive and selected IST end 23:59:59.000 inclusive; selected exact brand/country or All. These are current

outcomes of the original created-date cohort, not status as of cutoff. Same brand/country for both periods.

Edge cases: If either rate is null, Unavailable; If prior is zero: current nonzero -> New vs comparison; both zero -> No change; Count differences are absolute counts, not percentage growth; No comparison for Late resolutions or median currently shown.

Source refs: file: dist/portal.js; line: 291; symbol: metricDiff; file: dist/portal.js; line: 292; symbol: insightsPage

MET13 Missed in selected period

Grain: task

Baseline definition: Count tasks with coverage === 'complete' and firstMissedAt IST calendar date within selected from/to, once per task.

Filters: Task reportScope: selected brand/country on linked ticket; selected source type; selected responsible user equals ownerAtFirstMiss if present, otherwise current owner. Tasks without linked ticket remain for All brands/All countries but do not match a specific brand/country.

Edge cases: Task ticket-created date does not restrict cohort; Subsequent reschedule, reassignment, snooze or late completion does not remove first miss; Valid actual on-time completion can correct a provisional miss with an audit event; Unknown historical coverage excluded; cancellations with an observed miss remain included.

Source refs: file: dist/operations.js; line: 91; symbol: TaskBook.missed; file: dist/portal.js; line: 447; symbol: reportTasks

MET14 Completed after a miss

Grain: task

Baseline definition: Count MET13 cohort whose current status is completed.

Filters: Task reportScope: selected brand/country on linked ticket; selected source type; selected responsible user equals ownerAtFirstMiss if present, otherwise current owner. Tasks without linked ticket remain for All brands/All countries but do not match a specific brand/country.

Edge cases: Retained first miss is prerequisite through cohort; Not a completion-date cohort; completion can occur after selected period.

Source refs: file: dist/portal.js; line: 448; symbol: followupInsights

MET15 Overdue tasks now

Grain: task

Baseline definition: Count currently open tasks with dueAt < now and reportScope.

Filters: Task reportScope: selected brand/country on linked ticket; selected source type; selected responsible user equals ownerAtFirstMiss if present, otherwise current owner. Tasks without linked ticket remain for All brands/All countries but do not match a specific brand/country.

Edge cases: No selected first-miss-date restriction and no complete-history requirement; Responsible-person filter still uses ownerAtFirstMiss || ownerId; it is not purely current-owner attribution; Current operational count, distinct from missed-in-period.

Source refs: file: dist/portal.js; line: 401; symbol: isOverdue; file: dist/portal.js; line: 448; symbol: followupInsights

MET16 Linked tickets affected by missed follow ups

Grain: distinct ticket ID

Baseline definition: Count unique nonempty ticketId values in MET13 cohort.

Filters: Task reportScope: selected brand/country on linked ticket; selected source type; selected responsible user equals ownerAtFirstMiss if present, otherwise current owner. Tasks without linked ticket remain for All brands/All countries but do not match a specific brand/country.

Edge cases: Multiple missed tasks on same ticket count once here but separately in missed-task total; Unmatched parcel task may have no ticket; task remains counted but contributes no linked ticket.

Source refs: file: dist/portal.js; line: 448; symbol: followupInsights

MET17 Unknown earlier task history

Grain: task

Baseline definition: Count all tasks whose coverage !== 'complete' and reportScope.

Filters: Task reportScope: selected brand/country on linked ticket; selected source type; selected responsible user equals ownerAtFirstMiss if present, otherwise current owner. Tasks without linked ticket remain for All brands/All countries but do not match a specific brand/country.

Edge cases: Not limited by selected dates and not necessarily open; Displayed to explain exclusions; these tasks can still appear in current operational queues.

Source refs: file: dist/portal.js; line: 448; symbol: followupInsights

MET18 My work cards and personal queue

Grain: task or ticket as labelled

Baseline definition: Five cards: Open team tickets counts non-test tickets with status neither Resolved nor Closed; My next actions counts open tasks whose ownerId=actor; Overdue follow-ups counts open tasks with dueAt<now; Pending decisions counts open tasks with step==='decision'; Rejected — action needed counts open tasks with step==='rejection'.

Filters: TaskBook.open; dashboard ticket preview excludes Resolved/Closed and test tickets.

Edge cases: First card counts tickets; the other four count tasks. One ticket can have multiple obligations; Preview lists first six tickets and first four owned tasks; previews are not complete counts; Open queue sorted by due time; task navigation retains documented predicate.

Source refs: file: dist/portal.js; line: 51; symbol: dashboard; file: dist/operations.js; line: 60; symbol: TaskBook.open

Delivery plan and release acceptance

Stage 1 establishes the schema, real identity and permission service, audit, controlled catalogue, shared customers, intake, tickets and search. Stage 2 delivers revisioned approvals, tasks, RMA receipts and waivers, replacement and cancellation execution, refund authorisations and dispute handling. Stage 3 completes Insights, exports, imports and administration hardening. Stage 4 rehearses migration and runs full multi-role UAT, security, concurrency, recovery and accessibility verification.

Keep each vertical workflow usable before adding broad integration coverage. In particular, deliver approval through closure as one tested sequence with the exact item and amount records connected. Do not release an approval screen that creates work no eligible role can execute, or a closure button that trusts an unchecked front-end status.

The current prototype evidence records 17 corrected audit issues, 113 validation cases, 87 passing automated tests and 15 checked downloaded CSV files. This is useful regression evidence for the reviewed design. It does not certify production authentication, persistence, scheduling, concurrency, real payments or migration. Re-run adapted tests against the implemented production architecture.

Release requires traceability from every must-have requirement to implemented behaviour and an executed test. Test as CS Executive, country-scoped manager, Head of CS, Logistics, Finance, Data Steward and Super Admin, including users with multiple roles and reduced custom grants. An Ops Manager role is not a separately defined Version 6 default; agree its capability composition before provisioning it.

The product owner signs off workflow and open decisions; CS signs off intake and closure; Logistics signs off receipt/waiver and shipment controls; Finance signs off money, evidence and reconciliation; the technical owner signs off security, restore tests, performance and monitoring. Record remaining limitations explicitly. No unresolved critical authorisation, financial balance, receipt-history, data-loss or premature-closure defect may be accepted for launch.

Required test suite	Evidence needed before release
Module and role UAT	All module criteria with actor, precondition, input, observed state, evidence and result; positive and prohibited actions for each role.
End to end service paths	Return first, advance replacement and no-return exception; partial RMA waiver, receipt race, late arrival; cancellation success and fallback Return; refund retry and information/rejection cycles through closure.
Authorisation and concurrency	Direct API denial; revoked roles and disabled users; country mismatch; stale approval; duplicate command; simultaneous receipt/waiver; simultaneous refund; takeover and stranded-work races.
Data and downloads	Import idempotency and stale conflict; shared-contact ambiguity; archived references; all eight export datasets; full filtered rows, UTF-8, formula safety and protected links.
Operations and resilience	Closed-browser scheduler; worker failure and replay; restore drill; migration counts and balances; monitoring alerts and deployment rollback evidence.
Usability and accessibility	Indian CS users complete lookup/intake and manager checks without inventing labels; keyboard and screen-reader review; narrow layout; draft preservation and helpful errors.

Decision register

The named owners are recommended decision owners, not evidence that they have already approved these policies. Resolve each decision before implementing its affected behaviour. Preserve the agreed single-manager approval baseline while discussing any optional policy changes.

D 01 Approval separation and limits

Decision owner: Ramesh and Head of CS with Finance

Preserve one eligible manager decision for the launch baseline. Confirm the self-approval policy; Version 6 has role and country authority but no proven self-approval prohibition. Additional approvers or monetary thresholds are optional policy changes requiring explicit agreement, not prerequisites inferred from the prototype.

Recommendation: Consider independent requester and reviewer identities for financial and no-return exceptions. Do not change the single-decision approval flow or invent thresholds without agreement.

D 02 SLA and operating calendar

Decision owner: Head of CS and Country Managers

Agree response, resolution and task targets by query type, priority and country; business hours, holidays, pause conditions and timezone boundaries.

Recommendation: Keep task due dates and first misses separate from ticket SLA; do not silently pause or restart them on snooze or reassignment.

D 03 Operational scope model

Decision owner: Operations lead and Super Admin

Define country, warehouse and cross-market assignments, and whether an Ops Manager is a custom composition or a new explicit role. Version 6 All warehouses scope is broader than a real warehouse membership model.

Recommendation: Use explicit country and warehouse membership and stage-based owner eligibility; retain the requested shared customer and ticket read visibility.

D 04 Migration sources and depth

Decision owner: Data owner with CS and Finance

Identify actual files/systems, available order history, receipt and financial histories, volumes, timestamps and cutover dates; confirm whether executing customer merges is launch scope.

Recommendation: Ship audited import preview and publishing; keep irreversible merge execution disabled until identity and reversibility rules are approved.

D 05 Inspection and exception policy

Decision owner: Logistics and Head of CS

Agree acceptable inspection outcomes, damaged or excess receipt handling, return deadlines, waiver evidence and who may record evidence for another operator.

Recommendation: Keep actual receipt, inspection, collection waiver and remedy distinct; never infer receipt or customer acceptance from silence.

D 06 Finance currency and evidence

Decision owner: Finance lead

Agree supported currencies and minor units, external-reference uniqueness scope, reconciliation evidence, pending-result review times and refund amount revision policy.

Recommendation: Reserve unknown or pending amounts and require fresh approval for changed authority; do not retry money whose provider outcome is unknown.

D 07 Customer closure and disputes

Decision owner: Head of CS and Finance

Agree reviewed-closure evidence when the customer does not respond, chargeback deadlines and escalation, and dispute outcome policy. Any proposed exception for an unresolved dispute needs an explicit policy. Pending or unknown refund attempts continue to reserve money and always block resolution and closure.

Recommendation: Use explicit reviewed closure only after all obligations are cleared; no automatic closure or refund triggered solely by non-response. Reviewed closure must never waive receipt, payment or reconciliation obligations.

D 08 Privacy retention and exports

Decision owner: Business data owner and security or legal owner

Agree retention, deletion or anonymisation, sensitive-field access, evidence file limits, export expiry and permitted destinations for each operating jurisdiction.

Recommendation: Apply least privilege, protected downloads and purpose-limited logging; keep operational audit tamper resistant while implementing the approved retention policy.

D 09 Analytics policy

Decision owner: Management and Head of CS

Agree complaint tie ranking, reopening treatment, business-hour resolution calculations, historic state-at-cutoff needs, test-data exclusion and late-arriving imported data treatment.

Recommendation: Retain clearly labelled created-cohort metrics first; add event-based historical reporting only with a specified event model and reconciled definitions.

D 10 Availability and volume targets

Decision owner: Technical owner with operations

Confirm concurrent users, historical record counts, attachment/export sizes, expected peaks, uptime, RPO and RTO; ratify or replace proposed performance targets.

Recommendation: Load-test realistic data and rehearse restoration before release, rather than treating a browser prototype as a capacity benchmark.

D 11 Reference only legacy masters

Decision owner: Head of CS

Confirm whether Customer Outreach and Purchase Stores remain reference maintenance only or become active intake/order attributes; establish dependencies before making them mandatory.

Recommendation: Preserve legacy meanings visibly and add controlled consumers only where a defined workflow uses them.

D 12 Integrations and notifications

Decision owner: Ramesh and technical owner

Agree which email, Shopify/order, warehouse, carrier, payment and notification connections are included after manual launch and who owns credentials and reconciliation.

Recommendation: Phase integrations separately; validate idempotency, attribution, retries and conflicting events before enabling automatic state changes.

Detailed implementation clarifications

These details supplement the owner register above. Related entries may share the same owner decision; they are retained to avoid losing a module-specific dependency.

POL-ADM-01

May delegated user/role administrators assign Super Admin or grant capabilities they do not themselves hold?

Baseline: users can assign any listed role; roles can edit ordinary grant lists. No privileged delegation ceiling or separate privileged-role assignment permission exists.

Recommendation: Define explicit privileged delegation and escalation controls while preserving the user's confirmed full Super Admin authority.

POL-ADM-02

May a manager approve a service, waiver, refund or protected correction they requested or materially edited?

Baseline: Manager eligibility does not compare approver with requester. No universal self-approval prohibition exists.

Recommendation: Choose maker/checker rules and any emergency exception policy explicitly. Do not present separation of duties as already approved.

POL-ADM-03

Should high-value refunds, no-return exceptions or user privilege changes need a second approver?

Baseline: One eligible manager decision is sufficient; there is no threshold-based multi-approval engine.

Recommendation: Confirm thresholds and approval count before adding them; do not invent financial limits.

POL-ADM-04

Should All warehouses remain a global capability scope for Finance and other roles, or become a separate warehouse dimension?

Baseline: Shared operational eligibility treats it globally; cm managerial approval does not. This is tested behavior, not a normalized production authorization model.

Recommendation: Use distinct market and warehouse scopes if necessary, with an explicit migration from the baseline.

POL-ADM-05

Are Ops Manager and Logistics Manager distinct production roles or labels for Logistics?

Baseline: Only Logistics is a built-in logistics role; custom roles can compose capabilities but cannot gain manager authority by name.

Recommendation: Confirm role names, capabilities and any reserved authority before implementation.

POL-ADM-06

Which identity provider, invitation verification, recovery method and session policy will production use?

Baseline: Role switching, invitations, authentication and reset delivery are simulated.

Recommendation: Select an authenticated backend/identity-provider flow and define recovery evidence, expiry, reuse, MFA and session invalidation requirements without treating suggested settings as approved.

POL-ADM-07

How should shared contact visibility differ from bulk export permission and retention?

Baseline: read exposes shared customer/ticket data; export is separate, with no PII masking grant or row-level export country restriction.

Recommendation: Retain shared CS ticket visibility and decide field-level sensitive-data treatment, export audit/retention and any permitted dataset restrictions explicitly.

POL-ADM-08

May administrators create arbitrary query types, or only rename labels for fixed semantic types?

Baseline: The master permits added labels; complaint/reporting semantics are not fully data-driven.

Recommendation: Keep stable semantic type codes and approve any new workflow/reporting meaning before enabling it.

POL-ADM-09

Should Customer Outreach and Purchase Stores remain reference-only at launch?

Baseline: Neither is linked to a persisted current intake field or external integration.

Recommendation: Confirm any operational fields and mappings separately. Do not infer campaigns or order integration.

POL-ADM-10

May deleted unused SKUs/category names be reused, and what is the retirement policy for referenced countries, types and capacities?

Baseline: Deleted catalogue entries are excluded from uniqueness checks; generic reference masters lack in-use retirement safeguards.

Recommendation: Define identifier reuse and archival policies that preserve historical interpretation and current eligibility.

POL-ADM-11

Are the seeded default role grants the launch assignment policy for correction, import, export and account administration?

Baseline: The exact matrix above is the implemented baseline; manager authority alone does not grant correction/import/accounts/masters.

Recommendation: Confirm any changes to default compositions explicitly; preserve the confirmed full Super Admin and shared CS visibility requirements.

DEC-06 Refund eligibility cap

Specify shipping/tax/discount/currency/order limits and eligibility authority. Prototype technical cap and manual verification are not approved commercial refund policy.

Recommendation: Require verifiable order total and cumulative external refund cap; no automatic foreign-exchange conversion.

DEC-07 Currencies and references

Confirm supported currencies/exponents and whether attempt reference uniqueness must include provider/merchant account; prototype uses global case-insensitive reference uniqueness and 2-decimal money.

DEC-08 Multi-ticket/multiple remedies for one order

Specify controlled relinking/allocations when an existing authorisation is on a different ticket or linked service revision becomes superseded. Prototype blocks stale links and duplicate authorisations; no relink UI is claimed.

DEC-09 Replacement shipment expansion

Specify partial outbound shipments, substitutions, loss/damage in transit and additional replacements. V6 models one shipment per approved scope; further work needs separate request.

DEC-10 Receipt and waiver corrections

Specify approval needed to correct erroneous physical receipt, revoke mistaken final waiver or change completed provider outcome. V6 preserves final records; generic overwrite is not an approved correction policy.

DEC-11 Dispute outcomes

Define dispute intake, provider identifiers, evidence deadlines, won/lost/withdrawn outcomes, financial effects and reopening. V6 records only draft/submitted/provider confirmed closed.

DEC-14 Operational deadlines

Confirm SLAs by market/action and escalation calendar. V6 commonly defaults 24h, replacement customer delivery 48h, with future IST times; these are prototype defaults, not signed business SLA.

DEC-CS01 Shared customer directory

Status: Approved requirement

Decision: One shared directory across brands; related tickets retain brand/country. All authorised CS users need shared ticket/history visibility; write and export capabilities remain separate.

DEC-CS02 Manual, controlled intake

Status: Approved requirement

Decision: Manual entry first, guided customer lookup at the start, selected master-data values and preserved original customer wording. Option A shared workspace with guided checks for costly actions.

DEC-CS03 Complaint meaning

Status: Approved requirement

Decision: Top 10 means customer complaints, with 7/30/60-day rankings and custom equal-period comparisons.

DEC-CS04 SLA clocks and reopening

Status: Production decision required

Decision: Approve calendar/business-hours rules, country holidays, priority targets and reopen SLA policy. Prototype 24-hour initial action and 72-hour resolution defaults are elapsed-hour demonstrations; explicit reopen retains original resolutionDue.

DEC-CS05 Legacy migration

Status: Production decision required

Decision: Support both newly entered records and imported legacy history, but identify source files/systems, field mapping, order-history scope, deduplication, import idempotency and unknown-history handling. Current fixed three-row preview is not a real importer.

DEC-CS06 Customer merge

Status: Not implemented / separate future work

Decision: Baseline duplicate review can keep records separate. Real irreversible merge needs approved matching policy, source ownership, alias conflicts, relationship reassignment and audit/rollback strategy before implementation.

DEC-CS07 Email capture and ticket creation

Status: Future phase

Decision: Choose provider and permissions later. Require message IDs, thread association, attachment retention, duplicate prevention, reviewed identity matching and human review of classifications; never use email ingestion as automatic approval.

DEC-CS08 Historical analytics meaning

Status: Production decision required

Decision: Preserve current created-cohort metrics explicitly. Decide separately whether to add resolution-event cohorts, historical backlog snapshots, multi-cycle resolution, historical category-at-event reporting, first-response SLA, CSAT and sentiment.

DEC-CS09 Identity safety

Status: Production requirement

Decision: Broad partial phone search is discovery; identity resolution uses full canonical country-qualified values or exact full stored number. Never infer a person from phone suffix, household sharing or customer country alone. Retain original input/provenance.

DEC-CS10 Cross-country writes with shared reads

Status: Production decision required

Decision: Define a consistent mutation-scope policy for CS work while preserving shared visibility. Prototype stage-specific assignment/status/approval checks do not constitute universal server-side write scope enforcement.

DEC-CS11 Field privacy and export authorisation

Status: Production requirement

Decision: Use server projections/search whitelists and capability checks before returning data. Hidden PII, security notes, credentials and tokens must not leak through raw object search or exports. Authorise generation and download separately.

DEC-CS12 Stable export schemas and test records

Status: Production decision required

Decision: Version all eight schemas and decide whether nested evidence stays JSON or moves to linked child datasets. Apply a consistent test/archive exclusion rule to every related dataset; current raw-row exports have variable headers.

DEC-CS13 Audit truth and service integrations

Status: Production requirement

Decision: Keep actual event time, recorded time, actor identity and history coverage separate. Unknown legacy facts remain unknown. The prototype records actions/evidence; external mail, warehouse and refund execution requires separately authorised integrations and idempotency.

Administration implementation checklist

PROD-ADM-01 [Production requirement]: Enforce authorisation, scope and record versions at server transaction time. UI visibility and frontend helpers are not a security boundary.

PROD-ADM-02 [Production requirement]: Store durable immutable IDs for actors, roles, grants, countries, warehouses, products and categories. Preserve human-readable historical snapshots separately.

PROD-ADM-03 [Production requirement]: Persist administrative changes and failed protected attempts with current actor, timestamp, target, reason and before/after state. Protect audit records against ordinary edits.

PROD-ADM-04 [Production requirement]: Make last-admin, uniqueness, stale-record and open-work handover checks atomic under concurrent sessions. The current single-tab model cannot demonstrate those guarantees.

PROD-ADM-05 [Production requirement]: Keep real import, irreversible merge, notification delivery and credential recovery separate from their explicit demo placeholders. Do not report successful external actions before provider confirmation.

PROD-ADM-06 [Recommendation]: Use one shared authorization service for action validation, eligible-owner pickers and access-change impact analysis, with the same covered task-source mapping.

Traceability to the 17 corrected audit issues

These references connect the final prototype corrections to the development contract. They are a regression minimum, not the full acceptance scope. The source evidence remains the 30 October validation report and case register. All new production controls need fresh execution evidence.

Issue	Required regression	Acceptance references
ADMIN-01	Access changes require eligible handover	ADM-USERS-05, ADM-USERS-06, ADM-USERS-07, ADM-USERS-08, ADM-USERS-09, ADM-USERS-10, ADM-ROLES-05, ADM-ROLES-06, PLAT-03
ADMIN-02	Reopen requires ticket work and assign permission	MASTER-STATUSES-04, CS-M05-AC08, PLAT-02
ADMIN-03	Reset request must be reviewed again after changes	ADM-PROFILE-04, ADM-PROFILE-05, ADM-PROFILE-06, ADM-PROFILE-07, ADM-PROFILE-08, ADM-PROFILE-09
ADMIN-04	Query Statuses remain fixed and read only	MASTER-STATUSES-01, MASTER-STATUSES-02, MASTER-STATUSES-03
CS-01	Manager classification cannot be bypassed	CS-M04-AC07, CS-M04-AC08, CS-M04-AC09, CS06-04, CS06-06
CS-02	Protected identity correction checks historical aliases	CS-M03-AC01, CS-M03-AC02, CS-M03-AC03, CS-M03-AC04
CS-03	Ticket CSV includes complete fields and matching rows	CS09-01, CS09-02, CS09-03, CS09-06
CS-04	Phone search normalises international formatting	CS07-01, CS07-02
CS-05	Operational export search uses linked customer data	CS07-03, CS09-06
CS-06	Blank customer names are rejected	CS-M02-AC02
OPS-001	Dispatched cancellation supports linked Return fallback	CANCEL-01, CANCEL-02, CANCEL-03, CANCEL-04, CANCEL-05, CANCEL-06, CANCEL-07, CANCEL-08, CLOSE-01, CLOSE-07, PLAT-08
OPS-002	New refund authorisation and separate attempts	AUTH-01, AUTH-02, AUTH-03, AUTH-04, AUTH-05, AUTH-06, AUTH-07, AUTH-08, AUTH-09, AUTH-10, ATTEMPT-01, ATTEMPT-02, ATTEMPT-03, ATTEMPT-04, ATTEMPT-06, ATTEMPT-07, ATTEMPT-08, ATTEMPT-09, ATTEMPT-10, CLOSE-03, PLAT-05, PLAT-07
OPS-003	Whitespace-only financial evidence is rejected	ATTEMPT-05, DISPUTE-02
OPS-004	Unmatched parcel count updates after matching	PARCEL-01, PARCEL-02
TASK-01	Task owners qualify for the current stage and scope	ADM-SCOPE-02, ADM-SCOPE-03, ADM-SCOPE-04, ADM-SCOPE-05, ADM-SCOPE-07, ADM-SCOPE-08, ADM-SCOPE-09, ADM-SCOPE-10, PLAT-03
TASK-02	Reopening clears current resolution time and retains history	CS-M05-AC08, CS06-02, CS08-05
UI-01	Narrow layout retains full identity role and scope	PLAT-17

Source and verification references

Application baseline: published Version 6, commit a5b3b2b9be1fa3860382ec549beeaaad6c887cb0. Review URL: https://cs-portal-guided-review.ramesh-s325387.chatgpt.site/. Source package and test evidence: Support_Desk_V6_Developer_Handoff.zip and Support_Desk_Final_Validation_Report.pdf, dated 30 October 2026.

Repository references in module sections are relative to cs-portal-prototype. Primary sources include REVIEW-GUIDE.md, dist/data.js, dist/portal.js, dist/workflow-access.js, dist/operations.js, dist/replacements.js, dist/rma-waivers.js, dist/cancellations.js, dist/finance.js, dist/admin.js, dist/catalog.js and dist/cs-records.js, with their tests/*.test.mjs regression files.

Security verification recommendation: use the applicable OWASP ASVS 5.0.0 requirements as a review framework, selecting and documenting an assurance level with the security owner. This is not a certification statement. Official reference: https://asvs.owasp.org/.

Accessibility verification recommendation: evaluate the implementation against WCAG 2.2 AA, including keyboard access, focus visibility, error handling, target size and accessible authentication. Official reference: https://www.w3.org/TR/WCAG22/. Logging implementation reference: https://cheatsheetseries.owasp.org/cheatsheets/Logging_Cheat_Sheet.html.

Terminology

Term	Meaning
RMA	Return merchandise authorisation: the approved inbound collection and receipt record. Every warehouse supports RMA.
Return waiver	An explicit manager decision to waive outstanding collection quantities. It is neither a receipt nor a refund approval.
Approved scope	The exact request revision, item lines, quantities, arrangement and applicable amount that were decided.
Remedy confirmation	Evidence that the agreed customer remedy or replacement delivery was actually completed for the exact request revision.
Assigned to	The responsible ticket owner; separate from the person currently working and from stage-specific task owners.
Working now	The current collaboration claim, requiring explicit takeover where another person is working.
First missed deadline	The original due occurrence at which a task became late, with its owner then; later reschedules do not erase it.
Created cohort	Tickets selected by creation date whose outcomes are evaluated at report time; not historical state at the end of that range.
Authorised refund	Approved financial limit and scope. A payment is recorded separately as an externally executed attempt and verified result.
Reserved amount	Money in Pending or Outcome unknown attempts; unavailable for another attempt until a verified final result.
Source snapshot	Immutable historical values retained when current labels, roles, identities or catalogue records later change.
Acceptance criterion	An observable Given, When and Then contract to execute in the target implementation; it is not a claim of production certification.